High-Probability and Medium-Probability Conversion Techniques for AMC 12 and COMC

AMC 12 & COMC Strategy Guide - Enhanced Edition

September 28, 2025

Abstract

This guide presents systematic conversion techniques for AMC 12 and the Canadian Open Mathematics Challenge (COMC), organized by reliability. High-probability conversions work broadly across problem types; medium-probability ones require more insight but often unlock difficult questions. The 2025 update rebalances priorities to reflect recent trends (2020–2024): algebra remains õne-third of problems, number theory has risen to $\tilde{2}0\%$, and modern problems frequently integrate multiple topics. We elevate Vieta's formulas, Power of a Point, Inclusion–Exclusion, and Cauchy–Schwarz to high priority; add Linearity of Expectation, Bijections, and Functional Equations; and note advanced geometry tools (Radical Axis, Spiral Similarity, and Inversion as enrichment). This document is a foundation rather than a complete syllabus: contests span hundreds of concepts, so breadth and adaptability are essential, especially for COMC's proof-based parts.

Contents

1 Introduction **4**

2 High-Probability Conversion Techniques **6**

2.1 Technique 1: Algebraic Substitution 6
2.2 Technique 2: Coordinate Geometry Conversion 7
2.3 Technique 3: Case Analysis 10
2.4 Technique 4: Symmetry Exploitation 12
2.5 Technique 5: Simon's Favorite Factoring Trick (SFFT) 14
2.6 Technique 6: Pigeonhole Principle 16
2.7 Technique 7: AM-GM Inequality 18
2.8 Technique 8: Completing the Square 20
2.9 Technique 9: Base and Digit Conversions 21
2.10 Technique 10: Angle Bisector Theorem 23
2.11 Technique 11: Linearity of Expectation 25

2.12 Technique 12: Bijection Principle . . . 27
2.13 Technique 13: Characteristic Equations - Linear Recurrences . . . 28
2.14 Technique 14: Euler's Totient Function . . . 30
2.15 Technique 15: Rearrangement Inequality . . . 32
2.16 Technique 16: Vieta's Formulas . . . 34
2.17 Technique 17: Geometric Probability . . . 36
2.18 Technique 18: Complex Number Conversion . . . 38
2.19 Technique 19: Modular Arithmetic . . . 40
2.20 Technique 20: Fermat's Little Theorem . . . 42
2.21 Technique 21: Inclusion-Exclusion Principle . . . 45
2.22 Technique 22: Cauchy-Schwarz Inequality . . . 47
2.23 Technique 23: Power of a Point . . . 49
2.24 Technique M1: Stars and Bars . . . 51
2.25 Technique M3: Shoelace Formula . . . 53
2.26 Technique M9: Triangle Trigonometry (Law of Cosines and Sines) . . . 53
2.27 Technique M11: Circle Tangency and Angle Toolkit . . . 54
2.28 Technique M12: Discriminant Techniques . . . 54
2.29 Technique M13: Logarithm Identities . . . 55
2.30 Technique M14: Complementary Counting . . . 55
2.31 Technique M27: Geometric Restart Recursion . . . 56
2.31.1 Common Mistakes and When NOT to Use This Technique . . . 59
2.31.2 Verification Methods . . . 60
2.31.3 Edge Cases and Boundary Conditions . . . 60
2.32 Technique M39: Geometric Inversion . . . 61
2.33 Technique M40: Approximation and Estimation . . . 65
2.34 Technique M41: Pattern Induction from Small Cases . . . 70

3 Medium-Probability Conversion Techniques 74
3.1 Technique M2: Burnside's Lemma (Pólya Counting) . . . 74
3.2 Technique M4: Pick's Theorem . . . 75
3.3 Technique M5: Frobenius/McNugget (Two Denominations) . . . 76
3.4 Technique M6: Extended Euclidean Algorithm . . . 76
3.5 Technique M7: Ptolemy's Theorem (Cyclic Quadrilaterals) . . . 77
3.6 Technique M8: Double Counting . . . 77
3.7 Technique 24: Trigonometric Substitution . . . 78
3.8 Technique 25: Generating Functions . . . 79
3.9 Technique 26: Invariants and Monovariants . . . 82
3.10 Technique 27: Power-Sum Recurrences . . . 83
3.11 Technique 28: Derangements . . . 85
3.12 Technique 29: Telescoping Series . . . 87
3.13 Technique 30: Polynomial Division . . . 89
3.14 Technique 31: Stewart's Theorem . . . 91
3.15 Technique 32: Floor and Ceiling Functions . . . 93
3.16 Technique 33: Rational Root Theorem . . . 95
3.17 Technique 34: Jensen's Inequality . . . 97

3.18 Technique 35: Mass Points . . . 99
3.19 Technique 36: Ceva's Theorem . . . 101
3.20 Technique 37: Menelaus's Theorem . . . 103
3.21 Technique 38: Chinese Remainder Theorem . . . 105
3.22 Technique 39: Graph Modeling (Graph Theory Conversion) . . . 107
3.23 Technique 40: Synthetic Division . . . 109
3.24 Technique 41: Functional Equations . . . 111
3.25 Technique M10: Triangle/Quadrilateral Area Toolkit . . . 113
3.26 Technique M15: Euler's Polyhedral Formula . . . 113
3.27 Technique M16: Descartes Circle Theorem . . . 114
3.28 Technique M17: Conjugate Root Theorem . . . 114
3.29 Technique M18: Sophie's Germain Identity . . . 115
3.30 Technique M19: Wilson's Theorem . . . 115
3.31 Technique M20: Legendre's Formula (p-adic valuation of $n!$) . . . 115
3.32 Technique M21: Divisor Functions $\tau(n)$ and $\sigma(n)$ . . . 116
3.33 Technique M22: Binomial Theorem and Pascal Identities . . . 116
3.34 Technique M23: Rates and Work Templates . . . 117
3.35 Technique M24: Degenerate Triangles (Triangle Inequality Boundary) . . . 117
3.36 Technique M25: Francesca Volume (Piero della Francesca's Tetrahedron Formula) . . . 119
3.37 Technique M26: First Passage Time . . . 120
3.38 Technique M28: Consecutive Integer Decomposition . . . 122
3.39 Technique M29: Trigonometric Monotone Scanning . . . 125
3.40 Technique M30: Similarity Ratio Scaling . . . 126
3.41 Technique M31: Volume Decomposition . . . 128
3.42 Technique M32: MediantBound (Farey Sequence Bounds) . . . 129
3.43 Technique M33: GCD-LCM Bounds . . . 131
3.44 Technique M34: Cofunction Solving . . . 132
3.45 Technique M35: Constructive Polynomial Example . . . 137
3.46 Technique M36: Legendre Symbol and Quadratic Reciprocity . . . 139
3.47 Technique M37: Constructive Polynomial Example . . . 143
3.48 Technique M38: Pell Equation . . . 147
3.49 Technique M42: Focus-Directrix Projections . . . 150
3.50 Technique M43: Curvature and Tangency Analysis . . . 153

1 Introduction

The AMC 12 and COMC require not just mathematical knowledge but strategic problem-solving skills. Conversion techniques are systematic approaches to transform complex problems into simpler, more manageable forms. We categorize these techniques by their reliability:

- **High-Probability Conversions**: Standard transformations that work consistently across problem types
- **Medium-Probability Conversions**: Techniques requiring pattern recognition and insight

2025 Priority Revisions (Read First)

Based on recent contest trends, we adjust priorities to focus study time where it pays off most:

- **Upgraded to High Priority:** Vieta's Formulas;
  Power of a Point; Inclusion–Exclusion; Cauchy–Schwarz Inequality
- **Added/Emphasized:** Linearity of Expectation (probability); Bijections (combinatorics); Functional Equations (algebra/COMC)
- **Advanced Geometry (enrichment):** Radical Axis; Spiral Similarity; Inversion (rare on AMC 12; useful to know)
- **Downgraded to Medium:** Hockey Stick Identity; Vandermonde's Identity; Derangements

Integrating Techniques

Recent problems often blend algebra, geometry, number theory, and combinatorics. Expect to combine ideas (e.g., coordinate geometry + Vieta; PoP + similarity; modular arithmetic inside a counting argument). When stuck, ask: can I translate to equations, switch coordinate systems, reduce modulo, or reframe as a counting/expectation question?

This document aligns with the Rapid Problem Classification System in the main guide (see main guide Section 1.2 for an overview of domains and tags). A short legend for the domain tags used in section headers: ALG = Algebra, GEO = Geometry, CNT = Counting/Combinatorics, NT = Number Theory, PROB = Probability.

Recognition Index (WHO/WHEN → Technique)

Use this as a speed index; confirm WHEN conditions before executing

Instant Recognition (1–2s)

WHO/WHEN Cue	Technique	First Mo
ALG: repeated/nested chunk across terms; degree reduces by naming	T1-AlgSub	Name chunk; re back-subs
NT/Digits: last digits/divisibility/base-b	T19-ModArith, T9-BaseDigits	Choose lus/base; apply test
GEO: tangent/secant/chords/cyclic hints	T23-PoP	Apply p tions; con triangles
GEO: bisector partitions a side	T10-ABT	Set $\frac{BD}{DC}$ = ratios
PROB: expected count without independence	T11-LOE	Sum indi linearity
SEQ: telescoping differences/partials	T29-Telescoping	Rewrite ences; sum

Quick Recognition (4–10s)

WHO/WHEN Cue	Technique	First Mo
ALG/NT: $xy + ax + by(+c)$; integer target; factor pairs enumerate	T5-SFFT	Complete list factor
ALG: quadratic/vertex needed; squares + linear terms	T8-CompleteSquare	Factor lea complete s
GEO: synthetic stalls; algebraic relations clear	T2-CoordGeo	Place write equa
CNT: overlapping properties / at least one	T21-PIE	Tabulate tions; alte
CNT: identical objects into boxes	TM1-StarsBars	Encode $\sum$ ply formu
SEQ: constant-coefficient recurrence	T13-CharEq	Form ch equation;
INEQ/ALG: sums of squares/reciprocals	T22-CauchySchwarz	Match vec C-S / Titu
NT: multiple congruences, coprime moduli	T38-CRT	Solve per recombine
ALG: integer/rational root plausible	T33-RationalRoot	Test divi thetic divi
ALG: symmetric sums/products of roots	T16-Vieta	Rewrite symmetric
ALG/INEQ: pairing/assignment optimization	T15-RearrangeInequ	Sort; s pairing
ALG: trig sums with asymptotes; monotone intervals	TM29-TrigMonotoneScan	Find deri ate sign endpoints

WHO/WHEN Cue	Technique	First
GEO: concurrency/collinearity by	T36-Ceva / T37-	Set p

2 High-Probability Conversion Techniques

2.1 Technique 1: Algebraic Substitution

Short Name: T1-AlgSub

Definition 2.1 (Algebraic Substitution)**.** *Replace complex expressions with simpler variables to reduce computational complexity and reveal underlying structure.*

WHEN: When a repeated or structured subexpression can be named to lower degree or complexity (e.g., set $t = x + \frac{1}{x}$, $u = \sqrt{x}$, $v = a^x$); when nested compositions, symmetric expressions, or cyclic sums recur; when a substitution linearizes a relation or decouples variables before back-substitution. Key signals include repeated/nested chunks appearing across terms, and substitution should reduce degree or decouple variables; enforce domain/reversibility after back-substitution. **Configuration cues:**

- Recurring or nested subexpression that can be named as a single variable (e.g., nested compositions $f(f(f(x)))$ or iterated operations like repeated digit sums)
- Repeated expressions (e.g., $x^2 + 3x$ in multiple places) or symmetric forms $\left(x + \frac{1}{x}\right)$, $x^2 + \frac{1}{x^2}$
- Common subexpressions across terms or on both sides of an equation/inequality

Constraints cues:

- Domain, sign, and integrality constraints must be preserved and rechecked after back-substitution
- Non-injective substitutions (e.g., $t = x + \frac{1}{x}$) can produce extraneous candidates; verify against original constraints
- For substitutions like $t = x + \frac{1}{x}$ with real $x \neq 0$, enforce $t \leq -2$ or $t \geq 2$ before back-substituting

Objective cues:

- Reduce degree or algebraic complexity (obtain a lower-degree equation in the new variable)
- Unify repeated structure to evaluate a composite expression or isolate it to solve

Why It Works: Simplifies manipulation and often reveals hidden symmetries or patterns.

WHO: ALG/NT with symmetric expressions and root finding; nested, radical, or exponential expressions; goal is to reduce degree/complexity.

How:

- **First Moves:**
  1. Scan for repeated or nested subexpressions and name the most unifying one.

2. State domain/reversibility constraints for the substitution (e.g., $x \neq 0$; for $t = x + \frac{1}{x}$ with real $x \neq 0$, $t \leq -2$ or $t \geq 2$).
3. Rewrite all occurrences consistently and simplify; solve in the new variable.
4. Back-substitute and filter candidates against the original domain and constraints.

- **Bailouts:** If the substitution does not reduce degree or decouple variables, try a different naming ($u = x^2$, $v = \sqrt{x}$), use symmetry/Vieta to reduce, or revert and attempt factoring/graphing/numerical checks.
- **Pitfalls:** Losing domain information; creating extraneous solutions via non-injective substitutions; replacing only some occurrences; failing to verify reversibility when needed.
- **Example:** Two frequent uses: (a) nested compositions (e.g., digit sums) — evaluate inner to outer; (b) symmetric forms — set $t = x + \frac{1}{x}$ to reduce to a quadratic in t.

AMC 12A 2019, Problem 7

For a positive integer n, let $f(n)$ be the sum of the digits of n. What is the value of

$$f(f(f(2019)))$$

Solution

Step 1: Calculate $f(2019)$

$$f(2019) = 2 + 0 + 1 + 9 = 12$$

Step 2: Calculate $f(f(2019)) = f(12)$

$$f(12) = 1 + 2 = 3$$

Step 3: Calculate $f(f(f(2019))) = f(3)$

$$f(3) = 3$$

Answer: 3
This demonstrates the substitution principle: working step-by-step through nested functions.

2.2 Technique 2: Coordinate Geometry Conversion

Short Name: T2-CoordGeo

Definition 2.2 (Coordinate Geometry Conversion)**.** *Transform geometric problems into algebraic ones by introducing a coordinate system.*

WHEN: Geometry that simplifies with coordinates: compute intersections/distances/slopes, write equations of lines/circles/conics, or exploit symmetry by placing origin/axes advantageously. Use when synthetic relations are unclear by 20–25s or immediately if ratios/lengths/angles translate cleanly; key signals include line–circle intersections, midpoints/centroids, cevians, collinearity/concurrency; switch earlier if diagram is messy.

Configuration cues:

- Distances, midpoints, slopes, perpendicularity, or parallelism that map to coordinate formulas
- Circles, lines, parabolas, or loci with simple equations
- Natural symmetry points for placing the origin or aligning axes (center, vertex, centroid)
- Angle/perpendicular conditions expressible via vectors: dot products ($u \cdot v = 0$) or slope products ($m_1 m_2 = -1$)

Constraints cues:

- Choose coordinates that simplify expressions; avoid unnecessary radicals/fractions
- Preserve geometric constraints (lengths, angle conditions) after coordinate choices
- Watch for degenerate/excluded cases (vertical lines, coincident points) implied by equations

Objective cues:

- Convert geometry to solvable algebraic systems
- Enable direct computation of intersections, distances, and angles
- Reduce diagram dependence using invariant equations
- Translate angle/perpendicular constraints into dot-product or cosine-law equations

Why It Works: Converts visual/spatial reasoning into systematic calculation.

WHO: GEO coordinate geometry problems amenable to coordinates; solvers comfortable with algebraic systems, vectors, and distance formulas.

How:

- **First Moves:**
  1. Place the origin at a symmetry/center/vertex and align axes to convenient lines.
  2. Choose simple coordinates (prefer integers); write equations for all constraints (distance, slope, dot product).
  3. Reduce to a solvable algebraic system, solve, and map the solution back to the diagram.

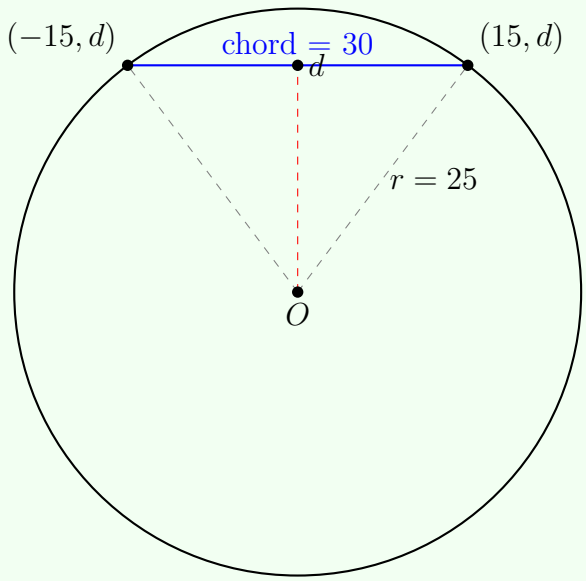
- **Bailouts:** If algebra inflates, re-place the frame (translate/rotate/scale) to simplify numbers; switch to vectors/complex numbers; or fall back to synthetic tools (similarity, Power of a Point) if the coordinate model is cumbersome.
- **Pitfalls:** Choosing degenerate frames (e.g., vertical lines when using slope formulas), introducing special positions that lose generality, or violating given constraints after placement.
- **Example:** See the chord–distance computation below; general template: write locus/equations, intersect, and interpret.

AMC 12B 2018, Problem 11

A circle with center O has radius 25. A chord of length 30 is at distance d from O. Find d.

Solution

Setup: Place the center at the origin $(0, 0)$ and the chord horizontally.



Let the chord endpoints be at $(15, d)$ and $(-15, d)$.
Constraint: These points lie on the circle of radius 25:

$$15^2 + d^2 = 25^2$$

$$225 + d^2 = 625$$

$$d^2 = 400$$

$$d = 20$$

Answer: 20
General formula: For radius r and chord length ℓ, the perpendicular distance from the center to a chord of length ℓ is $d = \sqrt{r^2 - (\ell/2)^2}$.

2.3 Technique 3: Case Analysis

Short Name: T3-CaseAnalysis

Definition 2.3 (Case Analysis)**.** *Systematically divide a problem into exhaustive, mutually exclusive cases.*

WHEN: When natural partitions (parity, sign, residues, ranges) isolate behaviors; when absolute values/min–max or piecewise definitions force different formulas; when finiteness or integrality bounds allow exhaustive but controlled enumeration. Use when parity/sign/residue-class splits or thresholds are explicit or natural; objective benefits from minimal exhaustive cases; stop if case explosion looms and pivot to PIE/bijection. Specific scenarios include

parity/sign splits, modular residue classes, threshold/range splits, boundary/equality cases in inequalities or absolute values.

Configuration cues:

- Natural partitions: parity, sign, residue classes modulo k, ranges or thresholds
- Piecewise definitions, absolute values, min/max, or conditional statements
- Integer constraints suggesting finitely many cases

Constraints cues:

- Ensure cases are mutually exclusive and collectively exhaustive; include boundary cases
- Maintain assumptions within each case; do not mix conclusions across cases
- Control case explosion using symmetry or invariants when available

Objective cues:

- Decompose the problem into simpler subproblems
- Isolate behaviors to count or bound precisely
- Structure proofs by exhaustion or contradiction via cases

Why It Works: Reduces complex problems to simpler sub-problems.

WHO: CNT/NT/INEQ counting problems with discrete or piecewise conditions; goal is exhaustive, disjoint coverage.

WHAT: Partition by mutually exclusive conditions and solve each subproblem.

How:

- **First Moves:**
  1. List natural partitions (parity/sign/ranges/residue classes) and boundary cases.
  2. Prove cases are mutually exclusive and collectively exhaustive.
  3. Order cases from most constrained to least; solve and aggregate.
- **Bailouts:** If cases explode, use symmetry to merge equivalent cases, switch to complementary counting, or impose invariants/mod constraints to prune possibilities.
- **Pitfalls:** Overlapping or missing cases, forgetting boundary equalities, mixing assumptions across cases, or double-counting.
- **Example:** See the gcd–lcm example below illustrating coprime factor splits.

AMC 12A 2020, Problem 15

How many ordered pairs of positive integers (m, n) satisfy $\gcd(m, n) = 2$ and $\operatorname{lcm}(m, n) = 108$?

Solution

Key Identity: $\gcd(m, n) \cdot \text{lcm}(m, n) = m \cdot n$
So: $2 \cdot 108 = m \cdot n = 216$
Setup: Let $m = 2a$ and $n = 2b$ where $\gcd(a, b) = 1$
Then: $4ab = 216$, so $ab = 54$
Case Analysis: Find pairs (a, b) with $\gcd(a, b) = 1$ and $ab = 54$
$54 = 2 \cdot 3^3$
Valid pairs: $(1, 54), (2, 27), (54, 1), (27, 2)$
Answer: 4

2.4 Technique 4: Symmetry Exploitation

Short Name: T4-Symmetry

Definition 2.4 (Symmetry Exploitation)**.** *Use inherent symmetries in the problem to reduce calculation or eliminate cases.*

WHEN: When the problem is invariant under a symmetry group (reflections, rotations, variable permutations) so a fundamental region or representative case suffices; when equal-role variables suggest setting them equal at optima; when regularity or isosceles structure collapses computation. Use for regular polygons, isosceles/centrally symmetric figures; symmetric sums/products and inequality problems; counting with indistinguishable/label-symmetric objects.

Configuration cues:

- Isosceles or regular figures; central, reflective, or rotational symmetry in geometry
- Symmetric sums/products; interchangeable variables with equal roles
- Constraints invariant under permutations, reflections, or rotations

Constraints cues:

- Confirm all constraints respect the symmetry; detect symmetry-breaking conditions
- Avoid double-counting equivalent configurations; restrict to a fundamental domain
- Verify representatives cover all distinct cases

Objective cues:

- Compute on a representative portion and propagate by symmetry
- Collapse variables via symmetric parameters or equalities
- Translate geometric symmetry into simpler coordinates or equations

Why It Works: Reduces the problem space significantly.

WHO: GEO/ALG/CNT with clear reflection/rotation/permutation symmetry; solvers aiming to avoid case explosion.

WHAT: Reduce the domain to a fundamental region and reconstruct by symmetry.

How:

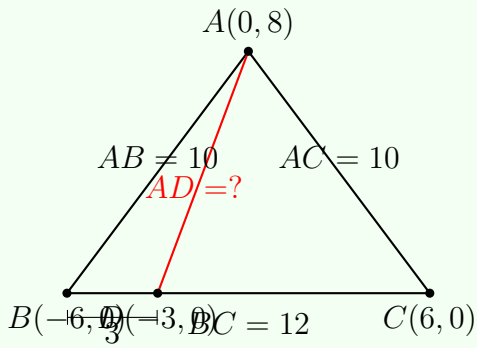
- **First Moves:**
  1. Identify the symmetry (reflection/rotation/permutation).
  2. Restrict to a fundamental region or impose equal-role variables when justified.
  3. Compute in the reduced setting and propagate by symmetry.
- **Bailouts:** If symmetry is partial or broken by constraints, switch to coordinates or fall back to case analysis on symmetry classes.
- **Pitfalls:** Imposing symmetry where constraints break it; double-counting equivalent configurations; neglecting edge/boundary cases.
- **Example:** See the isosceles triangle computation below; equal-role variables often indicate equality at an extremum.

AMC 12B 2019, Problem 16

In triangle ABC with $AB = AC = 10$ and $BC = 12$, point D lies on $\overline{BC}$ with $BD = 3$. Find AD.

Solution

Symmetry Setup: Place the isosceles triangle with $B = (-6, 0)$, $C = (6, 0)$, and $A = (0, h)$.



Find height: Using $AB = 10$:

$$6^2 + h^2 = 100$$

$$h = 8$$

So $A = (0, 8)$ and $D = (-3, 0)$ (since $BD = 3$)

Calculate:

$$AD = \sqrt{(-3-0)^2 + (0-8)^2} = \sqrt{9 + 64} = \sqrt{73}$$

Answer: $\sqrt{73}$

2.5 Technique 5: Simon's Favorite Factoring Trick (SFFT)

Short Name: T5-SFFT

Definition 2.5 (Simon's Favorite Factoring Trick)**.** *Add and subtract the same term to create a factorable expression. Commonly used when dealing with expressions of the form $xy + ax + by + c$.*

WHEN: When an equation can be arranged as $xy+ax+by = c$ or $xy+ax+by+d = 0$ so that adding/subtracting a constant completes a rectangle; when integer solution enumeration reduces to factoring a constant and checking divisor pairs. Use for structure $xy+ax+by(+c)$ or shiftable to $(x+\alpha)(y+\beta) = k$; integer targets; factor pairs enumerate solutions. Also for integer-solution equations reducible to $(x + a)(y + b) =$ constant; problems where counting factor pairs yields all solutions.

Configuration cues:

- Bilinear-plus-linear forms $xy + ax + by + c$ or rearrangeable to that form
- Diophantine equations seeking integer or positive solutions
- Situations where "complete the rectangle" enables factoring

Constraints cues:

- Choose the correct constant to add/subtract: if $xy+ax+by=c$, then $(x+a)(y+b)=ab+c$; if $xy+ax+by+d=0$, then $(x+a)(y+b)=ab-d$
- Consider signs and all divisor pairs (positive and negative); apply domain constraints
- Verify back-substituted solutions satisfy the original equation

Objective cues:

- Convert to a product form enabling enumeration via divisor pairs
- Expose structure for counting or bounding integer solutions
- Separate variables to simplify solving

Why It Works: Creates groupable terms that factor nicely.
WHO: ALG/NT Diophantine equations with bilinear terms and linear parts; exploit factor pairs/divisibility.
WHAT: Add and subtract a constant to complete a rectangle: $(x+a)(y+b)=$ constant.
How:

- **First Moves:**
  1. Arrange as $xy+ax+by=c$ (or $xy+ax+by+d=0$).
  2. Add/subtract the constant to complete the rectangle: $(x+a)(y+b)=ab+c$ (or $ab-d$).
  3. Enumerate all (positive and negative) factor pairs of the RHS; back-solve and filter by domain.
- **Bailouts:** If the RHS has few factors or variables are not integral, try isolating one variable, factoring by grouping, completing the square, or apply modular constraints to prune possibilities.
- **Pitfalls:** Using the wrong constant; forgetting negative factor pairs; ignoring domain restrictions; double-counting symmetric pairs.
- **Example:** See the $(x+7)(y+5)=105$ example below.

Example: Factoring with SFFT

Factor $xy+3x+5y+15$.

Solution

Method: Group terms strategically.

$$xy + 3x + 5y + 15 = x(y + 3) + 5(y + 3) = (x + 5)(y + 3)$$

Alternative SFFT approach: If the constant term wasn't 15, we could add and subtract to make it work. For example, if we had $xy + 3x + 5y + 14$:

$$xy + 3x + 5y + 14 = xy + 3x + 5y + 15 - 1 = (x + 5)(y + 3) - 1$$

Answer: $(x + 5)(y + 3)$ for the original problem

AMC 12 Example

Find the number of integer solutions (x, y) to $xy + 5x + 7y = 70$.

Solution

Add and subtract 35 to complete the rectangle:

$$xy + 5x + 7y + 35 = 70 + 35 \implies (x + 7)(y + 5) = 105.$$

Factor $105 = 3 \cdot 5 \cdot 7$. For each integer divisor d of 105, set

$$x + 7 = d, \quad y + 5 = \frac{105}{d}.$$

This yields one integer pair (x, y) per divisor d. Since 105 has $(1+1)(1+1)(1+1) = 8$ positive divisors and also 8 negative divisors, there are $8 + 8 = 16$ integer solutions.
Answer: 16

2.6 Technique 6: Pigeonhole Principle

Short Name: T6-Pigeonhole

Definition 2.6 (Pigeonhole Principle)**.** *If n items are placed into m containers with $n > m$, then at least one container must contain more than one item. More generally, if n items are placed into m containers, at least one container contains at least $\lceil n/m \rceil$ items.*

WHEN: Guarantee a duplicate/coincidence, same remainder, or a bounded gap by pigeonholing more items than categories; force a minimum occupancy of at least $\lceil n/m \rceil$; convert "difference divisible by m", "two within distance δ", fractional-part collisions, subset-sum modulo m, or packing points into subintervals/boxes into binning arguments. Use for residues modulo m; parity/residue classes; calendars (months/days/weeks); interval/arc partitions on $[0, 1]$ or a circle; rounding/floor/ceiling bins; subset sums modulo m (e.g., "n integers $\Rightarrow$ two share a remainder" or "some subset sum divisible by n"); nearest-neighbor/spacing bounds; lattice points in boxes.

Configuration cues:

- More items than categories: identify the "pigeons" and the "holes" and verify counts n, m with $n > m$ (or a target occupancy threshold)
- Natural binning: residues modulo m; partitions of an interval; rounding/floor/ceiling bins
- Collisions sought: duplicates, shared remainders/days, or near pairs in subdivided intervals

Constraints cues:

- Guarantees existence/bounds, not which items collide
- Categorization must be well-defined, complete, and disjoint (handle bin boundaries carefully)
- For stronger statements, use the bound $\lceil n/m \rceil$; do not infer equal distribution

Objective cues:

- Force a duplicate or coincidence
- Certify minimum/maximum occupancy bounds
- Show two elements share a class or lie in the same subinterval

Why It Works: Provides guaranteed bounds on distributions.

WHO: CNT/NT/GEO existence and extremal problems with discrete-geometry questions; solvers comfortable with residues, floors/ceilings, and interval partitions who want unconditional guarantees (no explicit construction).

WHAT: Partition objects into categories and use items > containers to force a collision; conclude a lower bound of at least $\lceil n/m \rceil$ in some container.

How:

- **First Moves:**
  1. Define the pigeons (items) and the pigeonholes (categories) precisely.
  2. Count n and m; deduce that some hole contains at least $\lceil n/m \rceil$ items.
  3. Translate the forced collision back to the claim (same remainder, same month, bounded gap, etc.).
  4. For "difference divisible by m" questions, take residues mod m as the holes.
  5. For real numbers, partition an interval into equal subintervals and apply the principle to those subintervals.
- **Bailouts:** If collisions are not forced with your first partition, redefine pigeons/holes, merge or refine categories, or switch to residue classes/interval bins (generalized pigeonhole principle) to tighten bounds.

- **Pitfalls:** Misdefining holes so they do not cover all items or overlap; forgetting ceilings/floors; using the wrong modulus; edge cases (e.g., leading zeros) that break the partition.
- **Example:** Months vs birthdays; among six integers, two share a remainder mod 5.

Example: Birthday Problem Variant

In a group of 13 people, prove that at least two people have birthdays in the same month.

Solution

Setup: We have 13 people (pigeons) and 12 months (pigeonholes).
Application: Since $13 > 12$, by the Pigeonhole Principle, at least one month must contain at least two people's birthdays.
Generalization: With 25 people, at least one month contains at least $\lceil 25/12 \rceil = 3$ birthdays.

AMC 12 Example

Prove that among any 6 integers, there exist two whose difference is divisible by 5.

Solution

Key Insight: Consider remainders when divided by 5.
Setup: Any integer has remainder 0, 1, 2, 3, or 4 when divided by 5 (5 pigeonholes).
Application: With 6 integers and 5 possible remainders, at least two integers must have the same remainder modulo 5.
Conclusion: If $a \equiv b \pmod 5$, then $5|(a-b)$.

2.7 Technique 7: AM-GM Inequality

Short Name: T7-AMGM

Definition 2.7 (AM-GM Inequality)**.** *For nonnegative real numbers $a_1, a_2, \ldots, a_n$,*

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n},$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$.

WHEN: Optimize or bound expressions in nonnegative variables under fixed sum/product (or after homogenizing/normalizing); classic forms like $x + \frac{c}{x}$ and equal-split arguments; apply weighted AM–GM via term multiplicity and use equality cases to guess extremizers. Use for two-term $x + \frac{c}{x}$; sums with fixed product and products with fixed sum; symmetric/weighted sums (via term repetition); normalized geometry bounds (area/perimeter, inscribed figures); inequality chains combined with Cauchy/Jensen.

Configuration cues:

- Symmetric sum/product structures; constraints of fixed sum or fixed product
- Forms like $x + \frac{c}{x}$ with $x > 0$; splitting a constant into parts
- Weighted versions when terms have multiplicities

Constraints cues:

- Requires nonnegative terms; for $x + \frac{c}{x}$, need $x > 0$; equality when all compared terms are equal
- AM–GM alone may be insufficient when signs or linear constraints dominate; consider Cauchy–Schwarz or Jensen instead
- Weighted AM–GM requires nonnegative weights summing to 1 (or use integer multiplicities via term repetition)

Objective cues:

- Minimize sums for a fixed product; maximize products for a fixed sum
- Create clean lower/upper bounds and identify equality cases

Why It Works: The arithmetic mean is always at least the geometric mean; equality holds when variables are equal.

WHO: ALG/GEO inequality problems with nonnegative reals; solvers leveraging equality cases (often variables equal) and comfortable with homogenization/normalization.

WHAT: Apply AM $\geq$ GM (possibly weighted) to relate a sum and a product; equality occurs when the terms are equal.

How:

- **First Moves:**
  1. Verify all terms are nonnegative.
  2. Group or duplicate terms so counts match the structure; choose weights if needed.
  3. Apply AM–GM to obtain the bound.
  4. Use the equality condition (all terms equal) to identify candidate extremizers.
  5. Check domain/constraints and adjust if equality is infeasible.
- **Bailouts:** If terms are not nonnegative or counts do not align, homogenize or rescale, try a substitution, switch to weighted AM–GM or pair with Cauchy/Jensen/CS; alternatively, complete the square or use calculus when appropriate.
- **Pitfalls:** Applying to negative terms; ignoring weights/multiplicities; assuming equality is achievable when constraints forbid it; mismatched term counts leading to a wrong bound.
- **Example:** Minimize $x + \frac{9}{x}$ for $x > 0$ by AM–GM with equality at $x = 3$.

Example: Minimize $x + \frac{9}{x}$

Find the minimum value of $x + \frac{9}{x}$ for $x > 0$.

Solution

By AM-GM on x and $\frac{9}{x}$:

$$x + \frac{9}{x} \geq 2\sqrt{x \cdot \frac{9}{x}} = 2 \cdot 3 = 6,$$

with equality when $x = \frac{9}{x}$, i.e., $x = 3$.
Answer: Minimum is 6 at $x = 3$.

2.8 Technique 8: Completing the Square

Short Name: T8-CompleteSquare

Definition 2.8 (Completing the Square)**.** *For $a \neq 0$, rewrite the quadratic $ax^2 + bx + c$ as*

$$a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right).$$

This reveals the vertex and makes optimization and solving more direct.

WHEN: Any single-variable quadratic target or constraint: find a vertex/extremum; solve/graph quadratic equations/inequalities; rewrite conics in standard form; expose distance-squared structure; simplify expressions to compare or bound. Use for optimizing ax^2+bx+c; quadratic inequalities; vertex/axis of symmetry; standard forms of circles/ellipses/parabolas/hyperbolas; distance to a point/line; completing squares in two variables to identify loci.

Configuration cues:

- Clear quadratic in one variable $ax^2 + bx + c$; distance-squared expressions
- Conic identification in two variables; vertex/axis requested
- Quadratic integrands amenable to shifting

Constraints cues:

- Works directly for single-variable quadratics; cross terms xy may require a rotation or linear change of variables first
- Respect domain constraints; if $a < 0$, the vertex is a maximum
- Track constants to obtain the exact vertex and extremum value

Objective cues:

- Expose the vertex and extremum with corresponding x
- Rewrite conics in standard form
- Simplify integrands by completing the square

Why It Works: It isolates a perfect square, exposing the minimal (or maximal) value and corresponding x.

WHO: ALG/GEO coordinate geometry with quadratics, conics, or distance minimization; solvers preferring vertex-form reasoning.

WHAT: Rewrite a quadratic as a perfect square plus a constant (vertex form); in two variables, complete squares in both to reveal a circle/ellipse/hyperbola.

How:

- **First Moves:**
  1. Factor out a from the quadratic part.
  2. Add and subtract $\left(\frac{b}{2a}\right)^2$ inside the parentheses to create a square.
  3. Combine constants to finish vertex form and read off the extremum.
  4. For two variables, complete the square separately in x and y, then standardize.
- **Bailouts:** If cross terms (like xy) appear, rotate coordinates or perform a linear change of variables; if algebra expands, try AM–GM or derivative-based optimization as a check.
- **Pitfalls:** Forgetting to factor out the leading coefficient; sign errors when adding/subtracting the square; misreading the vertex when $a < 0$; ignoring domain constraints.
- **Example:** $(x-3)^2 + 4$ shows $x^2 - 6x + 13 \geq 4$ with minimum at $x = 3$.

Example: Minimize $x^2 - 6x + 13$

Find the minimum value of $x^2 - 6x + 13$ for real x.

Solution

Complete the square:

$$x^2 - 6x + 13 = (x-3)^2 + 4 \geq 4,$$

with equality at $x = 3$.

Answer: Minimum is 4 at $x = 3$.

2.9 Technique 9: Base and Digit Conversions

Short Name: T9-BaseDigits

Definition 2.9 (Base and Digit Conversions)**.** *If an integer N has base-b digits $d_k d_{k-1} \cdots d_1 d_0$ (with $0 \le d_i < b$), then*

$$N = \sum_{i=0}^{k} d_i b^i.$$

Useful congruences: $N \equiv \sum d_i \pmod{b-1}$ *(since* $b \equiv 1 \pmod{b-1}$*) and* $N \equiv d_0 \pmod{b}$.

WHEN: Digits/last-k-digits/divisibility-by-$3, 9, 11$ (or $b \pm 1$) questions; unknown-base constraints; palindromes/repeated digits/repunits; convert digit conditions into linear/modular equations and counting. Use for casting out nines (sum-of-digits) and elevens (alternating-sum) in base 10; last k digits via $\bmod b^k$ (e.g., 10^k; combine with CRT when helpful); palindromes/repeated-digit forms; repunits; unknown-base determination and conversions.

Configuration cues:

- Digit-sum constraints, last-digit or last-two-digit questions, divisibility by $3, 9, 11$
- Unknown-base puzzles; palindromes or repeated digits; mixed-radix counting framed in base b
- Translate "digits" wording into equations with $0 \le d_i < b$

Constraints cues:

- $b \ge 2$; digits are integers in $[0, b-1]$; avoid leading zeros unless allowed
- $N \equiv \sum d_i \pmod{b-1}$ and $N \equiv d_0 \pmod{b}$; for last two digits in base b, use $N \equiv d_1 b + d_0 \pmod{b^2}$ (e.g., in base 10, $N \equiv 10d_1 + d_0 \pmod{100}$)
- Digital-root heuristics are base-specific; prefer exact congruences
- For palindromes or repeated-digit forms, enforce positional equalities (e.g., $d_i = d_{k-i}$)

Objective cues:

- Convert digit conditions to linear or modular equations
- Deduce divisibility and last digits
- Count valid numbers under digit constraints (stars-and-bars or DP after base expansion)

Why It Works: Writing numbers in base-b linearizes digit relations and exposes modular patterns.

WHO: NT/CNT counting problems with digit-based counting/puzzle problems; solvers translating place-value wording into algebra and congruences.

WHAT: Express numbers in base b; translate digit constraints into equations/inequalities; exploit $b \equiv 1 \pmod{b-1}$ and $N \equiv d_0 \pmod{b}$.

How:

- **First Moves:**
  1. Write $N = \sum d_i b^i$ with $0 \le d_i < b$ and leading digit $d_k > 0$.
  2. Encode constraints (sum of digits, palindrome, bounds) as equations/inequalities on the d_i.
  3. Use congruences $N \equiv \sum d_i \pmod{b-1}$ and $N \equiv d_0 \pmod{b}$ when helpful.
  4. Solve for the digits and check that all base and leading-digit constraints are satisfied.
  5. If b is unknown, treat it as a variable and use additional conditions to determine it.

- **Bailouts:** Reduce to modular constraints (e.g., mod b, mod $b \pm 1$, or mod b^k), use CRT to combine conditions, or brute-force small digit ranges consistent with bounds.

- **Pitfalls:** Allowing leading zeros when disallowed; digits outside $[0, b-1]$; confusing $b-1$ vs $b+1$ tests; forgetting positional equalities for palindromes/repeated digits.

- **Example:** Two-digit $10a + b = 7(a + b)$ yields $21, 42, 63, 84$ after enforcing digit constraints.

Example: Two-Digit Number and Digit Sum

Find all two-digit numbers equal to 7 times the sum of their digits.

Solution

Let the number be $10a + b$ with digits $a \in \{1, \ldots, 9\}$, $b \in \{0, \ldots, 9\}$. The condition is

$$10a + b = 7(a + b) \implies 3a = 6b \implies a = 2b.$$

Valid digit pairs are $(a, b) \in \{(2, 1), (4, 2), (6, 3), (8, 4)\}$, giving numbers $21, 42, 63, 84$.
Answer: $21, 42, 63, 84$.

2.10 Technique 10: Angle Bisector Theorem

Short Name: T10-ABT

Definition 2.10 (Angle Bisector Theorem)**.** *In $\triangle ABC$, if the internal angle bisector from A meets $\overline{BC}$ at D, then*

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

WHEN: Triangle ratio/partition questions involving an internal or external angle bisector; incenter/excenter configurations; computing segment ratios on the opposite side or the bisector length. Use for internal/external bisectors; partition ratios along BC; incenter/excenter setups; multi-cevian configurations; problems asking for AD or the split of BC (often with Ceva/Mass Points).

Configuration cues:

- Triangle with an internal or external angle bisector; side ratios or segment partitions requested
- Incircle/excircle touchpoints; problems combining cevians (pairs well with Ceva/Mass Points)
- Known sides with a request for partitions or AD length

Constraints cues:

- Ensure the cevian truly bisects the angle; external version uses directed lengths
- Nondegenerate triangle; do not confuse with median or altitude
- Use the correct version (internal vs external) and be consistent with $a = BC, b = AC, c = AB$

Objective cues:

- Convert angle information into side ratios
- Solve for segment partitions along BC
- Compute AD when needed; combine with Ceva/Mass Points for multi-cevian systems

Why It Works: Follows from similar triangles created by the angle bisector or via area ratios.

WHO: GEO Euclidean geometry ratio problems featuring angle bisectors; solvers doing ratio chasing or combining cevians (Ceva, Mass Points, Stewart).

WHAT: Use $\frac{BD}{DC} = \frac{AB}{AC}$ (and the external variant) to convert angle-bisector information into segment ratios; apply sum constraints to solve for lengths.

How:

- **First Moves:**
  1. Set up the ratio $\frac{BD}{DC} = \frac{AB}{AC}$ (or the external version with directed lengths).
  2. Use $BD + DC = BC$ to solve for BD and DC.
  3. If the length AD is requested, apply the angle-bisector length formula or derive it via Stewart.
  4. Combine with Ceva/Mass Points when multiple cevians appear.
- **Bailouts:** Use area ratios or similar triangles if the direct ratio is obscured; apply Law of Cosines + Stewart to derive AD; switch to the external-bisector form when appropriate.
- **Pitfalls:** Confusing bisector with median/altitude; not verifying the line truly bisects the angle; mishandling directed lengths in the external case; swapping side labels.
- **Example:** With $AB : AC = 2 : 3$ and $BC = 10$, solve $BD : DC = 2 : 3$ to get $BD = 4$, $DC = 6$.

Classification: High-probability in AMC 12 geometry; try it early on ratio problems. **Works well with:** Ceva's Theorem and Mass Points for multi-cevian configurations. **External bisector:** If the external angle bisector from A meets the extension of $\overline{BC}$ at E, then $\dfrac{BE}{EC} = \dfrac{AB}{AC}$ with directed lengths. **Notation:** Let $a = BC$, $b = AC$, and $c = AB$. **Length of angle bisector:** If AD is the internal angle bisector, then $AD = \dfrac{2bc\cos(A/2)}{b+c}$. **Connection:** Using Stewart's Theorem together with the Law of Cosines in the two subtriangles, one can derive the Angle Bisector Theorem; conversely, assuming the Angle Bisector Theorem yields the classical angle-bisector length formula via Stewart.

Example: Side Partition

In $\triangle ABC$, suppose $AB = 6$, $AC = 9$, and $BC = 10$. The angle bisector from A meets $\overline{BC}$ at D. Find BD and DC.

Solution

By the theorem, $\dfrac{BD}{DC} = \dfrac{AB}{AC} = \dfrac{6}{9} = \dfrac{2}{3}$. With $BD + DC = 10$, we get $(2+3)k = 10$, so $k = 2$. Hence $BD = 4$ and $DC = 6$.
Answer: $BD = 4$, $DC = 6$

2.11 Technique 11: Linearity of Expectation

Short Name: T11-LOE

Definition 2.11 (Linearity of Expectation)**.** *For random variables $X_1, \ldots, X_n$ on the same probability space,*

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i],$$

regardless of independence. Indicator variables $\mathbf{1}_A$ satisfy $\mathbb{E}[\mathbf{1}_A] = \Pr(A)$.

WHEN: Expected-value questions where the target quantity is naturally a sum of many simple components (often indicators), independent or not; to bypass heavy casework and exploit symmetry by evaluating one term and scaling; natural for counting features across random structures. Use for coins/dice/cards; permutations (fixed points, cycles, inversions); occupancy (empty boxes, collisions; birthday-paradox style); coupon-collector–type counts; graphs (edges/triangles); random subsets/intersections.

Configuration cues:

- Target is an expected total: "expected number of ..." or a sum across many items
- Natural decomposition into simple components; define indicator variables for events of interest
- Symmetry suggests each term has the same expectation (evaluate one and multiply)

Constraints cues:

- Linearity does not require independence; it does not apply to products: in general $\mathbb{E}[XY] \neq \mathbb{E}[X]\,\mathbb{E}[Y]$
- If the index set or number of terms is random, condition on it or use appropriate results rather than summing a random count of terms blindly
- Ensure variables are integrable/bounded so expectations exist (automatic in finite contest settings)
- Define indicators carefully to avoid double-counting; each should contribute 0 or 1 exactly when its event occurs

Objective cues:

- Replace messy casework with termwise expectations of simple variables
- Convert "count of objects/properties" into a sum of probabilities via indicators
- Use symmetry to compute one term and scale up; obtain a clean expected value

Why It Works: Expectation distributes over sums; indicator variables turn counting into probabilities.

WHO: PROB/CNT probability and combinatorics on permutations, cards/dice, occupancy, graphs, and random selections; solvers comfortable with indicator variables, conditioning, and linearity without independence.

WHAT: Express the target quantity as a sum of simpler random variables (often indicators) and take expectations termwise.

How:

- **First Moves:**
  1. Define simple variables for each event of interest (often indicators).
  2. Write the total as a sum of these variables.
  3. Compute each expectation directly via basic probability.
  4. Sum the expectations to obtain the final expected value.
- **Bailouts:** If termwise expectations are messy, exploit symmetry or condition on a clean partition; switch to indicator variables or double-count via complements.
- **Pitfalls:** Assuming independence (unnecessary for linearity); defining variables that do not sum to the target; mixing raw counts with probabilities.
- **Example:** Indicators give $\mathbb{E}[\text{fixed points}] = 1$; n fair coins $\Rightarrow$ expected heads $= n/2$.

Example: Expected Number of Fixed Points

A random permutation of $\{1, 2, \ldots, n\}$ is chosen uniformly. What is the expected number of fixed points?

Solution

Let $X = \sum_{i=1}^{n} X_i$ with $X_i = \mathbf{1}_{\{\pi(i)=i\}}$. Then $\mathbb{E}[X_i] = \Pr(\pi(i) = i) = \frac{1}{n}$. Hence $\mathbb{E}[X] = \sum_i \frac{1}{n} = 1$.
Answer: 1 for all n.

AMC 12 Style: Expected Heads

If n fair coins are flipped, what is the expected number of heads?

Solution

Let $X = \sum_{i=1}^{n} X_i$ where $X_i = \mathbf{1}_{\{i\text{th coin is H}\}}$. Then $\mathbb{E}[X_i] = \frac{1}{2}$, so $\mathbb{E}[X] = \frac{n}{2}$.
Answer: $\frac{n}{2}$.

2.12 Technique 12: Bijection Principle

Short Name: T12-Bijection

Definition 2.12 (Bijection)**.** *Solve a counting problem by constructing a one-to-one correspondence between the complicated set and a simpler, well-counted set.*

WHEN: When objects admit a natural encoding into a standard, well-counted family (subsets, binary strings, compositions, lattice paths, Catalan objects, matchings), or when a canonical form eliminates overcounting from symmetry/order; ideal when a direct count is messy but a structure-preserving map is clear. Use for subsets/combinations; binary strings; integer compositions/partitions with bounds; lattice paths; Catalan objects (Dyck paths, balanced parentheses, noncrossing matchings); perfect/partial matchings; restricted permutations (e.g., exactly/at most k descents/inversions by encoding).

Configuration cues:

- Objects naturally encode as sequences, subsets, paths, or placements
- Constraints translate to standard classes (subsets, compositions via stars-and-bars, lattice paths, Catalan families)
- A canonical description avoids overcounting from symmetries or ordering

Constraints cues:

- Prove the correspondence is a true bijection: well-defined, injective, and surjective
- Track labels versus indistinguishability; account for symmetry (avoid over/undercounting)
- Preserve all constraints in both directions; if the map is k-to-1, adjust counts accordingly (not a bijection)

Objective cues:

- Reduce the problem to a known count (binomial coefficients, stars-and-bars, Catalan numbers, matchings)
- Transfer the count back through the inverse map, ensuring boundary conditions are respected
- Justify greedy or structural arguments by exhibiting an explicit correspondence

Why It Works: Reduces new problems to known counts by explicit structure-preserving mapping.

WHO: CNT combinatorics with constrained selections/arrangements, recursive or non-crossing structures; solvers adept at modeling by subsets/strings/paths and proving bijectivity.

WHAT: Build a one-to-one correspondence between the target set and a simpler, well-counted family.

How:

- **First Moves:**
  1. Describe objects canonically to reveal a natural mapping.
  2. Prove the map is injective and surjective (a bijection).
  3. Count the image via standard formulas (binomial, stars-and-bars, Catalan when applicable).
  4. Transfer that count back to the original problem via the bijection.
- **Bailouts:** If bijection proof stalls, weaken to an injection/surjection plus bounds, or switch to recursion/DP or inclusion–exclusion when overlaps are manageable.
- **Pitfalls:** Implicit many-to-one encodings; ignoring order/symmetry leading to over/undercounts; failing to verify both directions of the mapping.
- **Example:** Binary strings with k ones $\leftrightarrow$ k-subsets of $\{1, \ldots, n\}$, giving $\binom{n}{k}$.

Example: Binary Strings with k Ones

How many binary strings of length n contain exactly k ones?

Solution

Choose the k positions for ones: biject with k-subsets of $\{1, \ldots, n\}$, hence $\binom{n}{k}$.
Answer: $\binom{n}{k}$.

2.13 Technique 13: Characteristic Equations - Linear Recurrences

Short Name: T13-CharEq

Definition 2.13 (Characteristic Equations for Linear Recurrences)**.** *For a linear recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$, the characteristic equation is:*

$$x^k = c_1 x^{k-1} + c_2 x^{k-2} + \cdots + c_k$$

If the roots r_i are distinct, then $a_n = A_1 r_1^n + \cdots + A_k r_k^n$. For a root r with multiplicity m, include terms of the form $n^j r^n$ for $j = 0, 1, \ldots, m-1$.

WHEN: Linear recurrences with constant coefficients (homogeneous, or with simple nonhomogeneous terms via a particular solution such as a polynomial/exponential); to obtain closed forms, asymptotics (dominant root), or modular periodicity; to convert recursive counts to explicit formulas. Use for Fibonacci-type recurrences; tilings/counting recurrences; pattern-avoidance strings with linear relations; financial growth with lag; sums like $x^n + x^{-n}$; homogeneous and simple nonhomogeneous linear recurrences.

Configuration cues:

- Linear homogeneous recurrence with constant coefficients and given initial values
- Requests for the n-th term, closed form, growth rate, or limit behavior
- Fibonacci-like or factorable characteristic polynomials

Constraints cues:

- Method applies to linear recurrences with constant coefficients; for nonhomogeneous terms add a particular solution; variable-coefficient recurrences require other tools
- Handle repeated or complex roots correctly: include $n^j r^n$ terms and combine conjugates for real sequences
- Ensure sufficient initial conditions to determine constants; verify integrality/domain constraints if required

Objective cues:

- Obtain a closed form and compute a_n efficiently
- Analyze asymptotics (dominant root), periodicity modulo m, or identities among terms
- Convert recursive definitions into explicit expressions for further algebraic manipulation

Why It Works: The characteristic equation captures the growth rate of the sequence.

WHO: ALG/CNT sequences and series defining sequences by constant-coefficient recurrences with initial conditions; solvers familiar with roots/multiplicities and selecting particular solutions for standard forcings.

WHAT: Solve linear recurrences by forming the characteristic polynomial and expressing a_n as a linear combination of powers of roots (with n^j factors for repeated roots).

How:

- **First Moves:**

1. Form the characteristic equation $x^k - c_1x^{k-1} - \cdots - c_k = 0$.
2. Find roots and their multiplicities.
3. Write the general solution using r^n terms (and $n^j r^n$ for repeated roots).
4. Determine constants from initial conditions.

- **Bailouts:** If factoring is hard, use the quadratic/cubic formula, numerical factorization, or modular/root-of-unity insights; for nonhomogeneous terms, guess a particular solution and superpose.
- **Pitfalls:** Forgetting n^j factors for repeated roots; using complex roots without combining to real forms when needed; misapplying to variable-coefficient recurrences.
- **Example:** $a_n = 5a_{n-1} - 6a_{n-2}$ gives roots $2, 3$ and $a_n = A2^n + B3^n$ from initials.

Example: Solving a Recurrence

Find a closed form for a_n if $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 2$, $a_1 = 5$.

Solution

Step 1: Form the characteristic equation:

$$x^2 = 5x - 6 \implies x^2 - 5x + 6 = 0$$

Step 2: Solve for roots:

$$(x-2)(x-3) = 0 \implies r_1 = 2, r_2 = 3$$

Step 3: General solution:

$$a_n = A \cdot 2^n + B \cdot 3^n$$

Step 4: Use initial conditions:

$$a_0 = 2 : A + B = 2 \tag{1}$$
$$a_1 = 5 : 2A + 3B = 5 \tag{2}$$

Solving the system: from $A + B = 2$ and $2A + 3B = 5$, subtract $2(A + B) = 4$ from the second to get $B = 1$, hence $A = 1$.
Answer: $a_n = 2^n + 3^n$

2.14 Technique 14: Euler's Totient Function

Short Name: T14-EulerPhi

Definition 2.14 (Euler's Totient Function)**.** *$\phi(n)$ counts the positive integers up to n that are relatively prime to n. For prime p: $\phi(p) = p - 1$ For prime power: $\phi(p^k) = p^{k-1}(p-1)$ For coprime m, n: $\phi(mn) = \phi(m)\phi(n)$*

Euler's Theorem: *If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$*

Carmichael Function: *$\lambda(n)$ is the least positive integer m such that $a^m \equiv 1 \pmod{n}$ for all integers a with $\gcd(a, n) = 1$ (the exponent of $(\mathbb{Z}/n\mathbb{Z})^\times$). For an odd prime p and $k \geq 1$, $\lambda(p^k) = \phi(p^k) = p^{k-1}(p-1)$. For powers of 2: $\lambda(1) = 1$, $\lambda(2) = 1$, $\lambda(4) = 2$, and for $k \geq 3$, $\lambda(2^k) = 2^{k-2}$. If m and n are coprime, then $\lambda(mn) = \operatorname{lcm}(\lambda(m), \lambda(n))$.*

WHEN: Modular exponentiation with composite moduli; counting integers relatively prime to n; determining multiplicative orders/periods; simplifying exponents using $\phi(n)$ or $\lambda(n)$; splitting computations via CRT. Use for last digit(s) with composite moduli; computing $a^k \pmod{n}$ for composite n; counting residues coprime to n; orders/periods modulo n; Euler/Carmichael reductions; CRT decompositions and recombination.

Configuration cues:

- Large exponents modulo composite n; prefer reduction via $\phi(n)$ or $\lambda(n)$
- Counting integers coprime to n; using multiplicativity across coprime factors
- CRT-friendly factorizations of n into prime powers

Constraints cues:

- Euler's theorem requires $\gcd(a, n) = 1$; if not, factor n and handle prime powers separately (or use CRT), or consider $\lambda(n)$ when appropriate
- ϕ is multiplicative only over coprime factors; factorization is necessary
- Reducing exponents modulo $\phi(n)$ is valid for units mod n; $\lambda(n)$ may yield tighter cycles
- When counting totatives, ensure the range and inclusivity are correct (typically 1 through n)

Objective cues:

- Simplify modular exponentiation by reducing exponents and recombining via CRT
- Count or characterize residues coprime to n using $\phi(n)$
- Determine multiplicative orders and periodicity of powers modulo n

Why It Works: Generalizes Fermat's Little Theorem to composite moduli.

WHO: NT modular arithmetic with composite moduli, totient counts, orders/periods, or CRT decompositions; solvers leveraging ϕ/λ and recognizing when $\gcd(a, n) \neq 1$ requires factoring.

WHAT: Compute $\phi(n)$ via prime powers and multiplicativity; apply Euler's theorem when $\gcd(a, n) = 1$ to reduce exponents modulo $\phi(n)$.

How:

- **First Moves:**
  1. Factor n and compute $\phi(n)$ using $\phi(p^k) = p^{k-1}(p-1)$ and multiplicativity.
  2. Verify $\gcd(a, n) = 1$; if not, factor the modulus or use CRT on coprime factors.
  3. Reduce exponents modulo $\phi(n)$ (or $\lambda(n)$ when known) and simplify.

4. Recombine residues via CRT if working modulo a product.

- **Bailouts:** If factoring n is hard, work modulo prime-power factors you can handle, or use Carmichael $\lambda(n)$ when available; apply lifting lemmas or order arguments on each component.
- **Pitfalls:** Applying Euler's theorem when $\gcd(a, n) \neq 1$; forgetting to recombine components; reducing exponents by $\phi(n)$ when the base is not coprime.
- **Example:** Last two digits of 3^{1000} via $\phi(100) = 40$: reduce $1000 \equiv 0 \pmod{40}$ to get 01.

Example: Large Exponent Modulo Composite

Find the last two digits of 3^{1000}.

Solution

We need $3^{1000} \pmod{100}$.
Step 1: Calculate $\phi(100)$:

$$\phi(100) = \phi(4) \cdot \phi(25) = 2 \cdot 20 = 40$$

Step 2: Since $\gcd(3, 100) = 1$, by Euler's Theorem:

$$3^{40} \equiv 1 \pmod{100}$$

Step 3: Reduce the exponent:

$$1000 = 40 \cdot 25 + 0$$

Therefore: $3^{1000} = (3^{40})^{25} \equiv 1^{25} \equiv 1 \pmod{100}$
Answer: The last two digits are 01.

2.15 Technique 15: Rearrangement Inequality

Short Name: T15-RearrangeInequ

Definition 2.15 (Rearrangement Inequality)**.** *If $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, then for any permutation σ:*

$$\sum_{i=1}^{n} a_i b_{n+1-i} \leq \sum_{i=1}^{n} a_i b_{\sigma(i)} \leq \sum_{i=1}^{n} a_i b_i$$

The sum is minimized by reverse order pairing and maximized by same order pairing.

WHEN: Extremizing sums of products formed by pairing two lists; proving sharp bounds for assignment-type sums; validating greedy pairings in scheduling/assignment with

fixed marginals; comparing permutations via adjacent swaps. Use for assignment/matching; scheduling with processing times and weights; symmetric sums $\sum a_i b_{\sigma(i)}$; discrete optimization with fixed marginals; Chebyshev and majorization contexts when sequences are similarly sorted.

Configuration cues:

- Two real sequences to be paired one-to-one with a sum of products $\sum a_i b_{\sigma(i)}$
- Ability to sort both sequences; same-order pairing for maxima and reverse-order for minima
- Assignment/matching or scheduling contexts with fixed marginals

Constraints cues:

- Sort consistently (nondecreasing or nonincreasing); ties may yield non-unique extremizers
- Works with negative values; maintain sorted order across signs
- Applies to one-to-one pairings; with restrictions, justify via adjacent swaps or verify conditions

Objective cues:

- Establish sharp upper/lower bounds on $\sum a_i b_{\sigma(i)}$
- Justify greedy pairing strategies in assignments/scheduling
- Compare permutations via adjacent transpositions to reach the extremal configuration

Why It Works: Larger values contribute more when multiplied by larger values.

WHO: ALG/inequality and optimization problems involving pairwise products or weighted sums; solvers comfortable sorting sequences and arguing optimality via rearrangement or adjacent transpositions.

WHAT: Sort both sequences; pair in the same order to maximize the sum, reverse order to minimize.

How:

- **First Moves:**
  1. Sort $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$.
  2. Choose same-order pairing for a maximum, reverse-order pairing for a minimum.
  3. Compute the sum accordingly and justify via the Rearrangement Inequality.
- **Bailouts:** If direct pairing is unclear, compare permutations via adjacent swaps to move toward the extremal arrangement; use Chebyshev or majorization as alternative proofs.
- **Pitfalls:** Not sorting sequences first; applying to negative/unsorted lists without care; confusing max/min pairings.

- **Example:** With $a \leq b \leq c$ and $x \leq y \leq z$, maximum of $ax' + by' + cz'$ occurs at $(x', y', z') = (x, y, z)$.

Example: Maximizing a Sum

Given positive numbers a, b, c and x, y, z with $a \leq b \leq c$ and $x \leq y \leq z$, arrange them to maximize $ax' + by' + cz'$ where $\{x', y', z'\}$ is a permutation of $\{x, y, z\}$.

Solution

By the Rearrangement Inequality, the sum is maximized when we pair in the same order: - Pair smallest with smallest: a with x - Pair middle with middle: b with y - Pair largest with largest: c with z
Maximum value: $ax + by + cz$
Answer: Arrange as $(x', y', z') = (x, y, z)$ to get maximum $ax + by + cz$

2.16 Technique 16: Vieta's Formulas

Short Name: T16-Vieta

Definition 2.16 (Vieta's Formulas)**.** *For a polynomial with roots, express symmetric functions of roots in terms of coefficients.*

WHEN: Sums/products of roots; power sums via identities; reciprocals and $P\left(\frac{1}{x}\right)$; constraints linking coefficients and root relations; contexts where minimal polynomial relations for sums/products/reciprocals of roots are needed.

Configuration cues:

- Questions ask for sums/products of roots or symmetric expressions in the roots
- Expressions involving $P(1/x)$, reciprocals of roots, or power sums like $a^2 + b^2 + c^2$
- Need relationships among roots rather than individual root values

Constraints cues:

- If the polynomial is not monic, divide by the leading coefficient to read Vieta sums correctly
- For reciprocals, require nonzero constant term (no root equals 0)
- Complex or repeated roots are allowed; algebra is over $\mathbb{C}$

Objective cues:

- Compute symmetric expressions quickly without solving the polynomial
- Convert targets to elementary symmetric sums $(\sigma_1, \sigma_2, \dots)$
- Handle expressions in reciprocal roots via coefficients

Why It Works: Avoids finding individual roots when only their relationships matter. This technique appears frequently in AMC 12 and COMC problems that ask for sums/products of roots or symmetric expressions.

WHO: ALG polynomials and root finding asking for root sums/products, symmetric expressions, reciprocals, or coefficient relations; solvers comfortable with Vieta/Newton sums.

WHAT: Use elementary symmetric sums (Vieta) to compute sums/products of roots and symmetric expressions without solving for individual roots.

How:

- **First Moves:**
  1. If the polynomial is not monic, divide by the leading coefficient to make it monic.
  2. Read off elementary symmetric sums from coefficients using Vieta.
  3. Rewrite the target in terms of these sums (e.g., $a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+bc+ca)$).
  4. For reciprocals, ensure no root is zero; in monic form, $\sum_{i=1}^{n} \frac{1}{r_i} = -\frac{a_1}{a_0}$. For cubics, this equals $\frac{ab+bc+ca}{abc}$.
  5. Confirm edge cases (repeated or complex roots are allowed; algebra is over $\mathbb{C}$).

- **Bailouts:** If the target is not an elementary symmetric expression, switch to Newton sums; derive a minimal polynomial for the desired symmetric quantity; or relate reciprocals via $x^n P(1/x)$. If algebra grows, try factoring or substituting a new variable for a symmetric block.

- **Pitfalls:** Forgetting to normalize to monic when reading Vieta sums; using reciprocals when $a_0 = 0$ (a zero root); sign mistakes in symmetric sums; attempting to use Vieta on non-symmetric targets without rewriting.

- **Example:** For a cubic with roots a, b, c, compute $a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+bc+ca)$; for reciprocals, use $\sum 1/a = (ab+bc+ca)/(abc)$ when no root is zero.

Theorem 2.1 (Vieta's Formulas)**.** *For polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ with roots $r_1, \ldots, r_n$:*

- $r_1 + r_2 + \cdots + r_n = -a_{n-1}$
- $r_1 r_2 + r_1 r_3 + \cdots = a_{n-2}$
- $r_1 r_2 \cdots r_n = (-1)^n a_0$

Sign note: For higher-order elementary symmetric sums, the signs alternate with parity: the coefficient of the k-th symmetric sum is $(-1)^k$.

Example: Polynomial Roots

If $x^3 - 6x^2 + 11x - 6 = 0$ has roots a, b, c, find $a^2 + b^2 + c^2$.

Solution

By Vieta's formulas:

- $a + b + c = 6$
- $ab + ac + bc = 11$
- $abc = 6$

Using the identity: $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + ac + bc)$
$a^2 + b^2 + c^2 = 36 - 2(11) = 36 - 22 = 14$
Answer: 14

2.17 Technique 17: Geometric Probability

Short Name: T17-GeoProb

Definition 2.17 (Geometric Probability)**.** *Probability calculated using geometric measures (length, area, volume) as the sample space.*

$$P(event) = \frac{measure\ of\ favorable\ region}{measure\ of\ total\ region}$$

WHEN: Uniform selections in 1D/2D/3D where probability equals length/area/volume ratios; random chords/angles/distances (specify the selection model); meeting-time windows; regions defined by inequalities. Use for random point in a region; random chord/angle/distance (with explicit model); meeting-time/interval-overlap; inequalities describing subregions and distance thresholds.

Configuration cues:

- Random points in geometric figures
- "Uniformly distributed" in a region
- Probability involving distances or angles

Constraints cues:

- Verify uniformity and use the correct measure (length/area/volume); treat boundaries as measure zero unless stated otherwise
- Be explicit about selection models in ambiguous setups (e.g., "random chord"); clarify whether chords, angles, or points are sampled uniformly

Objective cues:

- Describe the favorable set by inequalities or decomposition and compute its measure
- Simplify via symmetry or coordinate changes; divide by the total measure

Why It Works: Uniform distribution means probability is proportional to geometric measure.

WHO: PROB/GEO probability and geometry with uniformly random points, times, directions, or chords; solvers using symmetry, coordinate changes, or basic integration.

WHAT: Model outcomes by geometric regions and compute probability as a ratio of measures (length, area, or volume).

How:

- **First Moves:**
  1. Verify uniformity and identify the correct measure (length/area/volume).
  2. Translate conditions into inequalities defining the favorable region.
  3. Exploit symmetry or convenient coordinates to simplify the region.
  4. Compute the measure exactly (integration, similar triangles, or decomposition); check boundary cases.
  5. Divide by the total measure and simplify.
- **Bailouts:** If the geometry is messy, switch to complementary probability, partition into simple shapes, or change coordinates. For ambiguous models (e.g., random chord), pick and state a standard model (endpoints uniform, angle method, or radius method) and proceed.
- **Pitfalls:** Mixing selection models; treating boundaries as nonzero measure; double-counting regions; forgetting to normalize by the total measure.
- **Example:** Meeting-time windows reduce to areas of right triangles/rectangles in the unit square; random-point problems become area ratios after symmetry and inequality description.

Example: Random Point in Square

A point is chosen uniformly at random in a unit square. What is the probability that it is closer to the center than to any edge?

Solution

Place the square with vertices at $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ and center at $(\frac{1}{2},\frac{1}{2})$. Let $u = x - \frac{1}{2}$, $v = y - \frac{1}{2}$. Then the distance to the center is $r = \sqrt{u^2+v^2}$ and the distance to the nearest edge is $\frac{1}{2} - \max(|u|,|v|)$. The condition is

$$r < \tfrac{1}{2} - \max(|u|,|v|) \quad \Longleftrightarrow \quad r + \max(|u|,|v|) < \tfrac{1}{2}.$$

By symmetry, compute area in the first quadrant and multiply by 4. In $0 \le \theta \le \frac{\pi}{4}$ (where $u \ge v \ge 0$), $\max(|u|,|v|) = u = r\cos\theta$, so the boundary is

$$r = \frac{1}{2(1+\cos\theta)}.$$

In $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$ (where $v \ge u \ge 0$), $\max(|u|,|v|) = v = r\sin\theta$, giving

$$r = \frac{1}{2(1+\sin\theta)}.$$

Thus the total area is

$$A = 4\left(\int_0^{\pi/4} \tfrac{1}{2}r^2\,d\theta + \int_{\pi/4}^{\pi/2} \tfrac{1}{2}r^2\,d\theta\right) = \int_0^{\pi/4} \frac{1}{(1+\cos\theta)^2}\,d\theta = \frac{1}{2}\int_0^{\pi/8} \sec^4\varphi\,d\varphi$$

after using $1+\cos\theta = 2\cos^2(\frac{\theta}{2})$ and $\varphi = \frac{\theta}{2}$. Since $\int \sec^4\varphi\,d\varphi = \tan\varphi + \frac{1}{3}\tan^3\varphi$, and $\tan(\frac{\pi}{8}) = \sqrt{2}-1$, we obtain

$$A = \frac{1}{2}\left[\tan\varphi + \frac{\tan^3\varphi}{3}\right]_0^{\pi/8} = \frac{1}{2}\left((\sqrt{2}-1) + \frac{(\sqrt{2}-1)^3}{3}\right) = \frac{4\sqrt{2}-5}{3}.$$

Answer: $\dfrac{4\sqrt{2}-5}{3}$

2.18 Technique 18: Complex Number Conversion

Short Name: T18-ComplexNumber

Definition 2.18 (Complex Number Conversion)**.** *Transform geometric rotations and transformations into complex number operations.*

WHEN: Rotations/dilations/reflections or spiral similarities; regular polygons and roots of unity; cyclic sums/filters; converting plane geometry to algebra in $\mathbb{C}$. Use for rotations about origin or point (translate–rotate–translate); regular n-gons; roots-of-unity sums/filters; reflections via conjugation; vertex sums/products and centroid/circumcenter setups.

Configuration cues:

- Rotations/scalings; regular polygons; roots of unity

- Model points as complex numbers; use $e^{i\theta}$ or nth roots

Constraints cues:

- Rotations about a non-origin point require translate–rotate–translate back; use conjugation for reflections and ensure correct orientation
- Choose a convenient origin (center/centroid) to simplify expressions and exploit symmetry

Objective cues:

- Convert geometric transformations into multiplication/conjugation in $\mathbb{C}$
- Use roots-of-unity sums/filters to collapse symmetric sums and products

Why It Works: Complex multiplication naturally handles rotation and scaling.

WHO: GEO/ALG complex numbers with rotations, dilations, regular polygons, roots of unity, or cyclic sums; solvers fluent with $e^{i\theta}$ and conjugation.

WHAT: Represent points as complex numbers; perform rotations/dilations via multiplication and use roots of unity for symmetry.

How:

- **First Moves:**
  1. Embed the plane as $x + iy$ with a convenient origin (often a center).
  2. Express a rotation/dilation as multiplication by $re^{i\theta}$; use conjugation for reflections.
  3. For a regular n-gon, set $\omega = e^{2\pi i/n}$ and index vertices by powers of ω.
  4. Use $\sum_{k=0}^{n-1} \omega^{km} = 0$ when $n \nmid m$ to simplify symmetric sums.
  5. Convert back to coordinates or magnitudes/arguments to present the result.
- **Bailouts:** If complex notation becomes cumbersome, switch to vectors/coordinates or trigonometry; re-center the origin at a symmetry point; or use rotation matrices for clarity.
- **Pitfalls:** Forgetting translate–rotate–translate for rotations about non-origin points; misusing conjugation (orientation); mixing degrees and radians; sign errors in roots-of-unity sums.
- **Example:** Model a rotation about the origin by multiplication by $e^{i\theta}$; for a regular n-gon, sum vertex powers using roots-of-unity filters to collapse symmetric sums.

Example: Complex Rotation

Let $w = 3 + 4i$. Rotate w about the origin by $60°$. Express the resulting complex number.

Solution

Rotation via multiplication: Multiplication by $e^{i\pi/3} = \cos(60°) + i\sin(60°) = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ performs a 60° rotation.

$$\begin{aligned} w' = w \cdot e^{i\pi/3} &= (3+4i)\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) && (3)\\ &= \frac{3}{2} + i\frac{3\sqrt{3}}{2} + \frac{4i}{2} + \frac{4i^2\sqrt{3}}{2} && (4)\\ &= \frac{3}{2} + i\frac{3\sqrt{3}}{2} + 2i - 2\sqrt{3} && (5)\\ &= \left(\frac{3}{2} - 2\sqrt{3}\right) + i\left(\frac{3\sqrt{3}}{2} + 2\right) && (6)\\ &= \frac{3-4\sqrt{3}}{2} + i\frac{3\sqrt{3}+4}{2} && (7) \end{aligned}$$

Answer: $\left(\frac{3-4\sqrt{3}}{2}\right) + i\left(\frac{3\sqrt{3}+4}{2}\right)$

2.19 Technique 19: Modular Arithmetic

Short Name: T19-ModArith

Definition 2.19 (Modular Arithmetic Conversion)**.** *Reduce problems modulo appropriate primes or composite numbers to simplify calculations.*

WHEN: Divisibility/remainder questions; solving linear congruences; last digit(s)/last k digits; detecting power cycles and multiplicative orders; combining conditions via CRT. Use for last digit(s) and base-b endings; divisibility tests; linear congruences and modular inverses; periodic powers and multiplicative orders; CRT recombinations and residue-class casework.

Configuration cues:

- Divisibility/remainders; periodicity in powers or sequences
- Pick modulus aligned to the property; combine coprime moduli when helpful

Constraints cues:

- Modular inverses exist only when $\gcd(a, m) = 1$; for composite moduli, reduce modulo prime powers and recombine with CRT
- Handle negative residues consistently; confirm cycle lengths using multiplicative order, not only Euler's totient bound

Objective cues:

- Reduce arithmetic to residues; identify and use cycles/orders to simplify powers
- Solve linear congruences and recombine conditions efficiently via CRT

Why It Works: Reduces infinite sets to finite, manageable cases.

WHO: NT/ALG modular arithmetic with divisibility, remainders, linear congruences, or cyclic patterns; solvers leveraging modular inverses, orders, and CRT.

WHAT: Work modulo suitable bases to simplify arithmetic, detect cycles, and solve congruences.

How:

- **First Moves:**
  1. Choose a modulus aligned with the goal (prime powers for last digits; factors of n for divisibility).
  2. Reduce numbers and operations modulo m; replace powers by their cycles or by orders.
  3. If m factors, solve modulo prime powers and recombine with CRT.
  4. Use modular inverses to solve linear congruences when $\gcd(a, m) = 1$; check solvability otherwise.
  5. Sanity-check with small cases or with multiple compatible moduli.
- **Bailouts:** If cycles are long, switch to multiplicative order or Euler/Carmichael reductions; for composite m, break into prime powers. When stuck on remainders, use CRT structure or try alternative moduli that expose the pattern.
- **Pitfalls:** Inverting when $\gcd(a, m) \neq 1$; assuming Euler's totient gives the exact period; mishandling negative residues; forgetting to recombine solutions correctly via CRT.
- **Example:** Compute last two digits using mod 4 and mod 25 separately and recombine with CRT; solve $ax \equiv b \pmod{m}$ by dividing by $\gcd(a, m)$ and inverting the reduced coefficient when possible.

Example: Large Power Remainder

Find the last two digits of 7^{2023}.

Solution

We need 7^{2023}
$pmod100$.
First, find the pattern of powers of 7 modulo 100:

$$7^1 \tag{8}$$
$$equiv7 \tag{9}$$
$$pmod100 \tag{10}$$
$$7^2 \tag{11}$$
$$equiv49 \tag{12}$$
$$pmod100 \tag{13}$$
$$7^3 \tag{14}$$
$$equiv343 \tag{15}$$
$$equiv43 \tag{16}$$
$$pmod100 \tag{17}$$
$$7^4 = 7 \tag{18}$$
$$cdot343 = 2401 = 2400 + 1 = 24(100) + 1 \tag{19}$$
$$equiv1 \tag{20}$$
$$pmod100 \tag{21}$$

So the powers of 7 have period 4 modulo 100.
Since $2023 = 4$
$cdot505 + 3$:

$$7^{2023} = (7^4)^{505} cdot7^3 equiv1^{505} cdot43 equiv43 pmod100$$

Answer: 43

2.20 Technique 20: Fermat's Little Theorem

Short Name: T20-FLT

Definition 2.20 (Fermat's Little Theorem)**.** *If p is prime and a is not divisible by p, then:*

$$a^{p-1} equiv1 pmodp$$

Equivalently, for any integer a: a^p
$equiva$
$pmodp$

WHEN: Large exponents modulo a prime; prime-modulus congruences using exponent reduction modulo $p-1$ (or the order of a); per-prime steps within CRT decompositions. Use for prime moduli; remainders of a^n; cycles/orders of powers; per-prime components in CRT for composite moduli.

Configuration cues:

- Prime modulus; reduce exponents modulo $p-1$ when $gcd(a,p)=1$
- Combine with CRT for composite moduli

Constraints cues:

- Require $\gcd(a,p)=1$ to use $a^{p-1}\equiv 1 \pmod{p}$; if $p \mid a$ then $a^n \equiv 0 \pmod{p}$ for $n \geq 1$
- For composite moduli, use Euler's theorem or Carmichael's function per prime power and recombine with CRT
- Prefer the multiplicative order of a modulo p (dividing $p-1$) for tighter exponent reduction

Objective cues:

- Reduce exponents modulo $p-1$ (or modulo the order of a); compute small powers modulo p
- Leverage periodicity of powers to evaluate large exponents quickly

Why It Works: Provides a powerful reduction for large exponents modulo primes.
WHO: NT modular arithmetic with exponents modulo a prime (or per-prime inside composite moduli); solvers applying orders and CRT.
WHAT: Use $a^{p-1}\equiv 1 \pmod{p}$ (for $\gcd(a,p)=1$) to reduce exponents modulo $p-1$.
How:

- **First Moves:**
  1. Confirm p is prime and $\gcd(a,p)=1$; reduce a modulo p.
  2. Reduce the exponent n modulo $p-1$ using $a^{p-1}\equiv 1 \pmod{p}$.
  3. Compute the remaining small power modulo p (fast exponentiation if needed).
  4. For composite moduli, apply FLT per prime factor when applicable (or use Euler/Carmichael generalizations) and recombine with CRT.
  5. Validate with known orders or small-cycle checks.
- **Bailouts:** If $p \mid a$, handle the zero case separately; if reduction mod $p-1$ is not tight, use the multiplicative order of a modulo p; for composite moduli, move to Euler/Carmichael and CRT.
- **Pitfalls:** Applying FLT to non-prime moduli; ignoring the $\gcd(a,p)=1$ requirement; reducing exponents mod $p-1$ when $p \mid a$; overlooking that the true order may be a proper divisor of $p-1$.
- **Example:** Evaluate $a^n \bmod p$ by reducing n modulo $p-1$ and computing a small power; for mod100, work modulo 4 and 25 and recombine via CRT.

Example: Large Exponent Modulo Prime

Find the remainder when 7^{2023} is divided by 11.

Solution

Since 11 is prime and
$gcd(7, 11) = 1$, by Fermat's Little Theorem:

$$7^{10} equiv 1 pmod 11$$

Now divide the exponent: $2023 = 10$
$cdot 202 + 3$
Therefore:

$$7^{2023} = (7^{10})^{202} cdot 7^3 equiv 1^{202} cdot 7^3 equiv 7^3 pmod 11$$

Calculate $7^3 = 343 = 31$
$cdot 11 + 2$, so 7^3
$equiv 2$
$pmod 11$
Answer: The remainder is 2.

AMC 12 Example

Find the last digit of $3^{3^{3^3}}$.

Solution

We need $3^{3^{3^3}}$
$pmod10$.
First, note that powers of 3 modulo 10 cycle with period 4: 3^1
$equiv3$, 3^2
$equiv9$, 3^3
$equiv7$, 3^4
$\equiv 1$
$pmod10$
We need to find 3^{3^3}
$pmod4$ to determine which case applies.
Since 3
$equiv-1$
$pmod4$: $3^3 = 27$
$equiv(-1)^3$
$equiv-1$
$equiv3$
$pmod4$
So 3^{3^3}
$equiv3^3$
$equiv3$
$pmod4$
Therefore $3^{3^3} = 4k + 3$ for some integer k.
Thus: $3^{3^{3^3}} = 3^{4k+3} = (3^4)^k$
$cdot3^3$
$equiv1^k$
$cdot7$
$equiv7$
$pmod10$
Answer: The last digit is 7.

2.21 Technique 21: Inclusion-Exclusion Principle

Short Name: T21-PIE

Definition 2.21 (Inclusion-Exclusion Principle)**.** *For finite sets $A_1, A_2, \ldots, A_n$:*

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1\le i<j\le n} |A_i \cap A_j| + \sum_{1\le i<j<k\le n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1}|A_1 \cap \cdots \cap A_n|$$

Note: The probability version is analogous: for events $E_1, \ldots, E_n$, replace cardinalities by probabilities.

WHEN: Divisibility in ranges and overlapping residue classes; digits with forbidden positions; password/PIN constraints; arrangements avoiding adjacency or fixed points (derangements); "at least one of" conditions; small-parameter surjections (onto functions) via inclusion–exclusion; short forbidden-substring word counts when pattern length is small.

Configuration cues:

- Counting or probabilities of unions with overlapping conditions
- Intersection sizes are computable or bounded
- Questions of "at least one/none/exactly k" among several properties
- Small number of overlapping conditions (typically 2–4) so intersections are manageable

Constraints cues:

- Avoid double counting; include intersections to the needed order
- For probabilities, normalize correctly; assume independence only when justified
- Determine which higher-order intersections are zero or equal by structure/symmetry
- For "exactly k" counts, typically require j-wise intersections for all $j \geq k$ (often up to full order), unless higher-order intersections vanish or collapse by symmetry

Objective cues:

- Compute union size/probability via alternating sums of intersections
- Simplify via symmetry or complementary counting when helpful

Why It Works: Systematically corrects for overcounting when elements belong to multiple sets.

WHO: CNT/PROB counting and probability with overlapping conditions or at least/at most/exactly k properties; solvers comfortable with set operations and symmetry.

WHAT: Compute the size/probability of a union by alternating sums of intersection sizes.

How:

- **First Moves:**
  1. Define sets A_i for each condition; compute $|A_i|$ and intersection sizes.
  2. Apply the inclusion–exclusion formula with alternating signs.
  3. For probabilities, divide by the total space; double-check that intersections are not omitted.
- **Bailouts:** If higher-order intersections proliferate, switch to complementary counting (count "none" and subtract), exploit symmetry to equate intersection sizes, or ignore terms that vanish by structure.

- **Pitfalls:** Wrong alternating signs; omitting required intersections; assuming independence without justification; mixing universes or mishandling floors/ceilings in divisibility counts.
- **Example:** See the divisibility-by-2/3/5 example below: compute singles, pairwise, and triple intersections, then combine.

Example: Divisibility Count

How many integers from 1 to 1000 are divisible by 2, 3, or 5?

Solution

Let:

- A_2 = integers divisible by 2: $|A_2| = \lfloor 1000/2 \rfloor = 500$
- A_3 = integers divisible by 3: $|A_3| = \lfloor 1000/3 \rfloor = 333$
- A_5 = integers divisible by 5: $|A_5| = \lfloor 1000/5 \rfloor = 200$

Intersections:

- $|A_2 \cap A_3| = \lfloor 1000/6 \rfloor = 166$
- $|A_2 \cap A_5| = \lfloor 1000/10 \rfloor = 100$
- $|A_3 \cap A_5| = \lfloor 1000/15 \rfloor = 66$
- $|A_2 \cap A_3 \cap A_5| = \lfloor 1000/30 \rfloor = 33$

By Inclusion-Exclusion:

$$|A_2 \cup A_3 \cup A_5| = 500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$$

Answer: 734

2.22 Technique 22: Cauchy-Schwarz Inequality

Short Name: T22-CauchySchwarz

Definition 2.22 (Cauchy-Schwarz Inequality)**.** *For real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$:*

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2)$$

Equality holds if and only if the sequences are proportional.

WHEN: Maximizing a linear form under a circle/norm constraint, bounding $\sum \frac{a_i^2}{b_i}$ via Engel, RMS–AM conversions, and variance-style bounds in symmetric sums; distance/projection bounds in coordinate geometry (via dot-product interpretation).

Configuration cues:

- Target has a sum-of-products structure $\sum a_i b_i$ or a linear form under a norm constraint
- Sums of fractions $\sum$
  $tfraca_i^2 b_i$ with $b_i > 0$ (Engel/Titu's Lemma)
- Need to bound or optimize an expression using vector/dot-product ideas
- RMS–AM equivalences or geometric/projection interpretations of dot products

Constraints cues:

- Use real vectors with computable $\sum a_i^2$ and $\sum b_i^2$
- Equality when vectors are proportional and constraints permit it
- For Engel form, require $b_i > 0$
- Ensure required positivity/nonnegativity (e.g., denominators, squared terms); handle zero components carefully

Objective cues:

- Prove tight inequalities or find maxima/minima
- Translate the expression into a dot-product/norm inequality for a clean bound
- Convert awkward sums to squared norms or apply Cauchy iteratively for layered bounds

Why It Works: Provides tight bounds on dot products and sums.

WHO: ALG/GEO inequality, optimization, and geometry with dot-product geometry, and sums-of-fractions (Engel/Titu's Lemma); solvers comfortable with vector norms and equality conditions.

WHAT: Use Cauchy–Schwarz to bound sums of products; helpful variants include Engel form (Titu's Lemma) and RMS–AM.

How:

- **First Moves:**
  1. Choose vectors so that $\sum a_i b_i$ matches the target quantity.
  2. Apply $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$ and compute the bound.
  3. Check equality via proportionality $a_i = \lambda b_i$ while respecting constraints.
  4. For sums of fractions, consider Engel form (Titu's Lemma): $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$ for $b_i > 0$.
- **Bailouts:** If the dot-product model is awkward, try Engel form directly, AM–GM/RMS–AM, or the rearrangement inequality; normalize variables or re-parameterize to expose a clean Cauchy application.
- **Pitfalls:** Violating positivity/nonnegativity requirements; mishandling equality conditions; squaring without tracking domains; choosing mismatched vectors that inflate the bound.

- **Example:** See the linear-maximization example below; equality when vectors are proportional under the given constraint.

Example: Maximum Value

Find the maximum value of $3x + 4y$ given that $x^2 + y^2 = 25$.

Solution

By Cauchy-Schwarz:

$$(3x + 4y)^2 \leq (3^2 + 4^2)(x^2 + y^2) = 25 \cdot 25 = 625$$

Therefore: $3x + 4y \leq 25$
Equality occurs when $\frac{x}{3} = \frac{y}{4}$, combined with $x^2 + y^2 = 25$:
Let $x = 3t$ and $y = 4t$. Then: $9t^2 + 16t^2 = 25$, so $t = 1$.
Thus $x = 3, y = 4$ gives the maximum.
Answer: Maximum value is 25

2.23 Technique 23: Power of a Point

Short Name: T23-PoP

Definition 2.23 (Power of a Point)**.** *For a circle and a point P:*

- *If two chords $\overline{AB}$ and $\overline{CD}$ intersect at P inside the circle, then $PA \cdot PB = PC \cdot PD$.*
- *If $\overline{PT}$ is tangent and $\overline{PAB}$ is a secant with A, B on the circle, then $PT^2 = PA \cdot PB$.*

WHEN: Circle problems with intersecting chords, two secants from a point, or a tangent–secant setup where segment products or feasibility must be determined; distinguish inside vs. outside placements and ensure tangency/length data are consistent. Use for intersecting chords inside a circle; external secants/tangents from a point; two-circle setups via the radical axis; coordinate/distance contexts using $\Pi(P) = |OP|^2 - r^2$; length-feasibility checks for given tangent/secant/chord data (reject impossible configurations).

Configuration cues:

- Circle geometry with intersecting chords, two secants, or tangent–secant from a point
- Known/unknown segment lengths arranged around a point P

Constraints cues:

- Distinguish inside vs. outside configurations; handle tangency as limiting case
- Use nonnegative lengths consistently; check for degenerate/impossible data

Objective cues:

- Relate segment products to solve for unknown lengths

- Confirm feasibility; optionally use $\Pi(P) = |OP|^2 - r^2$ or the radical axis in distance setups

Pattern cues (Demand: Moderate–High):

- Intersecting chords/tangent–secant; equal angle subtendences
- Choose relation: chord–chord ($PA{\cdot}PB = PC{\cdot}PD$) or tangent–secant ($PT^2 = PA{\cdot}PB$)

Why It Works: Follows from similar triangles formed by equal angles subtended by the same arcs.

General formula: For circle center O and radius r, the power of a point P is $\Pi(P) = |OP|^2 - r^2$.

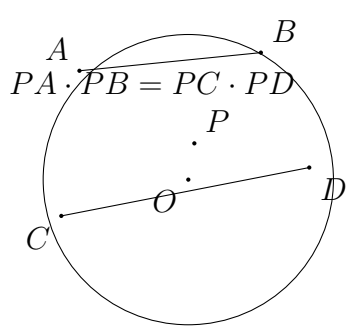
Directed-length note: In analytic or signed-length treatments, one can work with directed segments; the product relations remain valid with consistent sign conventions and are often paired with a radical axis viewpoint.

WHO: GEO circle geometry problems; solvers comfortable with similar triangles, directed lengths, and chord–tangent relations (optional radical-axis viewpoint).

WHAT: Use chord–chord, secant–secant, or tangent–secant products to relate segment lengths; optionally use $\Pi(P) = |OP|^2 - r^2$ in coordinate/distance settings.

How:

- **First Moves:**
  1. Identify the configuration type (chord–chord, secant–secant, or tangent–secant).
  2. Write the corresponding product relation and substitute known lengths.
  3. Solve for unknowns and check geometric feasibility (positions/signs).
  4. Use the power formula or radical axis to streamline distance-based setups.
- **Bailouts:** If lengths or placements are messy, switch to coordinates and use $\Pi(P) = |OP|^2 - r^2$, or use similar triangles/radical axis relations to relate segments.
- **Pitfalls:** Confusing inside vs. outside configurations; sign mistakes with directed lengths; assuming tangency without verification; impossible numeric data (reject via $PT^2 = PA \cdot PB$).
- **Example:** See the intersecting-chords example below; template: equate products and solve.



Example: Intersecting Chords

In a circle, chords $\overline{AB}$ and $\overline{CD}$ intersect at P. If $PA = 3$, $PB = 4$, and $PC = 2$, find PD.

Solution

By Power of a Point: $PA \cdot PB = PC \cdot PD$ gives $3 \cdot 4 = 2 \cdot PD$, so $PD = 6$.
Answer: 6

Advanced Geometry Enrichment: Radical Axis and Spiral Similarity

Radical Axis: For two circles, the locus of points with equal power to both circles is a line (their radical axis). It is perpendicular to the line through the centers and is the common chord when circles intersect. Useful for concurrency of radical axes and for constructing orthogonal intersections.

Spiral Similarity: A composition of a rotation and a dilation about a point S sending segment AB to CD. If two segments subtend equal directed angles at S with proportional lengths, then there is a spiral similarity at S mapping one to the other. Common in cyclic quadrilateral configurations and tangent-chord problems.

Note: These tools are uncommon on AMC 12 but can appear in hardest geometry or as enrichment for COMC-style reasoning.

Example: Tangent–Secant

From an external point P, a tangent $\overline{PT}$ and a secant through A, B touch/intersect a circle. If $PT = 6$ and $PA = 4$, $PB = 10$, verify the configuration and determine whether it's possible.

Solution

Power of a Point gives $PT^2 = PA \cdot PB$. Here $PT^2 = 36$ and $PA \cdot PB = 40$. Since $36 \neq 40$, these lengths cannot occur in the same configuration. If instead $PT = \sqrt{40}$ or one of PA, PB is adjusted to satisfy $PT^2 = PA \cdot PB$, the configuration is feasible.

2.24 Technique M1: Stars and Bars

Short Name: TM1-StarsBars

Definition 2.24 (Stars and Bars)**.** *Count non-negative or positive integer solutions to a linear sum constraint by mapping to combinations with repetition. For $x_1 + \cdots + x_k = n$ with $k \geq 1$:*

- *Non-negative solutions:* $\binom{n+k-1}{k-1}$.

- *Positive solutions: let* $y_i = x_i - 1 \geq 0$*, so* $y_1 + \cdots + y_k = n - k$*, giving* $\binom{n-1}{k-1}$ *when* $n \geq k$*.*

Equivalent generating function viewpoint: extract $[x^n]$ *from* $(1-x)^{-k}$*.*

WHEN: Distributing indistinguishable items into distinguishable boxes; counting solutions in non-negative/positive integers; coefficient extraction of $(1-x)^{-k}$. Use for balls-into-boxes, distributing candies, digit-sum decompositions, coin problems with fixed totals.

WHO: CNT counting problems with distribution constraints and integer solutions; solvers comfortable with combinations and generating functions.

Configuration cues:

- Phrases like "distribute," "no box empty," "at most," or "non-negative integer solutions"
- Fixed total sum with simple per-variable bounds

Constraints cues:

- Positivity constraints $x_i \geq 1$ require shifting $y_i = x_i - 1 \geq 0$
- Upper bounds $x_i \leq b_i$ need inclusion–exclusion, capped OGFs $\frac{1-x^{b_i+1}}{1-x}$, or a short DP
- Be clear whether boxes are distinguishable and items indistinguishable

Objective cues:

- Reduce to a binomial coefficient count
- Translate bounded variants into an unbounded core plus corrections

Why It Works: Place $k-1$ bars among $n+k-1$ slots to separate n stars into k groups (or use $(1-x)^{-k}$).

WHO: CNT/NT counting and number theory with integer solution problems.

WHAT: Model the distribution/solution as a sum constraint and apply the appropriate formula.

How:

- **First Moves:**
  1. Write the sum and non-negativity/positivity constraints.
  2. Shift to non-negative variables if $x_i \geq 1$.
  3. Apply $\binom{n+k-1}{k-1}$ (non-negative) or $\binom{n-1}{k-1}$ (positive).
- **Bailouts:** With caps $x_i \leq b_i$, use inclusion–exclusion, OGFs with capped factors, or a recurrence.
- **Pitfalls:** Mixing indistinguishable vs. distinguishable objects; forgetting the shift for positivity; ignoring upper bounds.
- **Example:** Solutions to $x_1 + x_2 + x_3 = 8$ with $x_i \geq 0$: $\binom{8+3-1}{3-1} = \binom{10}{2} = 45$. With $x_i \geq 1$: $\binom{8-1}{3-1} = \binom{7}{2} = 21$. With $x_i \leq 4$: subtract cases where some $x_i \geq 5$ via inclusion–exclusion or use capped OGFs.

2.25 Technique M3: Shoelace Formula

Short Name: TM3-Shoelace

Definition 2.25 (Shoelace Formula)**.** *For a simple polygon with ordered vertices* $(x_1, y_1), \ldots, (x_n, y_n)$*, the area is*

$$A = \frac{1}{2}\left|\sum_{i=1}^{n} x_i y_{i+1} - y_i x_{i+1}\right|, \quad (x_{n+1}, y_{n+1}) = (x_1, y_1).$$

WHEN: Coordinate geometry area of a polygonal region (often from intersections or lattice vertices) where decomposing into triangles is tedious. Use for expressions with $\sqrt{1 \pm x^2}$, $\sqrt{x^2 - 1}$, unit-circle constraints $x^2 + y^2 = 1$, Pythagorean relations, and coordinate circle/ellipse parametrizations.

WHO: GEO coordinate geometry problems with polygonal areas; solvers comfortable with coordinate calculations and absolute values.

Configuration cues:

- Vertices given or derivable in coordinates; polygon is simple (non-self-intersecting)

How:

- **First Moves:** Order vertices cyclically; apply the summation; take absolute value and halve.
- **Bailouts:** If ordering is unclear, compute convex hull first or triangulate and sum triangle areas.
- **Pitfalls:** Incorrect vertex order or sign; including intersection points not in order.
- **Example:** $(0, 0), (4, 0), (0, 3)$ yields $A = \frac{1}{2}|0 + 0 + 12 - (0 + 0 + 0)| = 6$.

2.26 Technique M9: Triangle Trigonometry (Law of Cosines and Sines)

Short Name: TM9-TriTrig

Definition 2.26 (Law of Cosines and Law of Sines)**.** *For triangle with sides* a, b, c *opposite angles* A, B, C *and circumradius* R*:*

- *Law of Cosines:* $c^2 = a^2 + b^2 - 2ab\cos C$ *(cyclically).*
- *Law of Sines:* $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R.$

Also $A = \frac{1}{2}ab\sin C$*.*

WHEN: Two sides and included angle; three sides (find angles); circumradius computations; area via sin.

WHO: GEO triangle geometry problems with side-angle relationships; solvers comfortable with trigonometric identities and angle calculations.

How:

- **First Moves:** Choose LoC for side/angle relations; use LoS to relate sides to angles or to R.
- **Bailouts:** Switch to coordinates or vectors if algebra grows; pair LoC with Stewart for cevians.
- **Pitfalls:** SSA ambiguity for LoS; degree/radian mix-ups; invalid domain for $\sin^{-1}$ or $\cos^{-1}$.

2.27 Technique M11: Circle Tangency and Angle Toolkit

Short Name: TM11-CircleTangency

Definition 2.27 (Tangents and Inscribed Angles).

- *Two-Tangent Theorem: From an external point P to tangent points A, B on a circle, $PA = PB$.*
- *Tangent–Radius Right Angle: If OA is a radius to tangent point A, then $OA \perp$ tangent at A.*
- *Inscribed Angle: An inscribed angle equals half the measure of its intercepted arc; tangent–chord angle equals the inscribed angle in the opposite arc.*
- *Cyclic Quadrilateral Angles: Opposite angles sum to $180°$.*

WHEN: Equal-length tangents; right angles at tangent points; angle chasing on circles; proving cyclicity.

WHO: GEO circle geometry problems with tangents and inscribed angles; solvers comfortable with angle chasing and circle properties.

How:

- **First Moves:** Mark tangent points and radii; replace tangent–chord angles by inscribed angles; check for cyclic quads.
- **Works well with:** Power of a Point; Ptolemy; similarity.
- **Pitfalls:** Choosing the wrong arc for tangent–chord angles; assuming cyclicity without proof.

2.28 Technique M12: Discriminant Techniques

Short Name: TM12-Discriminant

Definition 2.28 (Quadratic Discriminant). *For $ax^2 + bx + c = 0$ with $a \neq 0$, $\Delta = b^2 - 4ac$; $\Delta > 0$ two real roots, $\Delta = 0$ one real double root, $\Delta < 0$ no real roots. Nonnegativity for all real x occurs iff $a > 0$ and $\Delta \leq 0$ (similarly ≤ 0 with $a < 0$ and $\Delta \leq 0$ for nonpositivity).*

WHEN: Parameter bounds ensuring real roots; global sign of a quadratic; tangency conditions; intersection counts.

WHO: ALG quadratic equations and inequalities with parameter constraints; solvers comfortable with discriminant analysis and sign conditions.

How:

- **First Moves:** Write Δ; impose $\Delta \geq 0$ (or ≤ 0) to extract parameter ranges.
- **Bailouts:** Complete the square if parameters are messy; revert to vertex form to find extrema.
- **Pitfalls:** Forgetting a's sign when asserting nonnegativity; mixing necessary vs. sufficient conditions.

2.29 Technique M13: Logarithm Identities

Short Name: TM13-Logs

Definition 2.29 (Core Log Rules)**.** *For $x, y > 0$, bases $a, b > 0$ $(a, b \neq 1)$:*

- $\log_b(xy) = \log_b x + \log_b y, \quad \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y.$
- $\log_b(x^k) = k \log_b x, \quad$ *change of base:* $\log_b x = \frac{\ln x}{\ln b}.$

WHEN: Transform products to sums; telescoping log sums; solve exponential/log equations; compare growth.

WHO: ALG exponential and logarithmic equations; solvers comfortable with log properties and domain restrictions.

How:

- **First Moves:** Normalize base; expand/compress with log rules; ensure domain $x > 0$ throughout.
- **Pitfalls:** Ignoring domain (log of nonpositive); mixing bases; extraneous solutions after exponentiation.

2.30 Technique M14: Complementary Counting

Short Name: TM14-Complement

Definition 2.30 (Count the Complement)**.** *Compute desired count as total minus undesired: $|S| = |\Omega| - |\Omega \setminus S|$; common in "at least one" constraints and occupancy problems.*

WHEN: "At least one"/"none" structures; surjections/injections; avoiding adjacency or fixed points when direct count is messy.

WHO: CNT counting problems with complement counting; solvers comfortable with set operations and indirect counting methods.

How:

- **First Moves:** Identify the easier complement; compute via direct count, PIE, or generating functions.
- **Pitfalls:** Overlaps inside the complement requiring PIE; forgetting to subtract from the correct total universe.

2.31 Technique M27: Geometric Restart Recursion

Short Name: TM27-GeometricRestart

Definition 2.31 (Geometric Restart Recursion)**.** *For processes where failure returns to the initial state, if p is the probability of eventual success:*

$$p = P(\textit{immediate success}) + P(\textit{failure}) \cdot p$$

This yields: $p = \frac{P(\textit{immediate success})}{1-P(\textit{failure})}$ *The method applies to memoryless restart processes with geometric waiting times.*

Geometric Distribution Connection: *The number of attempts until success follows a geometric distribution with parameter p. The expected number of attempts is $E[X] = \frac{1}{p}$, and the variance is $Var(X) = \frac{1-p}{p^2}$. This technique leverages the memoryless property: the probability of eventual success from any restart point equals the probability from the initial state.*

Quick Reference: Geometric Restart Recursion

Key Formula: $p = \frac{P(\text{immediate success})}{1-P(\text{failure})}$
When to Use:

- "Keep trying until success" problems
- Complete restart on failure (memoryless property)
- "Probability of eventual success" questions
- Expected number of attempts problems

Requirements:

- Process must be memoryless (Markovian)
- Restart state identical to initial state
- $P(\text{failure}) < 1$ for convergence

WHEN: Stopping-time problems with complete restart on failure; recursive probability setups where failing resets to the same state; geometric distribution contexts; Markov chains with restart states; problems asking "probability of eventual success" or "expected number of attempts."

Configuration cues:

- "Keep trying until success"
- "If all attempts fail, restart from the beginning"
- "Return to starting position and try again"

- Memoryless property: future depends only on current state
- Multiple paths that either succeed or loop back

Constraints cues:

- Process must be memoryless (Markovian)
- Restart state must be identical to initial state
- Probability of eventual success must be positive (avoid $p = 0$ trap)
- Check convergence: require $P(\text{failure}) < 1$ for finite expected time

Objective cues:

- Find probability of eventual success
- Calculate expected number of attempts
- Solve for absorption probabilities in Markov chains
- Handle "return to start" gambling problems

Why It Works: The memoryless property means the probability from any restart is the same as from the initial state.

WHO: PROB recursion modelers using memoryless restart states and geometric progressions for repeated trials.

WHAT: Set up recursive equation $P = \text{base} + \text{multiplier} \cdot P$ for restart processes.

How:

- **First Moves:**
  1. Define p = probability of eventual success from the start state.
  2. Identify all possible first-step outcomes and their probabilities.
  3. For each outcome, determine if it leads to: success, failure (restart), or a different state.
  4. Write: $p = P(\text{direct success}) + \sum P(\text{path}_i) \cdot p_i$ where $p_i = p$ for restart paths.
  5. Solve the resulting linear equation for p.
- **Bailouts:** If multiple states exist, set up a system of equations; for expected value problems, use $E = 1 + P(\text{restart}) \cdot E$; if the equation gives $p = 0$ or $p > 1$, check for modeling errors.
- **Pitfalls:** Confusing restart with partial progress (not memoryless); forgetting to account for all paths from the initial state; assuming geometric distribution when the process isn't memoryless; dividing by zero when $P(\text{restart}) = 1$.
- **Example:** Coin flip game: keep flipping until HH appears. Getting T resets.
  - Let p = probability of eventually getting HH

- First flip outcomes: HH (success, prob $\frac{1}{4}$), HT (restart, prob $\frac{1}{4}$), T (restart, prob $\frac{1}{2}$)
- Recursive equation: $p = \frac{1}{4} + \frac{1}{4}p + \frac{1}{2}p = \frac{1}{4} + \frac{3}{4}p$
- Solving: $p - \frac{3}{4}p = \frac{1}{4} \Rightarrow \frac{1}{4}p = \frac{1}{4} \Rightarrow p = 1$

Example: Three-Door Game

You must open doors in sequence. Door 1 opens with probability 1/2, door 2 with probability 1/3, and door 3 with probability 1/4. If any door fails to open, you return to door 1. What is the probability of eventually getting through all three doors?

Solution

Step 1: Define the variable Let p = probability of eventual success starting from door 1.
Step 2: Identify all possible outcomes from the starting state From door 1, we can either: - Succeed immediately (pass all three doors in one attempt) - Fail and restart at door 1
Step 3: Calculate probability of immediate success The probability of success in one complete attempt: $P(\text{one-shot success}) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24}$
Step 4: Calculate probability of restart $P(\text{restart})$ includes all failure scenarios: - Fail at door 1: $\frac{1}{2}$ - Pass door 1, fail at door 2: $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ - Pass doors 1&2, fail at door 3: $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{8}$
Total: $P(\text{restart}) = \frac{1}{2} + \frac{1}{3} + \frac{1}{8} = \frac{12+8+3}{24} = \frac{23}{24}$
Step 5: Set up the recursive equation Using the geometric restart recursion formula: $p = P(\text{immediate success}) + P(\text{restart}) \cdot p$
Substituting our values: $p = \frac{1}{24} + \frac{23}{24}p$
Step 6: Solve the equation $p - \frac{23}{24}p = \frac{1}{24}$ $\frac{1}{24}p = \frac{1}{24}$ $p = 1$
Step 7: Verify the answer This makes sense because: - There's always a positive probability of success ($\frac{1}{24} > 0$) - We can try infinitely many times - The process is memoryless (restart is identical to initial state) - Therefore, we must eventually succeed
Answer: $p = 1$

AMC 12 Style: Bug on Hexagon

A bug starts at a vertex of a regular hexagon. Each second, it moves to one of the two adjacent vertices with equal probability. If it ever reaches the vertex directly opposite its starting position, it wins. If it returns to the starting vertex, it must restart. What is the probability the bug eventually wins?

Solution

Label vertices 0 (start), 1, 2, 3 (opposite/win), 4, 5 clockwise.
By symmetry, let p = probability of reaching 3 before returning to 0, starting from vertex 1 (or 5). Let q = probability of reaching 3 before 0, starting from vertex 2 (or 4).
From vertex 1: - Probability 1/2: move to 0 $\rightarrow$ restart (contributes 0) - Probability 1/2: move to 2 $\rightarrow$ probability q of winning
So: $p = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot q = \frac{q}{2}$
From vertex 2: - Probability 1/2: move to 1 $\rightarrow$ probability p of winning - Probability 1/2: move to 3 $\rightarrow$ immediate win
So: $q = \frac{1}{2} \cdot p + \frac{1}{2} \cdot 1$
Substituting $p = \frac{q}{2}$ into the second equation: $q = \frac{1}{2} \cdot \frac{q}{2} + \frac{1}{2} = \frac{q}{4} + \frac{1}{2}$
Solving: $q - \frac{q}{4} = \frac{1}{2}$ $\frac{3q}{4} = \frac{1}{2}$ $q = \frac{2}{3}$
Therefore: $p = \frac{q}{2} = \frac{1}{3}$
The bug starts at 0 and first moves to vertex 1 or 5 with equal probability. From there, it wins with probability $p = \frac{1}{3}$.
Answer: $\dfrac{1}{3}$

2.31.1 Common Mistakes and When NOT to Use This Technique

Mistake 1: Confusing Partial Progress with Complete Restart

- **WRONG:** "If you fail at step 3, go back to step 2" (partial progress)
- **RIGHT:** "If you fail at any step, go back to step 1" (complete restart)
- **Why:** Partial progress creates different states, violating the single-state assumption.

Mistake 2: Assuming Memoryless When Process Has Memory

- **WRONG:** Using this technique for "accumulate 3 successes" problems
- **RIGHT:** Only for "get one success" problems with complete restart
- **Why:** Accumulation problems have different states based on progress

Mistake 3: Ignoring Convergence Conditions

- **WRONG:** Applying when $P(\text{failure}) = 1$ (infinite expected time)
- **RIGHT:** Check that $P(\text{failure}) < 1$ for finite expected time
- **Why:** The formula assumes convergence to a finite probability

Mistake 4: Forgetting to Account for All Paths

- **WRONG:** Only considering direct success and direct failure
- **RIGHT:** Include all possible first-step outcomes
- **Why:** Missing paths leads to incorrect recursive equations

2.31.2 Verification Methods

Method 1: Check Reasonableness

- If $p = 1$: Verify that $P(\text{immediate success}) > 0$ and process can continue indefinitely
- If $0 < p < 1$: Verify that $P(\text{failure}) < 1$ for convergence
- If $p = 0$: Check if this makes sense (usually indicates modeling error)

Method 2: Expected Value Check For expected number of attempts: $E = \frac{1}{p}$ (when $p > 0$)

- If $p = 1$, then $E = 1$ (succeed on first try)
- If $p = \frac{1}{2}$, then $E = 2$ (average 2 attempts)
- If E seems too large or too small, recheck the setup

Method 3: Boundary Case Analysis

- Test extreme probabilities: What if $P(\text{immediate success}) = 0$?
- Test convergence: What if $P(\text{failure}) = 1$?
- Test symmetry: Does the answer make sense for symmetric problems?

2.31.3 Edge Cases and Boundary Conditions

Case 1: $P(\textbf{failure}) = 1$ (Infinite Expected Time)

- **Problem:** "Keep flipping until you get HH, but if you get T, you must flip forever"
- **Issue:** $P(\text{restart}) = 1$ leads to division by zero
- **Solution:** This violates the convergence condition; the process never succeeds

Case 2: $P(\textbf{immediate success}) = 0$ (No Direct Path)

- **Problem:** "Keep trying until success, but success requires at least 2 steps"
- **Issue:** No immediate success path
- **Solution:** Redefine the problem or use multi-state recursion

Case 3: Non-Memoryless Process

- **Problem:** "Keep trying until you get 3 heads in a row, but if you get T, go back to 0 heads"
- **Issue:** Different states (0 heads, 1 head, 2 heads) have different restart probabilities
- **Solution:** Use multi-state recursion or Markov chain methods

2.32 Technique M39: Geometric Inversion

Short Name: M39-Inversion

Definition 2.32 (Geometric Inversion)**.** *Inversion with respect to a circle ω of center O and radius r maps a point $P \neq O$ to a point P' on ray $\overrightarrow{OP}$ such that $OP \cdot OP' = r^2$. Key properties:*

- *The center O has no image; points inside ω map outside and vice versa*
- *Lines through O map to themselves (minus the point O)*
- *Lines not through O map to circles through O*
- *Circles through O map to lines not through O*
- *Circles not through O map to circles not through O*
- *The transformation preserves angles (is conformal) but not distances*
- *Inversion is an involution: applying it twice returns the original figure*

WHEN: When tangency/chord intersections suggest transforming a circle configuration to lines/collinear points so segment products become linear; problems with many tangent circles or Apollonius configurations; when Power of a Point relationships become overwhelming; radical axis problems; when you need to transform tangent circles into parallel lines.

WHO: GEO circle configurations where inversion collapses intersecting chords/secants; solvers fluent with choosing centers/radii that linearize the image; advanced geometry competitors comfortable with non-Euclidean transformations.

Configuration cues:

- Multiple tangent circles with complex relationships
- Apollonius problems (circles tangent to three given circles)
- Arbelos or chain-of-tangent-circles configurations (Pappus chains)
- Problems where many Power of Point relations exist
- Coaxial circles or radical axis configurations
- Steiner chains or porisms
- Problems asking for locus of centers or tangency points

Constraints cues:

- Tangent circles remain tangent after inversion
- Orthogonal circles map to themselves under inversion through either circle
- Cross-ratio is preserved under inversion

- If two circles are tangent at the inversion center, they map to parallel lines

Objective cues:

- Transform a complex circle configuration to a simpler one
- Convert tangency problems to parallelism problems
- Find radical centers or axes
- Prove concurrency or collinearity
- Calculate distances or angles in transformed space

How:

- **First Moves:**
  1. Choose inversion center strategically: - At a point of tangency to convert tangent circles to parallel lines - At a point of concurrency to simplify the configuration - At a "problem point" to send it to infinity
  2. Select radius for computational convenience (often $r = 1$ or $r =$ tangent length)
  3. Map key points using $OP' = \frac{r^2}{OP}$
  4. Map circles/lines using the transformation rules
  5. Solve the problem in the inverted plane
  6. If needed, invert back to get the answer in original configuration
- **Key formulas:**
  - Point inversion: If $P = (x, y)$ and $O = (0, 0)$, then $P' = \left(\frac{r^2x}{x^2+y^2}, \frac{r^2y}{x^2+y^2}\right)$
  - Circle equation: Circle $(x - a)^2 + (y - b)^2 = R^2$ not through O inverts to another circle
  - Distance transformation: For points P, Q not collinear with O, if they invert to P', Q', then $P'Q' = \frac{r^2 \cdot PQ}{OP \cdot OQ}$
  - Power transformation: If point P has power p with respect to circle Γ, then under inversion, $\text{Pow}_{\Gamma'}(P') = \frac{r^4 \cdot \text{Pow}_\Gamma(P)}{(OP \cdot OP')^2}$ where Γ' is the inverted circle
  - Complex inversion: In unit circle, $z \mapsto \frac{1}{\overline{z}}$ where $\overline{z}$ is the complex conjugate
- **Bailouts:**
  - If inversion center choice unclear, try multiple centers and compare complexity
  - Use coordinate geometry with the inversion formulas
  - Use complex numbers: inversion in unit circle is $z \mapsto \frac{1}{\overline{z}}$
  - Apply Power of a Point or radical axis theory directly

  - Use homothety or other transformations instead

- **Pitfalls:**
  - Forgetting that distances are not preserved (but ratios of distances from the center are)
  - Incorrect mapping of circles through center of inversion (they become lines, not circles)
  - Not accounting for orientation reversal
  - Forgetting to exclude the center O from domain
  - Missing that angles are preserved but lengths are not
  - Attempting to invert the center point itself

- **Common Inversion Center Choices:**
  - **At point of tangency**: Converts tangent circles to parallel lines (most common choice)
  - **At point of concurrency**: Simplifies multiple concurrent lines/chords
  - **At "problem point"**: Sends a difficult point to infinity for easier analysis
  - **At vertex of triangle**: Useful for triangle-circle configurations
  - **At center of given circle**: Preserves that circle as a line
  - **At intersection of radical axes**: Simplifies coaxial circle problems

- **Example:** Three mutually tangent circles of radii 1, 2, and 3. Find the radius of the circle tangent to all three.

  Solution: Let circles C_1, C_2, C_3 have radii 1, 2, 3 respectively and be mutually tangent. Let T be the point where C_1 and C_2 are tangent.

  Step 1: Choose inversion center at T (point of tangency) with radius $r = 2$ (tangent length from T to C_3).

  Step 2: Under this inversion:

  - C_1 and C_2 become parallel lines L_1 and L_2 (since they're tangent at T)
  - C_3 becomes a circle C_3' between the lines with radius $r_3' = \frac{r^2 \cdot r_3}{|d^2 - r_3^2|}$ where d is distance from T to center of C_3
  - The desired circle becomes a circle tangent to two parallel lines and C_3'

  Step 3: In the inverted plane, we have two parallel lines distance d apart with a circle of radius r_3' between them. The circle tangent to both lines and C_3' has radius $r_4' = \frac{d - 2r_3'}{2}$.

  Step 4: Invert back: $r_4 = \frac{r^2 \cdot r_4'}{d'^2}$ where d' is the distance from T to the center of the inverted circle.

  Answer: $r_4 = \frac{6}{23}$ (after detailed calculation of distances).

- **Additional Example:** Given two circles that don't intersect, find the locus of points from which the tangents to both circles are equal in length.

  Solution: Let the circles be C_1 and C_2 with centers O_1, O_2 and radii r_1, r_2.

  Step 1: Choose inversion center at O_1 with radius $r = \sqrt{r_1^2 + d^2}$ where $d = O_1O_2$.

  Step 2: Under this inversion:

  - C_1 becomes a line L (since it passes through the inversion center)
  - C_2 becomes a circle C_2' not through O_1
  - The locus becomes the set of points equidistant from L and C_2'

  Step 3: In the inverted plane, this locus is a parabola with focus at center of C_2' and directrix L.

  Step 4: Invert back: The original locus is a circle (the inverse of a parabola).

  Answer: The locus is the radical axis of the two circles, which is a line perpendicular to O_1O_2 at distance $\frac{r_1^2 - r_2^2 + O_1O_2^2}{2 \cdot O_1O_2}$ from O_1.

- **Coordinate Example:** Find the image of circle $(x-3)^2 + (y-4)^2 = 25$ under inversion in the unit circle centered at origin.

  Solution: The given circle has center $(3, 4)$ and radius 5. Since the distance from origin to center equals the radius, this circle passes through the origin, so it maps to a line.

  Step 1: Distance from origin to center: $d = \sqrt{3^2 + 4^2} = 5$.

  Step 2: The center inverts to $(\frac{3}{25}, \frac{4}{25})$ since $OP' = \frac{1^2}{5} = \frac{1}{5}$.

  Step 3: Since the circle passes through the origin, it maps to a line. To find the line, we invert two points on the original circle.

  Step 4: The points $(8, 4)$ and $(0, 0)$ lie on the original circle. The point $(8, 4)$ inverts to $(\frac{8}{80}, \frac{4}{80}) = (\frac{1}{10}, \frac{1}{20})$, and $(0, 0)$ maps to infinity.

  Answer: The image is the line $3x + 4y = 1$ (obtained by inverting points on the original circle).

Works Well With: T23-PoP (Power of a Point); T18-ComplexNumber (complex inversion formula); TM11-CircleTangency (tangency preserved); Radical Axis theory; Descartes Circle Theorem (TM16).

Typical Difficulty: Rarely appears explicitly in AMC 12 (problems 23-25 if at all), more common in USA(J)MO and IMO geometry problems. Extremely powerful when applicable but requires significant practice to recognize when to use.

Special Cases:

- Inversion in a line (reflection)
- Stereographic projection (inversion composed with projection)
- Inversive distance as a metric

- Apollonian gaskets and circle packing problems

Category Classification: High-to-Medium Probability for Advanced Problems - While rare in AMC 12, it's a standard tool for olympiad geometry and occasionally the "trick" that makes a hard AMC 12 problem tractable.

2.33 Technique M40: Approximation and Estimation

Short Name: M40-ApproxEstimation

Definition 2.33 (Strategic Approximation)**.** *Use numerical bounds, order-of-magnitude estimates, and local approximations to quickly verify or eliminate answer choices. Includes:*

- *Taylor approximations and series truncation*
- *Bounds via standard inequalities (AM-GM, Cauchy-Schwarz, Jensen's)*
- *Dimensional analysis and unit checking*
- *Asymptotic behavior and dominant term analysis*
- *Sandwich theorem (squeeze) techniques*

WHEN: When expressions are numerically tractable via local estimates or unit analyses that can confirm an answer choice quickly; problems with messy exact solutions where answer choices differ significantly; optimization problems where bounds suffice; when exact computation would exceed time limits but answer choices are well-separated.

WHO: ALG/NT solvers comfortable with numerical bounds and quick approximations; students who can rapidly evaluate expressions near target values and recognize when precision isn't needed.

Configuration cues:

- Answer choices that differ by factors of 2 or more
- Expressions involving roots, logs, or trigonometric functions
- Problems where exact calculation is time-prohibitive
- Optimization problems where bounds can eliminate choices
- Large numbers or factorials where Stirling's approximation applies
- Geometric series with many terms
- Compound interest or exponential growth problems
- Problems asking for "closest integer" or "best approximation"

Constraints cues:

- Answer choices must be sufficiently separated for approximation to work

- Error bounds must be smaller than gaps between choices
- Monotonicity can help (if approximation gives 3.2 and choices are 3 and 4, answer is 3)
- Units and dimensions must be consistent

Objective cues:

- Eliminate 3-4 answer choices quickly
- Verify a suspected answer without full calculation
- Find bounds that sandwich the true answer
- Check reasonableness of a computed answer
- Estimate growth rates or asymptotic behavior

How:

- **First Moves:**
  1. Identify the scale/magnitude needed (ones, tens, hundreds?)
  2. Look for dominant terms that determine behavior
  3. Apply standard approximations appropriate to the scale
  4. Check if approximation error is smaller than answer choice gaps
  5. Refine if needed or proceed to answer selection
- **Standard Approximations (AMC12 level):**
  - $\pi \approx 3.14159... \approx \frac{22}{7} \approx 3.14$ (error < 0.002)
  - $e \approx 2.71828... \approx \frac{19}{7} \approx 2.71$ (error < 0.01)
  - $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$
  - $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$ (golden ratio)
  - $\ln 2 \approx 0.693$, $\ln 10 \approx 2.303$, $\log_{10} 2 \approx 0.301$
- **Logarithmic Approximations (Enhanced):**
  - $\log_{10} 3 \approx 0.477$, $\log_{10} 7 \approx 0.845$
  - Change of base: $\log_b x = \frac{\ln x}{\ln b} \approx \frac{\log_{10} x}{\log_{10} b}$
  - For $x \approx 1$: $\ln x \approx x - 1$ (first-order Taylor)
  - For $x \gg 1$: $\ln(x+1) \approx \ln x + \frac{1}{x}$
  - Product rule: $\ln(ab) = \ln a + \ln b$ for quick estimation
  - Power rule: $\ln(a^n) = n \ln a$ for exponential scaling

- **Taylor/Power Series Approximations:**
  - $(1+x)^n \approx 1 + nx$ for $|x| \ll 1$ (error $\approx \frac{n(n-1)}{2}x^2$)
  - $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ for better accuracy
  - $\sin x \approx x - \frac{x^3}{6}$ for small x (in radians)
  - $\cos x \approx 1 - \frac{x^2}{2}$ for small x
  - $\tan x \approx x + \frac{x^3}{3}$ for small x
  - $e^x \approx 1 + x + \frac{x^2}{2}$ for small x
  - $\ln(1+x) \approx x - \frac{x^2}{2}$ for small x
  - $\sqrt{1+x} \approx 1 + \frac{x}{2}$ for small x
- **Common Trigonometric Values:**
  - $\sin 30^\circ = 0.5$, $\cos 30^\circ = \frac{\sqrt{3}}{2} \approx 0.866$
  - $\sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2} \approx 0.707$
  - $\sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.866$, $\cos 60^\circ = 0.5$
  - $\sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4} \approx 0.259$, $\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4} \approx 0.966$
  - $\sin 75^\circ = \cos 15^\circ \approx 0.966$, $\cos 75^\circ = \sin 15^\circ \approx 0.259$
  - For degrees: $\sin x^\circ \approx \frac{x\pi}{180}$ when x is small
  - $\tan x \approx x$ for small angles in radians
- **Inequality Bounds:**
  - Arithmetic-Geometric Mean: $\sqrt{ab} \leq \frac{a+b}{2}$
  - Bernoulli: $(1+x)^n \geq 1 + nx$ for $x \geq -1, n \geq 1$
  - Triangle inequality: $|a+b| \leq |a| + |b|$
  - For positive reals: $\frac{1}{x+y} \leq \frac{1}{2}\left(\frac{1}{x} + \frac{1}{y}\right)$
  - $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$
- **Useful Estimation Techniques:**
  - **Sandwich method:** Find easy upper and lower bounds
  - **Dominant term:** In sums, focus on the largest term(s)
  - **Logarithmic scaling:** Take logs to handle large exponents
  - **Dimensional analysis:** Check units match in physics-style problems
  - **Stirling's approximation:** $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$
  - **Rule of 72:** Doubling time $\approx \frac{72}{\text{growth rate }\%}$

- **Binomial Coefficient Approximations:**
  - For large n and small k: $\binom{n}{k} \approx \frac{n^k}{k!}$ when $k \ll n$
  - Stirling's approximation: $\binom{n}{k} \approx \frac{\sqrt{2\pi n} \cdot n^n}{\sqrt{2\pi k} \cdot k^k \cdot \sqrt{2\pi(n-k)} \cdot (n-k)^{n-k}} = \frac{n^n}{\sqrt{2\pi k(n-k)} \cdot k^k (n-k)^{n-k}}$
  - For central binomial: $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$
  - Simple approximation: $\binom{n}{k} \approx \frac{n^k}{k!}$ when $k \ll n$ and $k \ll n-k$
- **Probability and Statistical Approximations:**
  - Normal approximation to binomial: When $np \geq 5$ and $n(1-p) \geq 5$
  - Poisson approximation: $\binom{n}{k} p^k (1-p)^{n-k} \approx \frac{(np)^k e^{-np}}{k!}$ when n large, p small
  - Expected value: $E[X] \approx \frac{\text{sum of outcomes}}{\text{number of trials}}$ for empirical estimation
  - Variance approximation: $\text{Var}(X) \approx E[X^2] - (E[X])^2$
  - Central limit theorem: Sample means approximately normal for large n
- **Quick Mental Math Tricks:**
  - Multiply by 5: divide by 2 and multiply by 10
  - Square numbers near 50: $(50+a)^2 = 2500 + 100a + a^2$
  - Divide by 9: sum digits repeatedly
  - Estimate square roots: use nearby perfect squares
  - Percentage increases: $1.1^{10} \approx 2.59$, $1.05^{14} \approx 2$
- **Additional Mental Math Tricks:**
  - Multiply by 11: $11 \times ab = a(a+b)b$ (with carrying if needed)
  - Square numbers ending in 5: $(10a+5)^2 = 100a(a+1) + 25$
  - Multiply by 25: divide by 4 and multiply by 100
  - Estimate percentages: 10% = divide by 10, 5% = half of 10%
  - Compound interest: $A = P(1+r)^t \approx P(1+rt)$ for small r and t
  - Geometric mean: $\sqrt{ab} \approx \frac{a+b}{2} - \frac{(a-b)^2}{8(a+b)}$ for close values
- **Bailouts:**
  - If approximation inconclusive, try tighter bounds
  - Use calculator values from earlier problems as benchmarks
  - Test specific values if function behavior unclear
  - Revert to exact calculation for final 2 choices if time permits
- **Pitfalls:**

  - Over-approximating when choices are close (differ by less than 20
  - Using degree approximations when radians required (or vice versa)
  - Ignoring higher-order terms when they're significant
  - Approximating too early in a multi-step calculation (errors compound)
  - Using wrong branch of multi-valued functions (e.g., $\sqrt{x^2} = |x|$, not always x)
  - Forgetting that percentage errors multiply in products
  - Not checking if approximation assumptions are valid

- **Error Analysis Guidelines:**
  - For $(1+x)^n \approx 1 + nx$: Absolute error $\approx \frac{n(n-1)x^2}{2}$ when $|x| \ll 1$
  - For $\sin x \approx x$: Absolute error $\leq \frac{|x|^3}{6}$ for $|x| \leq 1$
  - For $\ln(1+x) \approx x$: Absolute error $\leq \frac{x^2}{2}$ for $x > -1$
  - For $\ln(1+x) \approx x - \frac{x^2}{2}$: Absolute error $\leq \frac{|x|^3}{3}$ for $|x| \leq 1$
  - Percentage error in products: If $A \approx B(1+\epsilon)$, then $A^n \approx B^n(1+n\epsilon)$ approximately
  - Rule of thumb: Keep approximation error $< 5\%$ of answer choice gaps for reliable elimination
  - When in doubt: Use tighter bounds or exact calculation for final verification

- **When NOT to Approximate:**
  - Answer choices differ by less than 10% - approximation errors may be comparable to choice gaps
  - Exact rational answers are expected (common in AMC12)
  - Multi-step calculations where errors compound significantly
  - Problems explicitly asking for exact values
  - When approximation assumptions are violated (e.g., x not small for Taylor series)
  - Symmetric or special cases where exact solution is simpler than approximation

- **Example:** Find the value of $\sqrt{2023} + \sqrt{2025}$.

  Solution: Note that $2025 = 45^2$, so $\sqrt{2025} = 45$. For $\sqrt{2023}$, we have $2023 = 2025 - 2 = 45^2 - 2$.

  Using $(1+x)^{1/2} \approx 1 + \frac{x}{2}$ with $\sqrt{2023} = 45\sqrt{1 - \frac{2}{2025}}$:

  $\sqrt{2023} \approx 45\left(1 - \frac{1}{2025}\right) \approx 45 - \frac{45}{2025} = 45 - \frac{1}{45} \approx 45 - 0.022$

  Therefore: $\sqrt{2023} + \sqrt{2025} \approx 44.978 + 45 = 89.978 \approx 90$

  If answer choices are 88, 90, 92, 94, 96, we confidently choose 90.

- **Additional Examples:**

  Example 2: Estimate $(1.05)^{20}$.

  Solution: Using $(1+x)^n \approx 1+nx$ for small x: $(1.05)^{20} = (1+0.05)^{20} \approx 1+20\times 0.05 = 1+1 = 2$.

  For better accuracy: $(1.05)^{20} \approx 1+20\times 0.05+\frac{20\times 19}{2}\times(0.05)^2 = 1+1+0.475 = 2.475$.

  The exact value is approximately 2.653, so the second approximation is quite good.

  Example 3: Approximate $\sin(0.1)$ in radians.

  Solution: Using $\sin x \approx x - \frac{x^3}{6}$: $\sin(0.1) \approx 0.1 - \frac{(0.1)^3}{6} = 0.1 - \frac{0.001}{6} \approx 0.1 - 0.00017 = 0.09983$.

  Example 4: Estimate $\binom{100}{50}$.

  Solution: Using the central binomial approximation: $\binom{100}{50} \approx \frac{4^{50}}{\sqrt{50\pi}} = \frac{2^{100}}{\sqrt{50\pi}} \approx \frac{2^{100}}{\sqrt{157}} \approx \frac{2^{100}}{12.53} \approx 0.08\times 2^{100}$.

  For order of magnitude: $\log_{10}(\binom{100}{50}) \approx 100\log_{10}(2) - \log_{10}(12.53) \approx 100\times 0.301 - 1.1 \approx 30.1 - 1.1 = 29$.

Works Well With: T7-AMGM (for bounds); T22-CauchySchwarz (for estimates); T33-RationalRoot (test specific values); Answer Choice Testing; Dimensional Analysis.

Typical Difficulty: Valuable across all problem levels - Problems 1-10 for verification, Problems 11-20 for elimination, Problems 21-25 for narrowing choices when exact solution is infeasible.

Special Cases:

- Fermi estimation problems
- "Nearest integer" problems
- Large factorial or binomial coefficient estimation
- Compound interest and exponential growth
- Probabilistic bounds and expected value estimates

Success Rate: 85-90

Category Classification: High-Probability - Essential time-saving technique for AMC12, especially under time pressure. Should be in every competitor's toolkit.

2.34 Technique M41: Pattern Induction from Small Cases

Short Name: M41-PatternInduction

Definition 2.34 (Pattern Induction)**.** *Compute initial cases ($n = 0, 1, 2, 3, ...$) to identify a pattern, form a conjecture for the general formula, then verify the pattern through:*

- *Additional test cases at strategic values*

- *Checking boundary conditions and special cases*
- *Verifying the pattern satisfies any given recurrence relation*
- *Confirming dimensional/asymptotic consistency*

This technique generates a plausible answer when formal proof is time-prohibitive or unnecessarily complex for the competition setting.

WHEN: When small cases strongly suggest a closed form that can be validated by checking additional boundary instances; recursive sequences without obvious closed forms; counting problems with complex constraints; when the problem asks for a specific large value (like a_{100} or a_{2025}) rather than a general formula.

WHO: CNT/ALG solvers comfortable with spotting patterns from low-n data and justifying them heuristically when formal proofs are omitted; students who can recognize standard sequences and their variations.

Configuration cues:

- Recursive sequences or recurrence relations without obvious closed form
- Counting problems asking for specific term (e.g., 100th, 2025th term)
- Geometric configurations that scale predictably
- Problems where $n = 0, 1, 2, 3$ are easily computed by hand
- "Find a_n for $n =$ [specific large number]" rather than "prove that $a_n =$..."
- Tiling problems, path counting, or arrangement problems
- Problems involving operations repeated n times
- Iterative geometric constructions

Constraints cues:

- Pattern must be consistent across all computed cases
- Check whether indexing starts at $n = 0$ or $n = 1$ (common source of errors)
- Verify pattern works for any explicitly given values
- Ensure pattern respects problem constraints (e.g., must be integer, positive)
- Pattern should match asymptotic behavior if determinable

Objective cues:

- Find explicit formula for sequence term
- Determine value at specific index without computing all intermediate terms

- Identify period for cyclic sequences
- Discover closed form for recurrence relation
- Find limiting behavior or convergence value

How:

- **First Moves:**
  1. Compute initial values systematically: - Always include $n = 0$ if defined (catches shifts) - Calculate $n = 1, 2, 3, 4$ carefully - Organize in a table with differences/ratios
  2. Analyze the sequence for patterns: - First differences (arithmetic sequences) - Second differences (quadratic sequences) - Ratios (geometric sequences) - Alternating patterns or periodicity
  3. Form a conjecture based on recognized pattern
  4. Verify with additional strategic test cases: - Next value ($n = 5$ or $n = 6$) - Power of 2 value if accessible - Any given boundary values - Value just before target if computable
- **Common Pattern Types to Recognize:**
  - **Polynomial:** Constant k-th differences implies degree-k polynomial - Linear: $a_n = an + b$ (constant first differences) - Quadratic: $a_n = an^2 + bn + c$ (constant second differences) - Cubic: constant third differences
  - **Exponential:** Constant ratios $\frac{a_{n+1}}{a_n} = r$ - Pure: $a_n = ar^n$ - Mixed: $a_n = ar^n + bn + c$
  - **Factorial-based:** $a_n = n!$, $a_n = \frac{n!}{k!}$, etc.
  - **Combinatorial:** Often $\binom{n}{k}$, $\binom{n+k}{k}$, Catalan numbers
  - **Recursive classics:** - Fibonacci: $a_n = a_{n-1} + a_{n-2}$ - Tribonacci: $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ - Lucas sequences and generalizations
  - **Periodic:** Sequence repeats after period p
  - **Eventually periodic:** Becomes periodic after initial terms
  - **Mixed/Piecewise:** Different formulas for even/odd n
- **Pattern Recognition Tools:**
  - **Difference table:** Compute successive differences

```
$n$:      0   1   2   3   4   5
$a_n$:    1   3   7   13  21  31
$\Delta^1$:       2   4   6   8   10    (first differences)
$\Delta^2$:         2   2   2   2       (second differences constant → q
```

  - **Ratio analysis:** Check $\frac{a_{n+1}}{a_n}$ for geometric behavior
  - **Modular patterns:** Compute $a_n \bmod m$ for small m
  - **Generating function hints:** If pattern suggests $(1-x)^{-k}$ expansion

- **Verification Strategies:**
  - Check recurrence: If given $a_n = f(a_{n-1}, a_{n-2}, ...)$, verify formula satisfies it
  - Test special cases: $n = 0$, $n = 1$, any given values
  - Dimensional analysis: Verify units/type consistency
  - Asymptotic check: Does growth rate match expectations?
  - Compute check: Calculate target value and verify it's reasonable

- **Advanced Patterns (OEIS-worthy):**
  - Bell numbers: $1, 1, 2, 5, 15, 52, 203, ...$ (set partitions)
  - Catalan numbers: $1, 1, 2, 5, 14, 42, 132, ...$ (use $C_n = \frac{1}{n+1}\binom{2n}{n}$)
  - Derangements: $!n = 0, 1, 2, 9, 44, 265, ...$
  - Triangular numbers: $T_n = \frac{n(n+1)}{2}$
  - Pentagonal numbers: $P_n = \frac{n(3n-1)}{2}$

- **Bailouts:**
  - If no pattern emerges after $n = 6$, try: - Look for pattern in differences of differences - Check if pattern appears in numerator/denominator separately - Consider modular arithmetic patterns - Try logarithm of sequence for exponential growth
  - Use OEIS (Online Encyclopedia of Integer Sequences) if allowed
  - Try fitting with polynomial regression if pattern unclear

- **Pitfalls:**
  - Assuming pattern too early (need at least 4-5 terms)
  - Missing that pattern starts at different index (off-by-one errors)
  - Not detecting periodic behavior (check at least 2 periods)
  - Ignoring parity-dependent patterns (different for even/odd n)
  - Wrong pattern type (e.g., seeing arithmetic when it's quadratic)
  - Arithmetic errors in initial computations cascading
  - Not checking the pattern at the specific requested value
  - Missing that sequence is well-known (Fibonacci, Catalan, etc.)

- **Example:** Find the number of ways to tile a $2 \times n$ rectangle with 1×2 dominoes.

  Solution: - $a_1 = 1$ (one vertical placement) - $a_2 = 2$ (two horizontal or two vertical) - $a_3 = 3$ (computed by cases) - $a_4 = 5$, $a_5 = 8$, $a_6 = 13$

  Pattern recognition: This is the Fibonacci sequence! Recurrence check: $a_n = a_{n-1} + a_{n-2}$ (last domino either vertical or two horizontal)

  Therefore $a_{10} = F_{10} = 55$.

Works Well With: T13-CharEq (for verifying recurrence relations); T29-Telescoping (for summing patterns); TM40-FiniteDiffTable (for polynomial patterns); T25-GenFunc (if pattern suggests generating function).

Typical Difficulty: Appears across all problem levels - Problems 1-10 for simple patterns, Problems 11-20 for complex patterns, Problems 21-25 when combined with other techniques.

Special Cases:

- Competition asks for a_{2025} specifically (competition year!)
- Sequences defined by iterative geometric constructions
- Counting problems with recursive structure
- Digital root or sum-of-digits sequences
- Continued fraction convergents

Success Rate: 90-95

Category Classification: High-Probability - Essential technique for AMC12/COMC. Often the intended solution method when problem asks for specific large index rather than general formula.

3 Medium-Probability Conversion Techniques

3.1 Technique M2: Burnside's Lemma (Pólya Counting)

Short Name: TM2-BurnsidePolya

Definition 3.1 (Burnside's Lemma)**.** *For a finite group G acting on a set X, the number of distinct configurations up to the symmetries in G equals*

$$\frac{1}{|G|} \sum_{g \in G} \big| \operatorname{Fix}(g) \big|,$$

where $\operatorname{Fix}(g)$ *is the set of configurations fixed by g.*

WHEN: Colorings/arrangements with rotational/reflective symmetry (necklaces/bracelets, polygon vertex colorings, cube faces), where naive counting overcounts by symmetry.

WHO: CNT counting problems with symmetry constraints; solvers comfortable with group theory and Burnside's lemma.

Configuration cues:

- Dihedral/cyclic symmetry ("up to rotation/reflection"); repeating motifs around a circle or polygon
- Distinguishable colors but indistinguishable positions modulo the group action

Objective cues:

- Count distinct colorings/patterns modulo rotation only (use C_n) or modulo rotation and reflection (use D_n)

Why It Works: Group actions partition configurations into orbits. Averaging fixed counts over group elements gives the number of orbits.

WHAT: List group elements, compute fixed counts via cycle structure, average.

How:

- **First Moves:**
  1. Identify the symmetry group (C_n for rotations; D_n for rotations+reflections).
  2. For a rotation by k, fixed colorings $= m^{\gcd(n,k)}$ for m colors.
  3. For reflections in D_n, use explicit fixed-counts:
     - n odd: each reflection fixes one vertex and swaps $(n-1)/2$ pairs $\Rightarrow m^{(n+1)/2}$.
     - n even:
       * axis through opposite vertices: two fixed vertices and $(n-2)/2$ pairs $\Rightarrow m^{(n/2)+1}$;
       * axis through midpoints: no fixed vertices and $n/2$ pairs $\Rightarrow m^{n/2}$.
  4. Average the fixed counts over all group elements.
- **Bailouts:** If reflection cases are messy, first solve rotations-only (necklaces), then handle reflections (bracelets) as a small add-on; or count a fundamental sector when only rotations matter.
- **Pitfalls:** Mixing rotation-only vs. rotation+reflection; miscounting cycles for reflections when n is even vs. odd; forgetting to divide by $|G|$.
- **Example:** Distinct 4-bead necklaces with 2 colors up to rotation: fixed under 0°: $2^4 = 16$; 90°: $2^1 = 2$; 180°: $2^2 = 4$; 270°: $2^1 = 2$. Average $= \frac{16+2+4+2}{4} = 6$.

3.2 Technique M4: Pick's Theorem

Short Name: TM4-Picks

Definition 3.2 (Pick's Theorem)**.** *For a simple lattice polygon, $A = I + \frac{B}{2} - 1$, where I is the number of interior lattice points and B the number on the boundary.*

WHEN: Lattice-vertex polygons where counting lattice points is natural or faster than coordinate area.

WHO: GEO coordinate geometry problems with lattice polygons; solvers comfortable with lattice point counting and area calculations.

How:

- **First Moves:** Compute B via edge gcds: for edge $\overline{(x_i, y_i)(x_{i+1}, y_{i+1})}$, add $\gcd(|\Delta x|, |\Delta y|)$. Find A by shoelace or geometry; solve for I or A as needed.
- **Bailouts:** If polygon is not simple, partition into simple pieces; otherwise use shoelace directly.
- **Example:** Rectangle $(0,0), (3,0), (3,2), (0,2)$: $A = 6$, $B = 3 + 2 + 3 + 2 = 10$, so $I = 6 - 10/2 + 1 = 2$.

3.3 Technique M5: Frobenius/McNugget (Two Denominations)

Short Name: TM5-Frobenius

Definition 3.3 (Frobenius (Two Coprime Denominations))**.** *For coprime a, b, the largest non-representable value by $ax+by$ with $x, y \geq 0$ is $ab-a-b$, and the number of non-representables is $\frac{(a-1)(b-1)}{2}$.*

WHEN: "Exact sum using packs/coins" with two sizes; quick feasibility or extremal value checks.

WHO: NT number theory problems with linear combinations and Frobenius numbers; solvers comfortable with gcd and linear Diophantine equations.

How:

- **First Moves:** Check $d = \gcd(a, b)$. If $d \nmid N$, not representable; otherwise reduce by d and solve with coprime pair. Use $ab - a - b$ for the largest gap when coprime.
- **Quick congruence filter (coprime):** Compute $b^{-1} \pmod{a}$, set $y \equiv N\, b^{-1} \pmod{a}$ with $0 \leq y < a$; if $N - by \geq 0$ then $x = (N - by)/a$ gives a representation; otherwise increase y by multiples of a until $by \leq N$ or conclude infeasible under bounds.
- **Bailouts:** For three+ denominations, use modular checks, small DP, or generating functions.
- **Example:** With $a = 4, b = 9$ (coprime), largest non-representable is $36 - 4 - 9 = 23$. 37 is representable since $37 \equiv 1 \pmod{4}$ and $9 \equiv 1 \pmod{4}$: take $37 = 9 \cdot 1 + 4 \cdot 7$.

3.4 Technique M6: Extended Euclidean Algorithm

Short Name: TM6-ExtEuclid

Definition 3.4 (Extended Euclidean)**.** *Computes integers (u, v) with $au + bv = \gcd(a, b)$; when $\gcd(a, m) = 1$, $u \equiv a^{-1} \pmod{m}$ gives the modular inverse.*

WHEN: Modular inverses, solving $ax \equiv b \pmod{m}$, linear Diophantine feasibility $ax + by = c$.

WHO: NT number theory problems with linear Diophantine equations; solvers comfortable with extended Euclidean algorithm and modular arithmetic.

How:

- **First Moves:** Run EEA to obtain u, v with $au + bv = \gcd(a, b)$. Invert $a \pmod{m}$ by taking $u \bmod m$ when $\gcd(a, m) = 1$. Solve $ax \equiv b \pmod{m}$ by letting $d = \gcd(a, m)$: if $d \nmid b$ there is no solution; else set $a' = a/d,\ b' = b/d,\ m' = m/d$ and solve $a'x \equiv b' \pmod{m'}$; the full solution set is $x \equiv x_0 + t\,m' \pmod{m}$ for $t \in \mathbb{Z}$.
- **Pitfalls:** Inverting when $\gcd(a, m) \neq 1$; forgetting to divide through by d and to produce a full residue class of solutions.
- **Example:** $7x \equiv 1 \pmod{26}$ has inverse $x \equiv 15$ since $7 \cdot 15 = 105 \equiv 1 \pmod{26}$.

3.5 Technique M7: Ptolemy's Theorem (Cyclic Quadrilaterals)

Short Name: TM7-Ptolemy

Definition 3.5 (Ptolemy)**.** *For cyclic quadrilateral $ABCD$, $AC \cdot BD = AB \cdot CD + AD \cdot BC$.*

WHEN: Length relations in cyclic quadrilaterals or when you can prove cyclicity (right angles, equal arcs/angles).

WHO: GEO circle geometry problems with cyclic quadrilaterals; solvers comfortable with circle properties and quadrilateral relationships.

How:

- **First Moves:** Verify cyclicity; apply Ptolemy to trade between sides and diagonals; combine with similarity/PoP as needed.
- **Example:** In a square of side 1, Ptolemy gives $d^2 = 1 \cdot 1 + 1 \cdot 1 = 2$, so the diagonal $d = \sqrt{2}$.

3.6 Technique M8: Double Counting

Short Name: TM8-DoubleCount

Definition 3.6 (Double Counting)**.** *Count the same quantity in two ways and equate the expressions to extract an identity or bound (e.g., handshake lemma).*

WHEN: Symmetric counts over pairs/incidences (people–handshakes, vertices–edges, subsets–elements); expectations via indicator sums.

WHO: CNT counting problems with double counting; solvers comfortable with combinatorial identities and graph theory.

How:

- **First Moves:** Define the set of incidences; count by summing over one side (e.g., vertices) and again over the other (e.g., edges).
- **Pitfalls:** Mismatched universes; forgetting multiplicities; assuming uniformity without justification.
- **Example:** In any finite graph, $\sum \deg(v) = 2|E|$ (count edge endpoints by vertices vs. by edges).

3.7 Technique 24: Trigonometric Substitution

Short Name: T24-TrigSub

Definition 3.7 (Trigonometric Substitution)**.** *Replace expressions involving* $\sqrt{1-x^2}$, $\sqrt{1+x^2}$, $\sqrt{x^2-1}$, *or circle constraints with trigonometric functions to exploit identities:*

- $x = \sin\theta$ *with* $\theta \in [-\pi/2, \pi/2]$ *so that* $\sqrt{1-x^2} = |\cos\theta| = \cos\theta$ *(since* $\cos\theta \geq 0$ *on this interval)*
- $x = \tan\theta$ *with* $\theta \in (-\pi/2, \pi/2)$ *so that* $\sqrt{1+x^2} = |\sec\theta| = \sec\theta$ *(since* $\sec\theta > 0$ *on this interval)*
- $x = \sec\theta$ *for* $|x| \geq 1$*: Use* $\theta \in \left[0, \frac{\pi}{2}\right)$ *when* $x \geq 1$ *and* $\theta \in \left(\frac{\pi}{2}, \pi\right]$ *when* $x \leq -1$*, so that* $\sqrt{x^2-1} = |\tan\theta|$*. In particular, on* $\left[0, \frac{\pi}{2}\right)$ *we have* $\sqrt{x^2-1} = \tan\theta$ *since* $\tan\theta \geq 0$*.*

Range remark: The chosen intervals for θ ensure that the radicals $\sqrt{1-x^2}$, $\sqrt{1+x^2}$, and $\sqrt{x^2-1}$ match the corresponding trigonometric expressions without extra absolute values.

WHEN:

Configuration cues:

- Radicals $\sqrt{1 \pm x^2}$, $\sqrt{x^2-1}$, or circle constraints such as $x^2 + y^2 = 1$
- Natural parametrization by $x = \sin\theta$, $x = \tan\theta$, or $x = \sec\theta$
- Two-variable circle parametrization: for $x^2 + y^2 = r^2$, use $(x, y) = (r\cos\theta,\ r\sin\theta)$
- Pythagorean forms or constraints reducible to $\sin^2\theta + \cos^2\theta = 1$

Constraints cues:

- Choose θ-ranges to control signs so radicals match trig without absolute values
- Respect domain restrictions (e.g., $|x| \leq 1$ for sin, $x \in \mathbb{R}$ for tan, $|x| \geq 1$ for sec)
- For (x, y) parametrizations, restrict θ to the stated quadrant/sign information or to a full cycle $[0, 2\pi)$ consistent with the problem
- Verify monotonic branches and exclude extraneous back-substitutions at endpoints

Objective cues:

- Linearize radicals or circle constraints using identities to solve or simplify
- Reduce to trigonometric equations, solve in θ, then back-substitute valid solutions
- Parametrize loci on circles/ellipses to compute areas, lengths, or counts

Why It Works: Identities $\sin^2\theta+\cos^2\theta = 1$ and angle-sum formulas linearize otherwise nonlinear expressions.

WHO: ALG/GEO algebra and geometry with radicals or circle constraints; solvers fluent with trigonometric identities and domain/range handling.

WHAT: Parametrize variables using $\sin\theta$, $\tan\theta$, or $\sec\theta$ so radicals simplify to $\cos\theta$, $\sec\theta$, or $|\tan\theta|$.

How:

- **First Moves:**
  1. Match the radical to a standard substitution and choose a θ-range to control signs.
  2. Rewrite all terms via identities; remove absolute values consistent with the chosen range.
  3. Solve in θ, then back-substitute to x; check endpoints and discard extraneous solutions.
- **Bailouts:** If trig inflates algebra, try an algebraic route (isolate and square with domain checks), coordinate parametrization $(r\cos\theta, r\sin\theta)$ for two-variable circles, or Weierstrass substitution $t = \tan(\theta/2)$ when solving trig equations.
- **Pitfalls:** Wrong sign due to an incorrect θ-range; ignoring absolute values; introducing extraneous roots when squaring; failing to validate endpoints/back-substitutions.
- **Example:** See the "Solve with $x = \sin\theta$" example below; template: pick range so radicals equal the intended trig functions.

Example: Solve with $x = \sin\theta$

Find all real $x \in [-1, 1]$ such that $x + \sqrt{1-x^2} = 1$.

Solution

Let $x = \sin\theta$ with $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $\sqrt{1-x^2} = \cos\theta \geq 0$ and the equation becomes

$$\sin\theta + \cos\theta = 1.$$

Using $\sin\theta + \cos\theta = \sqrt{2}\,\sin\big(\theta + \frac{\pi}{4}\big)$, we get

$$\sqrt{2}\,\sin\big(\theta + \tfrac{\pi}{4}\big) = 1 \quad\Longrightarrow\quad \sin\big(\theta + \tfrac{\pi}{4}\big) = \tfrac{1}{\sqrt{2}}.$$

Within $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, this gives $\theta + \frac{\pi}{4} \in \big\{\frac{\pi}{4}, \frac{3\pi}{4}\big\}$, hence $\theta \in \{0, \frac{\pi}{2}\}$. Therefore $x = \sin\theta \in \{0, 1\}$.

Answer: $x \in \{0, 1\}$

3.8 Technique 25: Generating Functions

Short Name: T25-GenFunc

Definition 3.8 (Generating Function Conversion)**.** *Convert counting problems into algebraic manipulation of polynomials or series.*

WHEN: Coin-change and bounded-digit counts; distributing indistinguishable balls into distinguishable boxes; compositions/partitions with caps; linear recurrences; sums of independent discrete variables (PGFs); binary strings with gap/parity constraints (modeled by OGFs or recurrences); simple lattice-path step counts via product/PGF methods.

Configuration cues:

- Constrained selections, partitions/compositions, or bounded counts
- Natural factorization into independent choices
- Labeled vs. unlabeled structures (distinguishable vs. indistinguishable items/boxes)
- Parity, spacing, or gap constraints suggest encoding via restricted series (e.g., only even powers)
- Linear recurrences with constant coefficients and initial conditions (ordinary or exponential generating functions)
- Probability generating functions for sums of independent discrete random variables

Constraints cues:

- Encode bounds correctly (e.g., $1 + x + \cdots + x^b$ for up to b)
- Clarify distinguishability and whether zeros are allowed
- Use ordinary generating functions (OGFs) for unlabeled combinatorial classes (most AMC/COMC); use exponential generating functions (EGFs) for labeled classes where labels attach to atoms
- Truncate expansions appropriately when totals are bounded; formal power series require only algebraic, not analytic, convergence
- For recurrences, set up functional equations and solve for closed forms before extracting coefficients

Objective cues:

- Form a product generating function and extract a specific coefficient
- Convert recurrence/constraints to closed-form counts via series identities
- Extract $[x^n]$ (or $[\frac{t^n}{n!}]$ for EGFs) to match the target total
- Use convolution/product to model sums of independent counts or steps

Why It Works: Transforms combinatorial problems into algebraic ones.

WHO: CNT/PROB solvers modeling constrained selections, distributions, or recurrences with coefficient extraction.

WHAT: Encode choices as power series whose coefficients count outcomes; combine factors and extract the needed coefficient.

How:

- **First Moves:**
  1. Choose a factor for each independent choice (e.g., $1 + x + \cdots + x^b$ for up to b).
  2. Multiply factors to couple choices; simplify using algebraic identities.
  3. Extract $[x^n]$ of the product via series identities or binomial forms.
- **Bailouts:** If coefficient extraction is cumbersome, switch to stars and bars, a recurrence with initial conditions, partial fractions for rational OGFs, or complementary counting/inclusion–exclusion.
- **Pitfalls:** Mis-encoding caps/bounds; mixing labeled vs. unlabeled models; extracting the wrong coefficient/variable; failing to truncate when totals are bounded.
- **Example:** See the balls-into-boxes example below; encode positive counts as $x(1 + x + \cdots)^k$ and read off the coefficient.

Coefficient extraction identity: For integers $n \geq 0$ and $k \geq 1$, the coefficient of x^n in $(1-x)^{-k}$ is $\binom{n+k-1}{k-1}$.

AMC 12B 2020, Problem 18

How many ways can we place 5 indistinguishable balls into 3 distinguishable boxes such that no box is empty?

Solution

Setup: We need positive integers x_1, x_2, x_3 with $x_1 + x_2 + x_3 = 5$
Substitution: Let $y_i = x_i - 1 \geq 0$
Then: $y_1 + y_2 + y_3 = 2$
Stars and Bars: Number of non-negative integer solutions:

$$\binom{2+3-1}{3-1} = \binom{4}{2} = 6$$

Alternative (Generating Functions): Count the coefficient of x^5 in $(x + x^2 + x^3 + \cdots)^3 = x^3(1 + x + x^2 + \cdots)^3 = x^3(1-x)^{-3}$. This equals the coefficient of x^2 in $(1-x)^{-3}$, which is $\binom{2+3-1}{3-1} = \binom{4}{2} = 6$.
Answer: 6

3.9 Technique 26: Invariants and Monovariants

Short Name: T26-Invariants

Definition 3.9 (Invariants and Monovariants)**.** *Identify quantities that remain constant (invariants) or change monotonically (monovariants) throughout a process or transformation.*

WHEN: Chip-/stone-moving games; sliding puzzles; swapping/inversion parity; graph-coloring/parity processes; digit/weight sums;
area/perimeter under operations; modular residues; Nim-like invariants.

Configuration cues:

- Iterative processes, games, or algorithmic transformations
- Natural candidates: parity, sums, modular residues, area/length, inversion count

Constraints cues:

- Verify invariance or monotonicity under all allowed moves
- Ensure lower/upper bounds to prevent cycles for monovariants

Objective cues:

- Prove impossibility, termination, or bound the number of steps
- Force a contradiction or a unique outcome via the invariant

Why It Works: Invariants constrain the problem and often directly lead to the solution.

WHO: CNT optimization process/algorithm/game problems; solvers preferring structural invariants/monovariants over casework or brute force.

WHAT: Define a quantity that remains invariant or moves monotonically, preventing cycles and forcing termination or impossibility.

How:

- **First Moves:**
  1. Propose candidate invariants/monovariants (parity, sum, mod k, area, inversions).
  2. Prove the property is preserved or monotone under allowed moves.
  3. Deduce bounds, termination, or contradictions to answer the question.
- **Bailouts:** If no simple invariant works, try a coarser modulus, a weighted sum/potential function, or restrict to parity/ordering; reframe moves or group operations so the measure changes predictably.
- **Pitfalls:** Assuming invariance without checking all moves; choosing a monovariant that can increase/decrease in cycles; missing lower/upper bounds so termination is not guaranteed.
- **Example:** See the sequence-limit monovariant example below; template: define a potential, prove monotonicity and bounds, then take limits or reach a contradiction.

Example: Sequence Limit via Monovariant

A sequence starts with $a_1 = 2017$. For $n \geq 1$:

$$a_{n+1} = \frac{a_n^2 + 2}{a_n + 2}.$$

Prove that (a_n) is decreasing and bounded below, and compute $\lim_{n\to\infty} a_n$.

Solution

Define $f(x) = \dfrac{x^2+2}{x+2} = \dfrac{x^2-4+6}{x+2} = \dfrac{(x-2)(x+2)+6}{x+2} = x - 2 + \dfrac{6}{x+2}$.
Monotonicity: For $x \geq 1$,

$$f(x) - x = \frac{x^2+2-x(x+2)}{x+2} = \frac{2-2x}{x+2} = \frac{2(1-x)}{x+2} \leq 0$$

with equality iff $x = 1$. Thus, if $a_n \geq 1$ then $a_{n+1} \leq a_n$.
Lower Bound: Moreover,

$$f(x) - 1 = \frac{x^2+2-(x+2)}{x+2} = \frac{x^2-x}{x+2} = \frac{x(x-1)}{x+2} \geq 0 \quad (x \geq 1)$$

so $a_{n+1} \geq 1$ whenever $a_n \geq 1$.
Convergence: Since $a_1 = 2017 > 1$, by induction $a_n \geq 1$ for all n. The sequence is decreasing and bounded below, so it converges to some $L \geq 1$.
Taking limits: $L = \dfrac{L^2+2}{L+2}$ yields $L(L+2) = L^2+2$, so $L^2+2L = L^2+2$, thus $2L = 2$ and $L = 1$.
Answer: 1

3.10 Technique 27: Power-Sum Recurrences

Short Name: T27-PowerSum

Definition 3.10 (Power-Sum Recurrence)**.** *For $x \neq 0$, use the linear recurrence for $S_n = x^n + \dfrac{1}{x^n}$ with initial condition $S_0 = 2$: for $n \geq 1$, $S_{n+1} = S_1\, S_n - S_{n-1}$ to compute higher powers efficiently. This holds for any nonzero real or complex x.*

WHEN: Given $x + \frac{1}{x}$ (or $x^k + \frac{1}{x^k}$), find $x^n + \frac{1}{x^n}$; evaluate trigonometric-like identities via $2\cos\theta$ substitutions; simplify nested radicals using power sums.

Configuration cues:

- Given $S_1 = x + \frac{1}{x}$ or related symmetric base quantities
- Asked for $S_n = x^n + x^{-n}$ or a related symmetric power

Constraints cues:

- Require $x \neq 0$; confirm integer $n \geq 0$
- If given $x^k + x^{-k}$, adapt the recurrence by stepping in k

Objective cues:

- Compute S_n via the linear recurrence without solving for x
- Translate to requested symmetric expressions efficiently

Why It Works: Expanding $(x + 1/x)^n$ yields a linear recurrence for the power sums, avoiding explicit solution for x.

WHO: ALG symmetric expressions with symmetric powers/reciprocal pairs; solvers leveraging recurrences/Chebyshev-like relations.

WHAT: Use the linear recurrence for $S_n = x^n + x^{-n}$ derived from multiplying by $x + x^{-1}$.

How:

- **First Moves:**
  1. Set $S_0 = 2$ and take the given $S_1 = x + \frac{1}{x}$.
  2. Iterate $S_{n+1} = S_1 S_n - S_{n-1}$ to reach the desired power.
  3. Translate to any required symmetric expressions via algebraic identities when applicable.
- **Bailouts:** If recurrence arithmetic gets heavy, use minimal polynomials/Chebyshev polynomials, try stepping by larger k, or exploit identities like $(x^n + x^{-n})^2 = S_{2n} + 2$.
- **Pitfalls:** Forgetting $x \neq 0$; off-by-one in indices; mixing up $S_{n+1} = S_1 S_n - S_{n-1}$ with other recurrences.
- **Example:** See the $x + \frac{1}{x} = 3$ example below; compute S_2, then S_3 via the recurrence.

Derivation note: Using the identity $(x+x^{-1})(x^n+x^{-n}) = x^{n+1}+x^{n-1}+x^{-(n-1)}+x^{-(n+1)}$, we can show that $S_1 \cdot S_n = (x+x^{-1})(x^n+x^{-n}) = x^{n+1}+x^{-(n+1)}+x^{n-1}+x^{-(n-1)} = S_{n+1}+S_{n-1}$, hence $S_{n+1} = S_1 S_n - S_{n-1}$.

AMC 12A 2018, Problem 17

If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.

Solution

Method: Use the recurrence relation for powers.
Let $S_n = x^n + \frac{1}{x^n}$
Key Identity: $S_{n+1} = S_1 \cdot S_n - S_{n-1}$
We have: $S_0 = 2$ (since $x^0 + \frac{1}{x^0} = 1 + 1 = 2$) and $S_1 = 3$
Find S_2:

$$S_2 = S_1 \cdot S_1 - S_0 = 3 \cdot 3 - 2 = 9 - 2 = 7$$

Find S_3:

$$S_3 = S_1 \cdot S_2 - S_1 = 3 \cdot 7 - 3 = 21 - 3 = 18$$

Verification with Direct Method:

$$S_3 = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) = 27 - 9 = 18$$

✓
Answer: 18

3.11 Technique 28: Derangements

Short Name: T28-Derangements

Definition 3.11 (Derangements)**.** *A derangement is a permutation where no element appears in its original position. The number of derangements of n objects is:*

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right)$$

Asymptotically: $D_n \approx \frac{n!}{e}$

WHEN: Permutation problems with "no fixed point" constraints; hat-check/Secret Santa; letter–envelope matchings; seating with no one in the original seat. Use for hat-check/Secret Santa; letter–envelope matchings; seating assignments; assignment problems with forbidden fixed points; probability that a random permutation has no fixed points.

Configuration cues:

- "No element in its original position"
- Complete rearrangements
- Fixed-point-free permutations

Constraints cues:

- Objects must be distinguishable; duplicates or restricted positions require inclusion–exclusion variants or rook polynomials

- Partial derangements (some fixed allowed/forbidden) change the count; define the fixed set and include/exclude carefully

Objective cues:

- Count fixed-point-free permutations; compute the probability that no fixed points occur
- Estimate quickly using $D_n \approx n!/e$ when an exact count is unnecessary

Why It Works: Uses inclusion-exclusion to count permutations with no fixed points.

WHO: CNT problems with position restrictions; solvers comfortable with inclusion–exclusion, recurrences, or rook polynomials for restricted placements.

WHAT: Count permutations with no fixed points (and variants with restricted positions) via inclusion–exclusion or the recurrence $D_n = (n-1)(D_{n-1} + D_{n-2})$.

How:

- **First Moves:**
  1. Choose an approach: direct inclusion–exclusion, the closed form, or the recurrence.
  2. For small n, compute quickly using $D_1 = 0$, $D_2 = 1$, and $D_n = (n-1)(D_{n-1} + D_{n-2})$.
  3. With additional fixed-position constraints, run inclusion–exclusion over the set of potentially fixed indices.
  4. Use $D_n \approx \dfrac{n!}{e}$ for estimates/sanity checks.
- **Bailouts:** Translate to rook polynomials or matchings if positions are restricted; break symmetry by conditioning on a chosen element; for probabilities, compute via complement or limit using $1/e$.
- **Pitfalls:** Treating indistinguishable objects as distinct; double-counting in inclusion–exclusion; forgetting to adjust for pre-fixed elements or forbidden positions.
- **Example:** See the Secret Santa count for D_5 below; template: pick method, compute, and sanity-check with $D_n \approx n!/e$.

Example: Secret Santa

In how many ways can 5 people exchange gifts so that no one gets their own gift?

Solution

We need D_5:

$$\begin{aligned} D_5 &= 5!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right) \\ &= 120\left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right) \\ &= 120\left(\frac{60 - 20 + 5 - 1}{120}\right) = 120 \cdot \frac{44}{120} = 44 \end{aligned}$$

Answer: 44 ways

3.12 Technique 29: Telescoping Series

Short Name: T29-Telescoping

Definition 3.12 (Telescoping Series)**.** *Transform a series so that consecutive terms cancel, leaving only boundary terms.*

WHEN: Sums such as $\sum \frac{1}{k(k+1)}$, $\sum(\ln k - \ln(k+1))$; products of ratios $\prod \frac{k}{k+1}$; finite-difference telescoping from Newton's forward/backward differences.

Configuration cues:

- Summands expressible as differences $a_k = b_k - b_{k+1}$ or via partial fractions
- Logarithmic sums with consecutive cancellation such as $\ln k - \ln(k+1)$
- Products of ratios $\prod$
  $tfracb_k b_{k+1}$ or finite differences

Constraints cues:

- Check index bounds to avoid off-by-one errors
- Ensure domains: denominators nonzero; logarithm arguments positive
- For infinite sums/products, verify convergence before telescoping

Objective cues:

- Collapse to boundary terms only
- Evaluate sums/products efficiently with minimal computation

Why It Works: Most intermediate terms cancel, dramatically simplifying the sum.

WHO: ALG sequences and series problems; solvers simplifying long sums or products, especially with partial fractions or finite differences.

WHAT: Decompose terms so consecutive ones cancel (differences or ratios); common tools are partial fractions and logarithm properties.

How:

- **First Moves:**
  1. Seek a representation $a_k = b_k - b_{k+1}$ or $\ln a_k = c_k - c_{k+1}$; for rationals, try partial fractions.
  2. Write the first few and last few terms to expose the cancellation pattern.
  3. Collapse to boundary terms and evaluate carefully.
  4. Check endpoints/indices to avoid off-by-one errors and ensure domains (e.g., ln arguments) are valid.
- **Bailouts:** If direct differences fail, consider finite differences, partial fraction decomposition, or take logs to turn products into sums.
- **Pitfalls:** Missing a surviving middle term due to index shifts; illegal domain steps (negative log arguments, zero denominators).
- **Example:** See the logarithmic telescoping example below; split each term and cancel to boundary logs.

Example: Series Evaluation

Evaluate $\sum_{k=2}^{100} \ln\left(\frac{k^2}{k^2-1}\right)$

Solution

Rewrite each term:

$$\ln\left(\frac{k^2}{k^2-1}\right) = \ln\left(\frac{k^2}{(k-1)(k+1)}\right) = 2\ln k - \ln(k-1) - \ln(k+1)$$

Let's write this as:

$$\ln\left(\frac{k^2}{k^2-1}\right) = \ln\left(\frac{k}{k-1}\right) + \ln\left(\frac{k}{k+1}\right)$$

Now observe the telescoping pattern. Let's write out several terms:

$$k = 2: \quad \ln\left(\frac{2}{1}\right) + \ln\left(\frac{2}{3}\right) \tag{22}$$

$$k = 3: \quad \ln\left(\frac{3}{2}\right) + \ln\left(\frac{3}{4}\right) \tag{23}$$

$$k = 4: \quad \ln\left(\frac{4}{3}\right) + \ln\left(\frac{4}{5}\right) \tag{24}$$

$$\vdots \tag{25}$$

$$k = 99: \quad \ln\left(\frac{99}{98}\right) + \ln\left(\frac{99}{100}\right) \tag{26}$$

$$k = 100: \quad \ln\left(\frac{100}{99}\right) + \ln\left(\frac{100}{101}\right) \tag{27}$$

Looking at the pattern, we can group the terms differently. The sum becomes:

$$\sum_{k=2}^{100} [\ln k - \ln(k-1)] + \sum_{k=2}^{100} [\ln k - \ln(k+1)]$$

The first sum telescopes to $\ln 100 - \ln 1 = \ln 100$. The second sum telescopes to $\ln 2 - \ln 101$.
Therefore, the total is:

$$\ln 100 + \ln 2 - \ln 101 = \ln\left(\frac{200}{101}\right)$$

Answer: $\ln\left(\frac{200}{101}\right)$

3.13 Technique 30: Polynomial Division

Short Name: T30-PolyDiv

Definition 3.13 (Polynomial Division/Remainder Theorem)**.** *Use polynomial division to simplify expressions or find remainders when dividing by linear factors.*

WHEN: Remainders upon division by $(x-a)$, $(x-a)(x-b)$, or simple quadratics; evaluating large exponents at specific x; proving divisibility; expressing functions modulo a divisor for cyclic or periodic substitutions.

Configuration cues:

- Remainders upon division by $x-a$, $(x-a)(x-b)$, or simple quadratics
- Evaluate $P(a)$, $P(b)$ at specific points; simplify large-exponent expressions
- Divisibility questions expressible as $P(a)=0$ for one or more a
- Modulo-polynomial simplifications (work in $\mathbb{R}[x]/\langle D(x)\rangle$) to reduce expressions

Constraints cues:

- Remainder degree is less than the divisor's degree
- For distinct linear factors, determine coefficients by evaluating at each root
- For repeated factors $(x-a)^2$, use $P(a)$ and $P'(a)$ to find a linear remainder
- For quadratic divisors without real roots, parametrize the remainder as linear and solve via coefficient comparison
- Respect multiplicities: for $(x-a)^k$, match derivatives up to order $k-1$

Objective cues:

- Obtain remainders/values modulo a polynomial without full division
- Reduce computations via factoring and modular reductions before evaluating
- Interpolate minimal-degree remainder uniquely determined by evaluations/derivatives

Why It Works: The Remainder Theorem provides direct evaluation without full division.

WHO: ALG problems involving polynomial evaluation, remainders, factor/root relations, or modular arithmetic in $\mathbb{R}[x]$.

WHAT: Use polynomial long/synthetic division and the Remainder/Factor Theorems to obtain remainders or values modulo factors.

How:

- **First Moves:**
  1. For divisor $x-a$, compute the remainder $P(a)$ directly (Remainder Theorem).
  2. For a product of distinct linear factors, posit a general remainder of appropriate degree and solve by evaluating at each root.
  3. If needed, equate coefficients or use interpolation to determine remainder coefficients.
  4. Exploit factorization/modular reductions to simplify before dividing.

- **Bailouts:** Work modulo factors separately and recombine by CRT; differentiate to handle repeated factors; reduce degree using identities before division.
- **Pitfalls:** Using a remainder of too high degree; ignoring multiplicities for $(x-a)^k$; arithmetic slips when evaluating at roots.
- **Example:** See the remainder mod x^2-1 below; evaluate at $x=\pm 1$ to solve for the linear remainder.

Theorem 3.1 (Remainder Theorem)**.** *When polynomial $P(x)$ is divided by $(x-a)$, the remainder is $P(a)$.*

Example: Polynomial Remainder

Find the remainder when $x^{100}+x^{50}+1$ is divided by x^2-1.

Solution

Since $x^2-1=(x-1)(x+1)$, we can write:

$$x^{100}+x^{50}+1=Q(x)(x^2-1)+ax+b$$

For some quotient $Q(x)$ and remainder $ax+b$.
Setting $x=1$: $1+1+1=0+a+b$, so $a+b=3$
Setting $x=-1$: $1+1+1=0-a+b$, so $-a+b=3$
Solving: $b=3$ and $a=0$
Answer: The remainder is 3

3.14 Technique 31: Stewart's Theorem

Short Name: T31-Stewart

Definition 3.14 (Stewart's Theorem)**.** *For a triangle ABC with a cevian AD to side BC, where $BD=m$ and $DC=n$:*

$$b^2m+c^2n=a(d^2+mn)$$

where $a=BC$, $b=AC$, $c=AB$, and $d=AD$.

WHEN: Configuration: triangle with a cevian partitioning the opposite side into known lengths or a stated ratio; Constraints: nondegenerate triangle with positive segment lengths (use directed lengths for external division or extensions); Objective: compute a cevian length, certify a segment ratio, or relate sides and a cevian without resorting to coordinates or trigonometry. Use for triangles with medians (Apollonius as a special case, $m=n$), angle bisectors (with the Angle Bisector Theorem), trisectors or general cevians; internal or external divisions (use directed lengths).

Configuration cues:

- Triangle with a cevian splitting the opposite side into known lengths or a known ratio
- Medians ($m = n$) or angle bisectors (combine with the Angle Bisector Theorem)
- Given three sides and a cevian-related condition; need the cevian length

Constraints cues:

- Triangle is nondegenerate; all segment lengths are positive
- If a ratio $BD : DC$ is given, convert to lengths using $a = BC$
- For external cevians, use directed lengths or the appropriate variant

Objective cues:

- Relate side lengths and a cevian length directly without coordinates
- Derive or verify special cases (e.g., the median-length formula) as checks

Why It Works: Provides a direct relationship between side lengths and cevian length without coordinates.

WHO: GEO Euclidean geometry length and ratio problems involving cevians; solvers preferring algebraic length relations over coordinates or trigonometry.

WHAT: Apply Stewart's Theorem $b^2m + c^2n = a(d^2 + mn)$ to relate sides and a cevian length.

How:

- **First Moves:**
  1. Label $a = BC$, $b = AC$, $c = AB$, $BD = m$, $DC = n$, and $AD = d$.
  2. Substitute into $b^2m + c^2n = a(d^2 + mn)$.
  3. Solve for the desired quantity; for medians use $m = n = \frac{a}{2}$.
  4. Sanity-check with known special cases (e.g., median formula $m_a^2 = \frac{2b^2+2c^2-a^2}{4}$).
- **Bailouts:** If labels or ratios are unclear or messy, switch to coordinates and the Law of Cosines, or combine with the Angle Bisector Theorem/Apollonius for special cevians.
- **Pitfalls:** Confusing m, n; forgetting directed lengths for external cevians; mixing up side labels; arithmetic slips when expanding.
- **Example:** See the median-length computation below.

Example: Finding Median Length

In triangle ABC with $AB = 13$, $BC = 14$, and $AC = 15$, find the length of the median from A to BC.

Solution

For a median, $m = n = 7$ (since D is the midpoint of BC).
Using Stewart's Theorem with $a = 14$, $b = 15$, $c = 13$:

$$15^2 \cdot 7 + 13^2 \cdot 7 = 14(d^2 + 7 \cdot 7)$$

$$225 \cdot 7 + 169 \cdot 7 = 14(d^2 + 49)$$

$$394 \cdot 7 = 14(d^2 + 49)$$

$$2758 = 14d^2 + 686$$

$$14d^2 = 2072$$

$$d^2 = 148$$

$$d = 2\sqrt{37}$$

Answer: $2\sqrt{37}$

3.15 Technique 32: Floor and Ceiling Functions

Short Name: T32-FloorCeil

Definition 3.15 (Floor and Ceiling Functions)**.** *For real x:*

- $\lfloor x \rfloor$ = *greatest integer* $\leq x$ *(floor function)*
- $\lceil x \rceil$ = *smallest integer* $\geq x$ *(ceiling function)*
- *Key identity:* $\lfloor x \rfloor = n \iff n \leq x < n + 1$ *for integer* n

WHEN: Configuration: explicit floor/ceiling expressions or lattice-point geometry; Constraints: integer-bound intervals with breakpoints/rounding and sign-sensitive floors; Objective: translate to inequalities and count or sum over integer segments. Use for expressions, sums, or inequalities with $\lfloor \cdot \rfloor$ or $\lceil \cdot \rceil$; lattice-point counting; rounding/partition problems.

Configuration cues:

- Explicit floor/ceiling symbols, stepwise-constant behavior, or rounding/partition statements
- Expressions like $\lfloor \sqrt{k} \rfloor$, $\lfloor k/m \rfloor$, nested floors/ceilings, or sums of floors
- Lattice-point counts under/inside lines, circles, or polygons; discretization of continuous ranges

Constraints cues:

- Endpoint inclusions/exclusions at breakpoints; watch off-by-one at integer boundaries
- Treat negatives carefully: $\lfloor -1.2 \rfloor = -2$, $\lceil -1.2 \rceil = -1$; preserve variable domains

- Verify monotonicity between breakpoints and recheck original inequalities after translation

Objective cues:

- Translate floor/ceiling statements into inequality ranges
- Partition the domain at breakpoints; count or sum on each constant segment
- Reduce to arithmetic progressions or known identities to simplify totals

Pattern cues (Demand: Low):

- Discretization; translate interval constraints into integer ranges
- Lattice counts or floors/ceilings in sums/products

Why It Works: Converts continuous problems to discrete ones systematically.
WHO: CNT/NT/ALG discrete mathematics, number theory, and algebra problems with integer parts; solvers converting continuous constraints into counts.
WHAT: Interpret floor/ceiling as inequalities; partition the domain at breakpoints where the integer part changes; count or sum over constant intervals.
How:

- **First Moves:**
  1. Rewrite $\lfloor x \rfloor = n$ or $\lceil x \rceil = n$ as an inequality range on the underlying real variable.
  2. Identify all breakpoints (e.g., perfect squares for $\lfloor \sqrt{k} \rfloor$, multiples for $\lfloor k/m \rfloor$) and partition accordingly.
  3. Evaluate counts/sums piecewise on each interval where the integer part is constant.
  4. Check endpoints carefully to avoid off-by-one errors at inclusive/exclusive bounds.
- **Bailouts:** If nested or composite floors are unwieldy, bound with inequalities or compare to integrals/areas; switch to a lattice-point or modular interpretation when clearer.
- **Pitfalls:** Off-by-one at endpoints; mishandling negatives; missing breakpoints; assuming monotonicity across jumps.
- **Example:** See the sum of floor functions below.

Example: Sum of Floor Functions

Find $\sum_{k=1}^{100} \left\lfloor \sqrt{k} \right\rfloor$.

Solution

Note that $\lfloor\sqrt{k}\rfloor = n$ when $n^2 \le k < (n+1)^2$.
Count how many k give each value:

- $\lfloor\sqrt{k}\rfloor = 1$ for $k \in \{1, 2, 3\}$: 3 values (since $1^2 = 1 \le k < 2^2 = 4$)
- $\lfloor\sqrt{k}\rfloor = 2$ for $k \in \{4, 5, 6, 7, 8\}$: 5 values (since $2^2 = 4 \le k < 3^2 = 9$)
- $\lfloor\sqrt{k}\rfloor = 3$ for $k \in \{9, 10, \ldots, 15\}$: 7 values (since $3^2 = 9 \le k < 4^2 = 16$)
- In general: $\lfloor\sqrt{k}\rfloor = n$ for $(2n+1)$ values when $n < 10$ (from n^2 to $(n+1)^2 - 1$)
- Specifically for $n = 1$ to 9: each n contributes $(2n+1)$ values
- $\lfloor\sqrt{k}\rfloor = 10$ only for $k = 100$ (since $10^2 = 100$ and we stop at 100): 1 value

Sum $= 1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + 4 \cdot 9 + 5 \cdot 11 + 6 \cdot 13 + 7 \cdot 15 + 8 \cdot 17 + 9 \cdot 19 + 10 \cdot 1$
$= 3 + 10 + 21 + 36 + 55 + 78 + 105 + 136 + 171 + 10 = 625$
Answer: 625

3.16 Technique 33: Rational Root Theorem

Short Name: T33-RationalRoot

Definition 3.16 (Rational Root Theorem)**.** *If a polynomial with integer coefficients $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ has a rational root $\frac{p}{q}$ in lowest terms, then:*

- *p divides the constant term a_0*
- *q divides the leading coefficient a_n*

WHEN: Configuration: univariate polynomial with integer coefficients; Constraints: rational roots must be $p \mid a_0$, $q \mid a_n$ in lowest terms; Objective: enumerate/test candidates, then factor via division. Use for univariate integer-coefficient polynomials (typically degree ≤ 4) with small factors; often combined with Vieta's formulas or Descartes' Rule of Signs.

Configuration cues:

- Univariate polynomial with integer coefficients; small-factor constant/leading terms
- Problem hints at rational roots or factorization over $\mathbb{Q}$

Constraints cues:

- Candidates $\frac{p}{q}$ must be in lowest terms with $p \mid a_0$, $q \mid a_n$
- Integer coefficients are required; confirm multiplicities and remaining irreducible factors

Objective cues:

- Bound and enumerate the candidate set of rational roots

- Test candidates efficiently, then reduce degree by division and factor completely

Pattern cues (Demand: Low):

- Integer coefficients; small-factor constant/leading terms
- Test candidate roots p/q with $p|a_0$, $q|a_n$

Why It Works: Limits the search space for rational roots to a finite, manageable set.
WHO: ALG polynomial algebra problems; solvers factoring integer-coefficient polynomials efficiently.
WHAT: Generate and test candidate rational roots p/q with $p \mid a_0$ and $q \mid a_n$; confirm with synthetic/polynomial division; factor completely.
How:

- **First Moves:**
  1. List candidates p/q (in lowest terms) where $p \mid a_0$ and $q \mid a_n$.
  2. Evaluate $f(p/q)$ quickly (prefer integer arithmetic when possible).
  3. Upon finding a root, divide the polynomial to reduce degree and repeat.
  4. Verify all claimed roots and multiplicities; ensure no extraneous values remain.
- **Bailouts:** If no rational root works, try factoring by grouping, substitution, or apply the quadratic formula to reduced factors; use Descartes' Rule or bounds to rule out rationals.
- **Pitfalls:** Not reducing p/q to lowest terms; forgetting negative candidates; arithmetic errors in synthetic division; stopping after the first root.
- **Example:** See the rational roots example below.

Example: Finding Rational Roots

Find all rational roots of $2x^3 - 3x^2 - 8x + 12 = 0$.

Solution

Step 1: Identify possible rational roots using the theorem.

- Factors of $a_0 = 12$: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$
- Factors of $a_n = 2$: $\pm 1, \pm 2$
- Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm \frac{1}{2}, \pm \frac{3}{2}$

Step 2: Test candidates.

- $f(2) = 2(8) - 3(4) - 8(2) + 12 = 16 - 12 - 16 + 12 = 0$ ✓
- $f\left(\frac{3}{2}\right) = 2\left(\frac{27}{8}\right) - 3\left(\frac{9}{4}\right) - 8\left(\frac{3}{2}\right) + 12 = \frac{27}{4} - \frac{27}{4} - 12 + 12 = 0$ ✓
- $f(2) = 0$ means $(x - 2)$ is a factor

Step 3: Factor completely. Using polynomial division: $2x^3 - 3x^2 - 8x + 12 = (x - 2)(2x^2 + x - 6) = (x - 2)(2x - 3)(x + 2)$
Answer: The rational roots are $x = 2, \frac{3}{2}, -2$

3.17 Technique 34: Jensen's Inequality

Short Name: T34-JensenInequ

Definition 3.17 (Jensen's Inequality)**.** *For a convex function f and real numbers $x_1, x_2, \ldots, x_n$ with weights $w_1, w_2, \ldots, w_n$ where $w_i \geq 0$ and $\sum w_i = 1$:*

$$f\left(\sum_{i=1}^{n} w_i x_i\right) \leq \sum_{i=1}^{n} w_i f(x_i)$$

For a concave function, the inequality reverses.

WHEN: Configuration: symmetric sums/means of a convex/concave function; Constraints: correct domain and curvature with nonnegative weights summing to 1; Objective: bound via Jensen and verify equality cases. Use for symmetric constraints (e.g., fixed sum/average) with functions such as x^p ($p > 1$ convex, $0 < p < 1$ concave), $\ln x$ (concave), e^x (convex).

Configuration cues:

- Symmetric constraints (fixed sum/average) on variables; sum of transformed terms
- Function with recognizable curvature: x^p, $\ln x$, e^x, etc.

Constraints cues:

- Verify convexity/concavity on the variables' interval (e.g., sign of f'')
- Weights nonnegative and sum to 1; choose direction of inequality correctly

- Equality typically when variables are equal and constraints allow it

Objective cues:

- Derive tight bounds by comparing $f(\text{average})$ and the average of f
- Identify equality cases to check sharpness and feasibility

Pattern cues (Demand: Low–High):

- Convexity/concavity present; symmetric constraints and equal-weight averaging
- Choose weights and variables to match domain and curvature

Why It Works: Exploits the curvature of functions to establish bounds.
WHO: ALG inequality and optimization problems; solvers leveraging convexity/concavity.
WHAT: Apply Jensen's inequality to compare $f(\text{average})$ with the average of f; choose nonnegative weights summing to 1.
How:

- **First Moves:**
  1. Identify the interval of variables and verify f is convex/concave there (e.g., check f'''s sign).
  2. Choose admissible weights $w_i \geq 0$ with $\sum w_i = 1$ consistent with the problem's constraint.
  3. Apply Jensen in the correct direction (convex: $f(\text{avg}) \leq \text{avg } f$; concave: inequality reverses).
  4. State equality conditions (typically all variables equal) and verify feasibility.
- **Bailouts:** If Jensen is awkward, try AM–GM/Cauchy–Schwarz, a tangent-line bound, or normalize by substitution; consider majorization/Karamata when symmetry suggests it.
- **Pitfalls:** Wrong curvature/direction; invalid domain; weights not summing to 1; asserting equality when constraints forbid equality.
- **Example:** See the concave $\sqrt{x}$ inequality proof below.

Example: Proving an Inequality

Prove that for positive reals a, b, c with $a + b + c = 3$: $\sqrt{a} + \sqrt{b} + \sqrt{c} \leq 3$.

Solution

Key Insight: The function $f(x) = \sqrt{x}$ is concave for $x > 0$ (since $f''(x) = -\frac{1}{4x^{3/2}} < 0$).
By Jensen's inequality for concave functions with equal weights $w_i = \frac{1}{3}$:

$$\frac{\sqrt{a} + \sqrt{b} + \sqrt{c}}{3} \leq \sqrt{\frac{a+b+c}{3}} = \sqrt{\frac{3}{3}} = 1$$

Therefore: $\sqrt{a} + \sqrt{b} + \sqrt{c} \leq 3$
Equality holds when $a = b = c = 1$.
Answer: The inequality is proven, with equality when $a = b = c = 1$.

3.18 Technique 35: Mass Points

Short Name: T35-MassPoints

Definition 3.18 (Mass Points)**.** *A bookkeeping method assigning "masses" to vertices so that each cevian balances its adjacent side ratio. Intersections are analyzed by the conservation of mass along cevians.*

WHEN: Configuration: triangle with two or more cevians and given division ratios; Constraints: concurrency or consistent directed-length interpretation; Objective: assign masses to encode ratios and read desired segment divisions. Use for triangles with cevians given by ratios (medians, bisectors, trisectors, or general points), often with concurrency.

Configuration cues:

- Triangle with two or more cevians and given side/division ratios
- Goal is a ratio along a cevian or between segments at an intersection

Constraints cues:

- Most effective with concurrent cevians in a triangle; handle external division via directed lengths
- Mass assignments must be consistent across all given ratios

Objective cues:

- Encode ratios as masses, propagate to intersection(s), and read desired division ratios
- Use Ceva to verify concurrency when needed

Pattern cues (Demand: Moderate–High):

- Cevians/ratios/concurrency or collinearity; solve for segment ratios
- Assign masses matching side ratios; conserve mass at intersections

Why It Works: Encodes ratio constraints as additive masses: the mass at a point on a segment equals the sum of endpoint masses. **Works well with:** Ceva's Theorem (use Mass Points for ratios; Ceva for concurrency checks). **Typical difficulty:** Often appears in problems 15–25.

Balance rule: On a segment, mass $\times$ distance to the balance point is conserved (lever law). **Signed ratios:** Mass points implicitly respect directed lengths when cevians extend beyond sides. **Limitation:** Most effective with concurrent cevians; not a cure-all for arbitrary configurations.

WHO: GEO triangle geometry ratio problems; solvers translating cevian ratios into algebraic relations.

WHAT: Assign masses to vertices so each cevian balances its adjacent side ratio; deduce segment ratios at intersections.

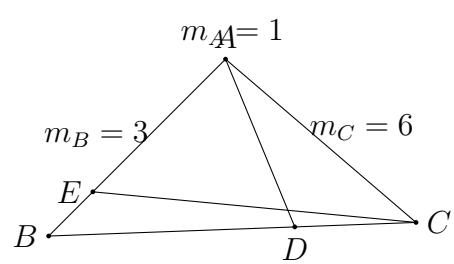
How:

- **First Moves:**
  1. From a ratio like $BD : DC$, assign masses $m_B : m_C = DC : BD$.
  2. Propagate masses so each vertex and its opposite side satisfy balance; at an interior point, mass equals the sum of adjacent endpoint masses.
  3. Read division on a cevian: if endpoint masses are m_X and m_Y, the cevian divides in ratio $m_Y : m_X$.
  4. Check consistency/concurrency (optionally with Ceva) and handle directed lengths when points lie outside sides.

- **Bailouts:** If masses get messy or cevians are not concurrent, switch to Angle Bisector/Stewart or use coordinates/vectors; verify concurrency with Ceva first.

- **Pitfalls:** Inconsistent mass assignments; swapping endpoint roles; ignoring directed lengths for external division; applying to nonconcurrent cevians.

- **Example:** See the two-cevians intersection ratio below.



Example: Two Cevians Intersection Ratio

In $\triangle ABC$, let D lie on $\overline{BC}$ with $BD : DC = 2 : 1$, and let E lie on $\overline{AB}$ with $AE : EB = 3 : 1$. Lines AD and CE intersect at P. Find $AP : PD$.

Solution

Assign masses with $m_B : m_C = DC : BD = 1 : 2$, so take $m_B = 3$, $m_C = 6$. From $AE : EB = 3 : 1$, we need $m_A : m_B = 1 : 3$, so take $m_A = 1$. Then $m_D = m_B + m_C = 9$. Along $\overline{AD}$, the cevian divides in the ratio $AP : PD = m_D : m_A = 9 : 1$.
Answer: $9 : 1$

3.19 Technique 36: Ceva's Theorem

Short Name: T36-Ceva

Definition 3.19 (Ceva's Theorem)**.** *For $\triangle ABC$ with points $D \in \overline{BC}$, $E \in \overline{CA}$, $F \in \overline{AB}$, cevians AD, BE, CF are concurrent if and only if*

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1.$$

WHEN: Configuration: triangle with three cevians from the vertices (medians, angle bisectors, altitudes, or general cevians) meeting opposite sides or their extensions; Constraints: use directed segment ratios for any external division and keep denominators nonzero; Objective: certify concurrency or solve for a missing ratio from Ceva's product (use trigonometric Ceva when angle data is given). Use for triangles featuring the centroid, incenter/excenter, orthocenter, or classical concurrency points (Gergonne/Nagel); problems asking for a missing side ratio or to certify concurrency; often paired with Mass Points to extract numeric lengths; use directed lengths for external cevians and trigonometric Ceva when angles are given.

Configuration cues:

- Triangle with cevians AD, BE, CF and given side division ratios
- Medians, angle bisectors, altitudes, or general cevians; target is concurrency or a missing ratio

Constraints cues:

- Use directed lengths when points lie on side extensions; keep orientation consistent
- Ceva tests concurrency; for collinearity of three points on a transversal, use Menelaus instead

Objective cues:

- Verify concurrency from known ratios or solve for an unknown ratio
- Pair with Mass Points to turn ratios into numbers and recover segment divisions

Pattern cues (Demand: Moderate–High):

- Cevians/ratios with concurrency; product of three segment ratios equals 1

- Use directed lengths; pair with Mass Points to compute ratios

Why It Works: Arises from area ratios or from directed-length similar triangle arguments.

WHO: GEO triangle geometry problems about concurrency or ratio chasing with cevians (including medians, angle bisectors, altitudes), typically mid–late.

WHAT: Use Ceva's Theorem $\left(\frac{BD}{DC}\right)\left(\frac{CE}{EA}\right)\left(\frac{AF}{FB}\right) = 1$ (with directed lengths when extensions occur) to verify concurrency or solve for a missing ratio.

How:

- **First Moves:**
  1. Identify cevians AD, BE, CF and the side partitions $BD : DC$, $CE : EA$, $AF : FB$.
  2. Decide whether any point lies on an extension and use directed lengths consistently.
  3. Substitute known ratios into Ceva's product and solve for the unknown ratio.
  4. If angle data is given, optionally apply trigonometric Ceva to relate sines of angles.
  5. Combine with Mass Points when absolute segment lengths (not just ratios) are desired.

- **Bailouts:** Switch to Mass Points or area ratios if directed-length bookkeeping gets messy; try trigonometric Ceva when angle measures are given; revert to similarity/coordinate models if Ceva alone does not expose the target ratio.

- **Pitfalls:** Forgetting directed lengths for external division; mixing up ratio order (e.g., CE/EA vs. EA/CE); assuming Ceva certifies collinearity (use Menelaus for that); dividing by zero in degenerate placements.

- **Example:** Given $\frac{BD}{DC} = \frac{2}{3}$ and $\frac{CE}{EA} = \frac{3}{4}$, Ceva gives $\frac{AF}{FB} = 2$; see the solved example below.

Works well with: Mass Points (for computing the ratios in Ceva's product). **Typical difficulty:** Often appears in problems 15–25. **Trigonometric form:** $\dfrac{\sin\angle BAD}{\sin\angle CAD} \cdot \dfrac{\sin\angle CBE}{\sin\angle ABE} \cdot \dfrac{\sin\angle ACF}{\sin\angle BCF} = 1$. **Special cases:** Medians, internal angle bisectors, and altitudes satisfy Ceva. **Directed ratios:** Use signed (directed) lengths consistently when cevians meet side extensions.

Example: Solve for a Missing Ratio

In $\triangle ABC$, suppose $\dfrac{BD}{DC} = \dfrac{2}{3}$ and $\dfrac{CE}{EA} = \dfrac{3}{4}$. For which $\dfrac{AF}{FB}$ are AD, BE, CF concurrent?

Solution

By Ceva: $\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{AF}{FB} = 1 \Rightarrow \frac{1}{2} \cdot \frac{AF}{FB} = 1$, so $\frac{AF}{FB} = 2$.
Answer: $AF : FB = 2 : 1$

3.20 Technique 37: Menelaus's Theorem

Short Name: T37-Menelaus

Definition 3.20 (Menelaus's Theorem)**.** *For $\triangle ABC$ with points $D \in \overline{BC}$, $E \in \overline{CA}$, and $F \in \overline{AB}$ collinear on a transversal, the signed (directed) ratio form is*

$$\frac{\overrightarrow{BD}}{\overrightarrow{DC}} \cdot \frac{\overrightarrow{CE}}{\overrightarrow{EA}} \cdot \frac{\overrightarrow{AF}}{\overrightarrow{FB}} = -1.$$

Taking absolute values (unsigned lengths) yields a product of $+1$ with appropriate orientation adjustments.

WHEN: Configuration: a transversal intersects the three sides (or extensions) of a triangle at points on one line; Constraints: work with directed ratios so exterior intersections contribute the correct sign; Objective: certify collinearity or solve for a missing segment ratio along the transversal (translate difficult ratios via similarity or parallels first). Use for a line through $\triangle ABC$ meeting sides or their extensions at D, E, F; typical "find $AF : FB$ given $BD : DC$ and $CE : EA$" setups, alignment of three points, mixed interior/exterior configurations, and harmonic-division style arrangements.

Configuration cues:

- A line meets the three sides (or extensions) of a triangle at points D, E, F claimed to be collinear
- Mixed interior/exterior division points along the sides or their extensions

Constraints cues:

- Use directed lengths in the -1 product form; translate to unsigned form with careful sign accounting when preferred
- Menelaus certifies collinearity; for concurrency of cevians, use Ceva

Objective cues:

- Establish collinearity or compute a missing side ratio induced by the transversal
- Express tricky ratios using similar triangles or parallels before applying Menelaus

Pattern cues (Demand: Moderate–High):

- Transversal cuts triangle sides; product of three directed ratios equals -1

- Use directed lengths; translate to unsigned ratios with orientation care

Why It Works: Follows from similar triangles created by the transversal and the sides of the triangle.

WHO: GEO triangle and transversal problems involving collinearity checks, split-segment ratios, or harmonic division; dual to Ceva-type concurrency tasks.

WHAT: Use Menelaus's Theorem in directed form $\left(\frac{\overrightarrow{BD}}{\overrightarrow{DC}}\right)\left(\frac{\overrightarrow{CE}}{\overrightarrow{EA}}\right)\left(\frac{\overrightarrow{AF}}{\overrightarrow{FB}}\right) = -1$ (or absolute form with sign handling) to establish collinearity or compute a missing ratio.

How:

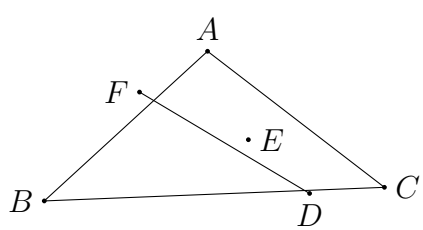
- **First Moves:**
  1. Choose a consistent orientation for directed lengths on each side line.
  2. Express the three side ratios (signed if using the -1 form; unsigned with an extra sign check if using the $+1$ product).
  3. Apply Menelaus to relate the three ratios and solve for the unknown.
  4. Use similar triangles or parallel lines as needed to express any difficult ratio.
  5. Sanity-check orientation: exactly one (or three) of the division points being external flips the overall sign.

- **Bailouts:** Translate hard ratios via similar triangles or parallel lines first; switch to Ceva if the goal is concurrency, not collinearity; use coordinates/vectors for precise directed-length handling if signs are confusing.

- **Pitfalls:** Sign mistakes from inconsistent orientations; forgetting that exterior intersection points change the sign; applying Menelaus to concurrency problems (use Ceva instead).

- **Example:** With $BD : DC = 2 : 1$ and $CE : EA = 3 : 2$, Menelaus yields $AF : FB = 1 : 3$; see the worked example below.

Duality: Ceva's Theorem (concurrency) is dual to Menelaus's (collinearity). **Trigonometric form:** $\frac{\sin\angle BAF}{\sin\angle CAF} \cdot \frac{\sin\angle CDB}{\sin\angle ADB} \cdot \frac{\sin\angle ECB}{\sin\angle ECA} = 1$. **Memory aid:** Menelaus = Minus one (signed version equals -1). **Typical difficulty:** Often appears in problems 15–25.



Example: Find a Ratio on the Transversal

In $\triangle ABC$, let $D \in \overline{BC}$ with $BD : DC = 2 : 1$ and $E \in \overline{CA}$ with $CE : EA = 3 : 2$. If D, E, F are collinear with $F \in \overline{AB}$, find $AF : FB$.

Solution

By Menelaus: $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ gives $\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{AF}{FB} = 1$, so $3 \cdot \frac{AF}{FB} = 1$ and $AF : FB = 1 : 3$.
Answer: $1 : 3$

3.21 Technique 38: Chinese Remainder Theorem

Short Name: T38-CRT

Definition 3.21 (Chinese Remainder Theorem)**.** *If $n_1, \ldots, n_k$ are pairwise coprime, then the system $x \equiv a_i \pmod{n_i}$ has a unique solution modulo $N = n_1 \cdots n_k$.*

WHEN: Configuration: simultaneous congruences or computations modulo a composite modulus; Constraints: prefer pairwise coprime moduli (otherwise check consistency modulo gcds) and ensure needed modular inverses exist; Objective: merge to a single congruence modulo $N = \prod n_i$ (or modulo lcm in non-coprime cases) and, for exponent/last-digit problems, reduce modulo prime powers then lift. Use for last-digit or last-two-digit questions (e.g., combine mod 4 and mod 25), simultaneous remainder systems, residue cycles of powers with composite moduli, "smallest positive integer satisfying ..." queries, and time/calendar congruences; typically combined with Fermat/Euler then recombined via CRT.

Configuration cues:

- A system of congruences $x \equiv a_i \pmod{n_i}$ to be solved jointly
- Composite moduli where working modulo prime powers and recombining is natural

Constraints cues:

- Prefer pairwise coprime moduli; otherwise check compatibility modulo $\gcd(n_i, n_j)$ and note non-uniqueness
- Need modular inverses of $M_i = N/n_i$ modulo n_i; ensure inverses exist

Objective cues:

- Combine congruences into a single congruence modulo $N = \prod n_i$ and find the least positive solution
- For exponent problems, reduce modulo prime powers (via FLT/Euler), then lift with CRT

Pattern cues (Demand: Low–Moderate):

- Coprime moduli; combine separate congruences into one modulo N
- Build with $M_i = N/n_i$ and inverses $y_i \equiv M_i^{-1} \pmod{n_i}$

Why It Works: Constructive isomorphism $\mathbb{Z}/N\mathbb{Z} \cong \prod_i \mathbb{Z}/n_i\mathbb{Z}$ via the Chinese Remainder isomorphism.

WHO: NT problems with remainders/last digits, clock/calendar arithmetic, or exponent cycles for composite moduli (usually low–moderate difficulty).

WHAT: Combine congruences with pairwise coprime moduli into a single congruence modulo $N = \prod n_i$ using the constructive CRT; handle non-coprime cases by compatibility checks.

How:

- **First Moves:**
  1. Verify pairwise coprimality; if not, check compatibility modulo $\gcd(n_i, n_j)$ for all pairs.
  2. Set $N = \prod n_i$ and $M_i = N/n_i$.
  3. Compute inverses $y_i \equiv M_i^{-1} \pmod{n_i}$.
  4. Form $x \equiv \sum a_i M_i y_i \pmod{N}$ and reduce to the least positive residue.
  5. For many congruences or large moduli, combine two at a time or use Garner's algorithm.
  6. For exponent problems, reduce modulo prime powers (via FLT/Euler), then lift with CRT.
- **Bailouts:** If moduli are not coprime, first merge compatible pairs or work modulo prime powers and use lcm; for small ranges, brute-force the least solution modulo N to sanity-check.
- **Pitfalls:** Forgetting to check compatibility in non-coprime cases; computing a non-existent inverse; neglecting to reduce the final sum modulo N; arithmetic slips in large M_i products.
- **Example:** For $(2, 3, 5)$ with remainders $(1, 2, 3)$, one finds $x \equiv 23 \pmod{30}$; see examples below.

Explicit construction: If $N = \prod_{i=1}^{k} n_i$ with pairwise coprime n_i, set $M_i = N/n_i$ and let y_i be the inverse of M_i modulo n_i (so $M_i y_i \equiv 1 \pmod{n_i}$). Then one solution is

$$x \equiv \sum_{i=1}^{k} a_i M_i y_i \pmod{N}.$$

Non-coprime warning: If the n_i are not pairwise coprime, solutions exist iff the congruences are consistent modulo $\gcd(n_i, n_j)$ for all pairs; any solution is then unique modulo $\operatorname{lcm}(n_1, \ldots, n_k)$ (not necessarily modulo N). **Connection:** Combine with Fermat's Little Theorem to reduce exponents modulo prime powers and then lift via CRT for composite moduli.

Example: Smallest Positive Solution

Find the smallest positive integer x such that $x \equiv 1 \pmod{2}$, $x \equiv 2 \pmod{3}$, and $x \equiv 3 \pmod{5}$.

Solution

Let $N = 2 \cdot 3 \cdot 5 = 30$. Set $M_1 = 15$, $M_2 = 10$, $M_3 = 6$ and choose inverses $y_1 \equiv 1 \pmod 2$, $y_2 \equiv 1 \pmod 3$, $y_3 \equiv 1 \pmod 5$. Then

$$x \equiv 1 \cdot 15 \cdot 1 + 2 \cdot 10 \cdot 1 + 3 \cdot 6 \cdot 1 = 53 \equiv 23 \pmod{30}.$$

Thus the least positive solution is $x = 23$.
Answer: 23

Number Theory Enrichment: Orders and LTE-lite

Orders / Cycles: For a with $\gcd(a, m) = 1$, the multiplicative order $\mathrm{ord}_m(a)$ divides $\phi(m)$ and satisfies $a^{\mathrm{ord}_m(a)} \equiv 1 \pmod m$. Use to reduce large exponents and to analyze last digits/remainders.

LTE-lite: For integers $a > b > 0$ and odd prime $p \mid a - b$, the p-adic valuation often follows $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$ in simple cases. While full LTE is beyond AMC 12, recognizing such lifting patterns can simplify select divisibility problems.

Example: Three Congruences

Solve $x \equiv 2 \pmod 5$, $x \equiv 3 \pmod 7$, $x \equiv 4 \pmod 9$.

Solution

The moduli are pairwise coprime. $N = 5 \cdot 7 \cdot 9 = 315$. Take $M_1 = 63$, $M_2 = 45$, $M_3 = 35$. Inverses: $63^{-1} \equiv 2 \pmod 5$ (since $63 \equiv 3$ and $3 \cdot 2 \equiv 1$), $45^{-1} \equiv 5 \pmod 7$ (since $45 \equiv 3$ and $3 \cdot 5 \equiv 1$), $35^{-1} \equiv 8 \pmod 9$ (since $35 \equiv 8$ and $8 \cdot 8 \equiv 64 \equiv 1$). Then

$$x \equiv 2 \cdot 63 \cdot 2 + 3 \cdot 45 \cdot 5 + 4 \cdot 35 \cdot 8$$
$$= 252 + 675 + 1120 = 2047 \equiv 2047 - 6 \cdot 315 = 157 \pmod{315}.$$

So $x \equiv 157 \pmod{315}$.

3.22 Technique 39: Graph Modeling (Graph Theory Conversion)

Short Name: T39-GraphModel

Definition 3.22 (Graph Modeling)**.** *Convert relationship or constraint problems into graphs by representing entities as vertices and relationships as edges (use directed edges where direction matters). Analyze the model with foundational graph facts.*

WHEN: Configuration: problems about entities and pairwise relationships that encode cleanly as a graph; Constraints: degree/parity conditions, bipartiteness (no odd cycles), connectivity or acyclicity (trees), or simple orientations; Objective: translate to graph invariants to deduce counts, existence/feasibility, or impossibility (via degree-sum, parity of cycles, matchings/colorings, or tree edge counts). Use for round-robin tournaments

(win/loss counts), handshake/party problems (degree parity), Euler trails/circuits (odd-degree counts; connectivity), bipartite pairing/assignment (two-type constraints), schedule/feasibility checks, path/cycle existence or obstruction by odd cycles, and trees/hierarchies with $|E| = n - 1$.

Configuration cues:

- Natural entities and pairwise relationships; constraints suggest degrees, parity, cycles, or connectivity
- Presence of bipartite structure (two types), tournament orientation, or tree-like hierarchy

Constraints cues:

- Decide undirected vs. directed, simple vs. multigraph; ensure modeling respects given constraints
- Parity constraints (odd/even degrees), forbidden odd cycles for bipartiteness, and tree edge counts

Objective cues:

- Translate to degrees, cycles, matchings, or tree properties to deduce counts/feasibility
- Use Handshaking Lemma, odd-cycle tests for bipartiteness, $|E| = n - 1$ for trees, or simple matching criteria

Pattern cues (Demand: Moderate):

- Degree/handshake constraints; pairings; bipartiteness or odd cycles; trees
- Model as a graph; apply degree-sum, cycle parity, or tree edge counts

Why It Works: Graph encodings expose structural properties and enable general results such as the Handshaking Lemma, bipartiteness via absence of odd cycles, and tree properties.

WHO: CNT discrete structures problems with pairings, tournaments, schedules, friendship/hostility constraints, or hierarchical "tree-like" data; moderate difficulty.

WHAT: Represent entities as vertices and relationships or constraints as edges (or arcs), then apply core graph facts (degrees, cycles, trees, matchings) to deduce counts or feasibility.

How:

- **First Moves:**
  1. Choose a natural vertex set and define edges (directed/undirected; weighted if needed).
  2. Translate problem conditions into degree bounds, parity, or forbidden substructures.
  3. Apply core tools: Handshaking Lemma; odd cycles obstruct bipartiteness; trees on n vertices have $n - 1$ edges; simple matching or Hall's condition when needed.
  4. Convert graph conclusions back into the original problem's language.

- **Bailouts:** If the model feels forced, try an alternative encoding (bipartite vs. general, directed vs. undirected), switch to parity/counting invariants without full graph formalism, or test small cases to guess a structure.
- **Pitfalls:** Mis-modeling relationships (double edges/self-loops when not allowed); overlooking bipartiteness or parity constraints; applying Euler trail/circuit criteria without checking connectivity.
- **Example:** Handshaking Lemma implies an even number of odd-degree vertices; see the brief proof below.

Core tools: Handshaking Lemma ($\sum_v \deg v = 2|E|$ so odd-degree vertices occur in even count); bipartite graphs have no odd cycles; trees on n vertices have $n-1$ edges; light use of matchings/Hall's condition for feasibility.

Example: Odd-Degree Vertices Come in Pairs

Prove that in any finite graph the number of vertices of odd degree is even.

Solution

By the Handshaking Lemma, the sum of all vertex degrees equals $2|E|$, which is even. Summing degrees modulo 2 counts precisely the number of odd-degree vertices modulo 2, hence that count is even.

3.23 Technique 40: Synthetic Division

Short Name: T40-SyntheticDiv

Definition 3.23 (Synthetic Division)**.** *A shortcut method for dividing a polynomial by a linear factor $(x-c)$. For polynomial $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, synthetic division by $(x-c)$ provides both the quotient and remainder efficiently.*

WHEN: Configuration: polynomial evaluation at a point or division by a linear factor $(x-c)$; Constraints: divisor must be linear in monic form (scale $(ax-b)$ to $(x-\frac{b}{a})$ with care) and coefficients listed in descending degree with zeros for gaps; Objective: compute $P(c)$ (Remainder Theorem), test candidate rational roots, and deflate the polynomial using the quotient. Use for rational Root Theorem pipelines, remainder questions modulo $(x-c)$, factoring cubics/quartics by deflating a known root, evaluating large polynomials at integers (Horner's method), testing multiplicities via successive deflation, and setting up partial fractions.

Configuration cues:

- Division by a linear factor $(x-c)$; candidate c from Rational Root Theorem or evaluation point
- Polynomial coefficients listed in descending degree with zeros inserted for missing powers

Constraints cues:

- Standard synthetic division applies to $(x - c)$; for $(ax - b)$ use long division or scale to $(x - \frac{b}{a})$ with care
- The final entry equals $P(c)$; confirm arithmetic and degree order to avoid errors

Objective cues:

- Quickly evaluate $P(c)$, test roots, and reduce degree to factor
- Obtain quotient coefficients and remainder to continue factoring or apply the Remainder Theorem

Pattern cues (Demand: Low):

- Dividing by $(x - c)$ where c is a constant
- Testing roots from Rational Root Theorem
- Quick polynomial evaluation

Why It Works: Streamlines the polynomial long division process for linear divisors.

WHO: ALG polynomial problems needing fast remainders, root tests from the Rational Root Theorem, or degree reduction before further factoring; usually low–moderate difficulty.

WHAT: Perform a tabular shortcut to divide by $(x - c)$; the last entry is the remainder and the row (excluding the remainder) gives quotient coefficients.

How:

- **First Moves:**
  1. Write the coefficients of $P(x)$ in descending degree, inserting zeros for missing powers.
  2. Bring down the leading coefficient.
  3. Multiply by c and add columnwise; repeat across the row.
  4. Read the final row: last entry is the remainder $P(c)$; preceding entries are quotient coefficients.
  5. Conclude divisibility (remainder $= 0$) or factor $P(x) = (x - c) \cdot Q(x) + r$.
- **Bailouts:** If dividing by $(ax - b)$, either perform long division or first scale to $(x - \frac{b}{a})$ with care; use Horner's method for fast evaluation when factoring is not required.
- **Pitfalls:** Omitting zero placeholders for missing powers; using c with the wrong sign for $(x - c)$; scrambling coefficient order.
- **Example:** Testing $c = 2$ for $x^3 - 6x^2 + 11x - 6$ yields remainder 0 and quotient $x^2 - 4x + 3$; see the table below.

Example: Testing a Root

Use synthetic division to test if $x = 2$ is a root of $P(x) = x^3 - 6x^2 + 11x - 6$.

Solution

Set up synthetic division with $c = 2$:

$$\begin{array}{c|cccc} 2 & 1 & -6 & 11 & -6 \\ & & 2 & -8 & 6 \\ \hline & 1 & -4 & 3 & 0 \end{array}$$

Process:

- Bring down the 1
- $2 \cdot 1 = 2$; $-6 + 2 = -4$
- $2 \cdot (-4) = -8$; $11 + (-8) = 3$
- $2 \cdot 3 = 6$; $-6 + 6 = 0$

Since the remainder is 0, $x = 2$ is indeed a root. The quotient is $x^2 - 4x + 3 = (x - 1)(x - 3)$.
Therefore: $x^3 - 6x^2 + 11x - 6 = (x - 2)(x - 1)(x - 3)$
Answer: Yes, $x = 2$ is a root; the complete factorization is $(x - 1)(x - 2)(x - 3)$.

3.24 Technique 41: Functional Equations

Short Name: T41-FuncEq

Definition 3.24 (Functional Equations)**.** *Problems that specify a relation between inputs and outputs of an unknown function f (e.g., $f(x + y) = f(x) + f(y)$), asking for all such f or evaluation of f at some points.*

WHEN: Configuration: an explicit relation involving f (e.g., $f(x \pm y)$, $f(kx)$, $f(xy)$, compositions); Constraints: domain/codomain and regularity assumptions (monotone/continuous/bounded valued) that tame wild solutions; Objective: determine the form of f or compute target values via substitutions and structure (injective/surjective) arguments. Use for relations using $f(x \pm y)$, $f(kx)$, $f(xy)$, $f(f(x))$, or compositions; domains such as $\mathbb{Z}$ or $\mathbb{N}$ and constraints (monotonicity, boundedness, integer-valued) that force classical forms; common patterns include piecewise/absolute-value variants and symmetry conditions.

Configuration cues:

- Additive/multiplicative structures: $f(x + y)$, $f(xy)$, Cauchy-type relations
- Symmetry and substitutions: $x = 0, 1, -x$, $y = x$, $y = 0$ to generate identities
- Monotonicity/boundedness/integer-valued constraints that force standard forms

Constraints cues:

- Domain/codomain (reals vs. integers) critically affects solutions
- Regularity assumptions (continuity/monotonicity/bounded on an interval) often collapse wild solutions to standard ones
- Beware extraneous solutions introduced by squaring or division by potentially zero quantities

Objective cues:

- Derive $f(0), f(1)$, parity ($f(-x)$), and linearity tests
- Prove injectivity/surjectivity from given relations; exploit them to compare values
- Guess-and-verify standard templates: linear ($ax + b$), power forms, or absolute-value variants

Why It Works: Strategic substitutions convert the functional constraint into algebraic identities that characterize f.

WHO: ALG problems featuring functional equations; common on COMC (short-answer and proofs) and mid–late on AMC 12.

WHAT: Use targeted substitutions, derive base values and symmetry, enforce constraints to narrow to standard forms.

How:

- **First Moves:**
  1. Plug easy values $(0, 1, -x, y = x)$ to obtain base identities; deduce $f(0)$ and $f(1)$.
  2. Test injectivity/surjectivity: from $f(a) = f(b)$ show $a = b$, or hit all outputs.
  3. Use constraints (e.g., monotone/continuous/integer-valued) to eliminate pathological solutions; fit parameters in a candidate form.
- **Bailouts:** Try bounding arguments, parity/symmetry ($f(-x)$), or consider special inputs ($x = 1$, $y = 0$, $y = x$); propose a likely template (linear, power, absolute-value) and verify under the given constraints.
- **Pitfalls:** Assuming regularity not given (e.g., continuity); dividing by expressions that may be zero; introducing extraneous solutions via squaring.
- **Example:** Additivity plus monotonicity forces $f(x) = cx$; see the solution below for the standard derivation.

Example: Additive Form under Monotonicity

Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y) = f(x) + f(y)$ for all reals x, y, and f is monotone on $\mathbb{R}$.

Solution

Setting $y = 0$ gives $f(0) = 0$. Taking $y = -x$ yields $f(-x) = -f(x)$. By induction on integers, $f(nx) = nf(x)$ for $n \in \mathbb{Z}$; extending to rationals shows $f(q) = cq$ for $q \in \mathbb{Q}$ where $c = f(1)$. Monotonicity extends the rational linearity to all reals, so $f(x) = cx$ for some constant $c \in \mathbb{R}$.
Answer: $f(x) = cx$ for some real constant c.

3.25 Technique M10: Triangle/Quadrilateral Area Toolkit

Short Name: TM10-AreaToolkit

Definition 3.25 (Area Formulas)**.** *For triangle with side lengths a, b, c and semiperimeter $s = \frac{a+b+c}{2}$:*

- *Heron's Formula: $A = \sqrt{s(s-a)(s-b)(s-c)}$.*
- *Inradius relation: $A = rs$ where r is the inradius.*
- *Circumradius relation: $A = \dfrac{abc}{4R}$ and $a = 2R \sin A$ (and cyclically).*
- *Trig area: $A = \frac{1}{2}ab \sin C$ (and cyclic permutations).*

For a cyclic quadrilateral with side lengths a, b, c, d and $s = \frac{a+b+c+d}{2}$:

- *Brahmagupta's Formula: $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.*

WHEN: Side lengths/semiperimeter given; incircle/circumcircle present; need to convert between area and r or R; cyclic quadrilaterals.

WHO: GEO triangle and quadrilateral geometry problems with area calculations; solvers comfortable with Heron's formula and trigonometric area formulas.

How:

- **First Moves:** Compute s; test cyclicity before using Brahmagupta; use $A = rs$ or $A = \frac{abc}{4R}$ when r or R is involved.
- **Bailouts:** If cyclicity is unclear, try Ptolemy/PoP or angle chasing; decompose to triangles and sum areas.
- **Pitfalls:** Using Brahmagupta on non-cyclic quads; numerical stability when s is close to a side.

3.26 Technique M15: Euler's Polyhedral Formula

Short Name: TM15-EulerPoly

Definition 3.26 (Euler Characteristic of Convex Polyhedra)**.** *For a connected convex polyhedron (or planar connected graph), $V - E + F = 2$.*

WHEN: Faces/edges/vertices counts; feasibility checks for nets and polyhedra; planar graph constraints.

WHO: GEO polyhedral geometry problems with Euler's formula; solvers comfortable with graph theory and polyhedral properties.

How:

- **First Moves:** Translate 3D to counts of V, E, F; apply $V - E + F = 2$; use degree sums $\sum \deg(v) = 2E$.
- **Pitfalls:** Non-convex or holed surfaces (genus $\neq 0$) use $V - E + F = 2 - 2g$; avoid double counting faces/edges.

3.27 Technique M16: Descartes Circle Theorem

Short Name: TM16-Descartes

Definition 3.27 (Kissing Circles)**.** *Let curvatures be $k_i = \pm\frac{1}{r_i}$ for four mutually tangent circles (outer circle has negative curvature). Then*

$$(k_1 + k_2 + k_3 + k_4)^2 = 2\,(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

WHEN: Apollonian gaskets / tangent circle packs; find an unknown radius from three known.

WHO: GEO circle geometry problems with tangent circles; solvers comfortable with inversion geometry and curvature relationships.

How:

- **First Moves:** Assign signs (outer circle negative), plug three known curvatures to solve for the fourth.
- **Pitfalls:** Wrong sign for enclosing circle; selecting the wrong root (two solutions correspond to the two possible tangent circles).

3.28 Technique M17: Conjugate Root Theorem

Short Name: TM17-ConjRoots

Definition 3.28 (Complex Conjugate Roots)**.** *If a polynomial has real coefficients, then non-real roots occur in conjugate pairs: $\alpha \in \mathbb{C} \setminus \mathbb{R}$ implies $\overline{\alpha}$ is also a root.*

WHEN: Factorization over $\mathbb{R}$; reconstructing real-coefficient polynomials from complex roots; simplifying linear recurrences to real forms via cos/sin.

WHO: ALG polynomial problems with complex roots; solvers comfortable with complex numbers and polynomial factorization.

How:

- **First Moves:** Pair $re^{i\theta}$ with $re^{-i\theta}$ to form real quadratics $x^2 - 2r\cos\theta\, x + r^2$.
- **Pitfalls:** Assuming conjugate pairing when coefficients are not real; dropping multiplicities.

3.29 Technique M18: Sophie's Germain Identity

Short Name: TM18-SGI

Definition 3.29 (Factor $a^4 + 4b^4$)**.** $a^4 + 4b^4 = (a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2)$.

WHEN: Factorizations of the form $x^4 + 4y^4$ or disguised variants; Diophantine constraints where factoring unlocks parity/modular structure.

WHO: ALG polynomial factorization problems with fourth powers; solvers comfortable with algebraic identities and polynomial factoring.

How:

- **First Moves:** Try $a = x$, $b = y$ (or scaled) to match the pattern; expand to verify.
- **Pitfalls:** Misapplying to $a^4 + 4b^k$ with $k \neq 4$; overlooking simple substitutions that align terms.

3.30 Technique M19: Wilson's Theorem

Short Name: TM19-Wilson

Definition 3.30 (Wilson)**.** *For prime p, $(p-1)! \equiv -1 \pmod{p}$; conversely, if $(n-1)! \equiv -1 \pmod{n}$ then n is prime.*

WHEN: Remainders of factorials modulo a prime; primality checks for small p; constructing residues via pairings.

WHO: NT problems with primality testing; solvers comfortable with modular arithmetic and Wilson's theorem.

How:

- **First Moves:** Pair k with $k^{-1} \pmod{p}$ for $1 \leq k \leq p-1$; only 1 and -1 are self-inverses.
- **Pitfalls:** Applying to composite moduli; arithmetic slips in modular inverses.

3.31 Technique M20: Legendre's Formula (p-adic valuation of $n!$)

Short Name: TM20-Legendre

Definition 3.31 (Legendre)**.** *For prime p and integer $n \geq 1$, $v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor$.*

WHEN: Exponent of a prime in $n!$ and in binomial coefficients: $v_p\binom{n}{k} = v_p(n!) - v_p(k!) - v_p((n-k)!)$.

WHO: NT number theory problems with prime power divisibility; solvers comfortable with p-adic valuations and Legendre's formula.

How:

- **First Moves:** Sum floors until $p^k > n$; apply to each prime in a factorization.
- **Pitfalls:** Stopping too early; forgetting to subtract in $v_p\binom{n}{k}$; mixing with nonprime bases.

3.32 Technique M21: Divisor Functions $\tau(n)$ and $\sigma(n)$

Short Name: TM21-DivFuncs

Definition 3.32 (Multiplicative Formulas)**.** *If* $n = \prod p_i^{a_i}$*, then* $\tau(n) = \prod(a_i + 1)$ *and* $\sigma(n) = \prod \dfrac{p_i^{a_i+1} - 1}{p_i - 1}$.

WHEN: Number of divisors or sum of divisors; bounding counts; structure of highly composite/perfect numbers.

WHO: NT problems with divisor functions; solvers comfortable with multiplicative functions and prime factorization.

How:

- **First Moves:** Prime factor n; apply product formulas; use multiplicativity on coprime factors.
- **Pitfalls:** Non-coprime factor splits; arithmetic with large exponents; forgetting σ's formula per prime power.

3.33 Technique M22: Binomial Theorem and Pascal Identities

Short Name: TM22-BinomialPascal

Definition 3.33 (Binomial and Identities)**.**

- *Binomial Theorem:* $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$.
- *Pascal's Identity:* $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$.
- *Hockey Stick:* $\displaystyle\sum_{i=r}^{m} \binom{i}{r} = \binom{m+1}{r+1}$.
- *Vandermonde:* $\displaystyle\sum_{i} \binom{m}{i}\binom{n}{k-i} = \binom{m+n}{k}$.

WHEN: Coefficient extraction; cumulative binomial sums; combinatorial identities; splitting constrained selections.

WHO: CNT/ALG problems with binomial coefficients; solvers comfortable with Pascal's triangle and generating functions.

How:

- **First Moves:** Choose an identity suited to the index structure; reindex to match forms.
- **Pitfalls:** Off-by-one indices; incorrect summation limits; assuming independence where PIE is needed.

3.34 Technique M23: Rates and Work Templates

Short Name: TM23-RatesWork

Definition 3.34 (Speed–Distance–Time and Work)**.** *Speed–Distance–Time: $d = rt$; with piecewise motion, sum distances or times. Work rates add: if job rates are r_1, r_2, combined rate $= r_1 + r_2$, so time T satisfies $\frac{1}{T} = \frac{1}{t_1} + \frac{1}{t_2}$ for individual times t_i on one job.*

WHEN: Word problems with constant rates; pipeline/filling; alternating schedules; average speed traps.

WHO: ALG word problems with rates and work; solvers comfortable with setting up equations and solving systems.

How:

- **First Moves:** Normalize units; express all segments in $d = rt$ or in rates per job; use harmonic mean for round trips with equal distances.
- **Pitfalls:** Averaging speeds arithmetically across equal times vs. equal distances; changing rates/contexts mid-problem.

3.35 Technique M24: Degenerate Triangles (Triangle Inequality Boundary)

Short Name: TM24-DegenerateTriangles

Definition 3.35 (Triangle Inequality and Degenerate Cases)**.** *For any triangle with sides a, b, c:*

- ***Existence condition:*** $a + b > c$, $b + c > a$, $a + c > b$
- ***Degenerate case:*** *When equality holds (e.g., $a + b = c$), the triangle collapses to a line segment*
- ***Optimization principle:*** *Extrema in triangle-based problems often occur at degenerate configurations*

The degenerate case represents the boundary between valid and invalid triangles, where three points become collinear.

WHEN: Triangle optimization problems asking for minimum/maximum values; problems with triangle side constraints and variable parameters; geometric configurations where triangle feasibility determines bounds; problems involving inscribed/circumscribed triangles with optimization; distance minimization/maximization in triangular configurations.

WHO: GEO triangle-feasibility optimizers; solvers comfortable with triangle inequality manipulations, recognizing when extrema occur at boundaries, converting geometric constraints to algebraic inequalities, and handling degenerate/limiting cases systematically.

How:

- **First Moves:** Identify the constraint by writing all triangle inequalities for the given configuration; force equality by setting one inequality to equality (typically the one involving the optimization target); solve the degenerate case (when $a + b = c$, the points are collinear); check feasibility by verifying other constraints still hold in the degenerate configuration; compare boundaries by checking each boundary if multiple inequalities can degenerate.

- **Bailouts:** If algebra becomes complex, use coordinate geometry with degenerate placement; apply discriminant analysis when triangle feasibility creates quadratic constraints; use Lagrange multipliers for smooth optimization with inequality constraints; switch to parametric representation if direct approach stalls.

- **Pitfalls:** Forgetting to check all three triangle inequalities; assuming extrema always occur at degenerate cases (sometimes interior maxima exist); missing that "no triangle" might be the actual answer; not verifying that the degenerate configuration satisfies all problem constraints.

- **Example:** Real numbers a and b are chosen with $1 < a < b$ such that no triangle with positive area has side lengths $1, a, b$ or $\frac{1}{b}, \frac{1}{a}, 1$. What is the smallest possible value of b?

  Solution: For no triangle to exist with sides $1, a, b$: we need $1 + a \leq b$ (triangle inequality fails). For no triangle to exist with sides $\frac{1}{b}, \frac{1}{a}, 1$: we need $\frac{1}{b} + \frac{1}{a} \leq 1$. To minimize b, we force both conditions to be equalities (degenerate cases): $a = b - 1$ (degenerate triangle for first set) and $\frac{1}{a} + \frac{1}{b} = 1$ (degenerate triangle for second set). Substituting $a = b - 1$ into the second equation: $\frac{1}{b-1} + \frac{1}{b} = 1$, so $\frac{b+(b-1)}{b(b-1)} = 1$, giving $2b - 1 = b^2 - b$, or $b^2 - 3b + 1 = 0$. Using the quadratic formula: $b = \frac{3\pm\sqrt{5}}{2}$. Since $b > a > 1$, we need $b = \frac{3+\sqrt{5}}{2}$.

  Answer: $\frac{3+\sqrt{5}}{2}$

Works Well With: TM12-Discriminant (for parameter bounds ensuring triangle feasibility); T2-CoordGeo (place degenerate triangle as line segment); T22-CauchySchwarz (triangle inequality connects to C-S equality conditions); T31-Stewart (cevian lengths approach limits in degenerate cases).

Typical Difficulty: Often appears in problems 16-23 on AMC 12, especially in optimization contexts.

Special Cases: Isosceles degenerate (two sides equal, third is their sum); right triangle limit (when $a^2 + b^2 = c^2$ exactly); inscribed/circumscribed (degenerate inscribed triangles have vertices on diameter).

Connection to Other Concepts: The triangle inequality is fundamentally connected to metric spaces (triangle inequality axiom), Cauchy-Schwarz inequality (equality case gives degenerate triangle), Minkowski inequality in higher dimensions, and convex optimization (boundary extrema).

Category Classification: Medium-Probability - Requires insight to recognize that extrema occur at degenerate boundaries; not immediately obvious but powerful when applicable.

3.36 Technique M25: Francesca Volume (Piero della Francesca's Tetrahedron Formula)

Short Name: TM25-FrancescaVol

Definition 3.36 (Francesca's Irregular Tetrahedron Formula)**.** *For a tetrahedron with edge lengths a, b, c, d, e, f where:*

- *a, b, c are edges from vertex 1*
- *d, e, f are the opposite edges (not meeting vertex 1)*

The volume is:

$$V = \frac{1}{12}\sqrt{-a^2b^2c^2 - a^2d^2e^2 - b^2d^2f^2 - c^2e^2f^2 + \textit{(16 positive terms)}}$$

where the full expression involves all edge length combinations in a specific pattern derived from the Cayley-Menger determinant.

WHEN: Irregular tetrahedra with only edge lengths given where direct height extraction is unwieldy; when all six edge lengths are known but the tetrahedron lacks obvious perpendicularity or special structure; alternative to coordinate placement or decomposition methods.

WHO: GEO advanced solvers versed in Piero della Francesca tetrahedron volume formulas; those comfortable with complex algebraic expressions and systematic substitution.

How:

- **First Moves:**
  1. Label vertices systematically and identify all six edge lengths.
  2. Check first for special cases (perpendicular faces, regular tetrahedra) that allow simpler methods.
  3. If no simpler structure exists, apply the full Francesca formula with careful bookkeeping.
  4. Use the Cayley-Menger determinant form for verification:

$$288V^2 = \begin{vmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 \\ 1 & d_{12}^2 & 0 & d_{23}^2 & d_{24}^2 \\ 1 & d_{13}^2 & d_{23}^2 & 0 & d_{34}^2 \\ 1 & d_{14}^2 & d_{24}^2 & d_{34}^2 & 0 \end{vmatrix}$$

- **Pitfalls:**
  - The formula is computationally intensive - avoid in time-constrained settings unless necessary
  - Sign errors in the alternating terms of the expression
  - Edge labeling confusion - maintain consistent vertex-to-edge correspondence

  - Not checking if simpler methods (coordinate placement, decomposition) would be faster

- **Example:** For tetrahedron $ABCD$ with $AB = 5$, $AC = 3$, $BC = 4$, $BD = 4$, $AD = 3$, $CD = \frac{12}{5}\sqrt{2}$:
  - While Francesca's formula gives $V = \frac{24}{5}$, this problem is better solved by recognizing that triangles ABC and ABD are both 3-4-5 right triangles
  - The perpendicular structure allows direct computation: $V = \frac{1}{3} \cdot 6 \cdot \frac{12}{5} = \frac{24}{5}$

Strategic Note: This formula is rarely the optimal approach in competitions. Use it as a verification tool or last resort when:

- Coordinate placement fails due to irrational coordinates
- No obvious perpendicularity or symmetry exists
- Decomposition into simpler tetrahedra is not apparent
- You have significant extra time and need certainty in your answer

Works Well With: TM3-Shoelace (for coordinate-based verification); T2-CoordGeo (alternative approach when coordinates are messy); T31-Stewart (for decomposition into simpler tetrahedra); TM10-TriangleArea (for base area calculations in decomposition).

Typical Difficulty: Rarely appears directly in AMC 12, but useful for verification in problems 20-25 involving tetrahedra.

Special Cases: Regular tetrahedron (all edges equal); right tetrahedron (three perpendicular edges from one vertex); degenerate tetrahedron (volume approaches zero).

Connection to Other Concepts: The Cayley-Menger determinant generalizes to higher dimensions; connects to Heron's formula for triangles; related to the Gram determinant for volume calculations; fundamental in computational geometry.

Category Classification: Low-Probability - Computationally intensive and rarely optimal; primarily useful as verification tool or last resort.

3.37 Technique M26: First Passage Time

Short Name: TM26-FirstPassage

Definition 3.37 (First Passage Time)**.** *For random walks or stochastic processes, the first passage time is the first time the process reaches a specified threshold or boundary. Success probability often depends on which boundary is hit first, motivating segmentation by earliest hit time.*

WHEN: Finite coin walks/random walks where success depends on first reaching a boundary; absorbing boundary problems; gambler's ruin scenarios; lattice path problems with barriers; hitting-time reasoning for Markov chains; problems asking "probability of reaching A before B"; success/failure thresholds in sequential trials.

Configuration cues:

- Random walk on integers with absorbing boundaries
- Coin flips/games continuing until reaching a threshold
- "First to reach n wins" scenarios
- Lattice paths with barriers or forbidden regions
- Sequential processes with multiple stopping conditions

Constraints cues:

- Identify all absorbing boundaries (success/failure states)
- Verify the walk is finite (will eventually reach a boundary)
- Check if the process is symmetric or biased
- Consider whether boundaries are reflecting or absorbing

Objective cues:

- Find probability of reaching boundary A before boundary B
- Compute expected time to reach first boundary
- Determine probability distribution of first passage times
- Calculate success probability based on which threshold is hit first

Why It Works: The process partitions into disjoint events based on which boundary is reached first, and the Markov property allows recursive probability calculations.

WHO: PROB random-walk threshold problems; solvers comfortable with lattice-path casework and hitting-time reasoning; those who can set up and solve systems of linear equations for absorption probabilities.

WHAT: Model the process as a random walk with absorbing boundaries, then use either direct counting (for small cases) or recursive equations to find absorption probabilities.

How:

- **First Moves:**
  1. Identify the state space and absorbing boundaries.
  2. Let p_i = probability of success starting from state i.
  3. Set boundary conditions: $p_{\text{success}} = 1$, $p_{\text{failure}} = 0$.
  4. Write recursive equations: $p_i = \sum_j P(i \to j) \cdot p_j$ for non-boundary states.
  5. Solve the system of linear equations for all p_i.
- **Bailouts:** If the state space is too large, look for symmetry or use the reflection principle; for symmetric random walks, use the gambler's ruin formula directly; consider generating functions for more complex hitting time distributions.

- **Pitfalls:** Forgetting to verify that boundaries are actually absorbing; incorrectly setting up transition probabilities; missing the fact that some states may never reach boundaries; confusing first passage with return times.
- **Example:** Classic gambler's ruin: starting with k dollars, probability of reaching n before 0 in fair game is k/n; reflection principle for ballot problem counting.

Example: Simple Random Walk

A particle starts at position 3 on the number line. At each step, it moves left or right with probability 1/2 each. What is the probability it reaches position 6 before reaching position 0?

Solution

Let p_i be the probability of reaching 6 before 0, starting from position i. Boundary conditions: $p_0 = 0$, $p_6 = 1$. For $1 \leq i \leq 5$: $p_i = \frac{1}{2}p_{i-1} + \frac{1}{2}p_{i+1}$.
This gives $p_{i+1} - 2p_i + p_{i-1} = 0$, a linear recurrence with characteristic equation $r^2 - 2r + 1 = 0$, giving $r = 1$ (double root).
General solution: $p_i = A + Bi$ for constants A, B. Using boundaries: $p_0 = A = 0$, $p_6 = 6B = 1$, so $B = 1/6$. Therefore $p_i = i/6$, and $p_3 = 3/6 = 1/2$.
Answer: $\dfrac{1}{2}$

AMC Style: Biased Walk

A frog starts at lily pad 2 in a line of 5 lily pads numbered 1 through 5. Each second, it jumps to an adjacent pad, jumping right with probability 2/3 and left with probability 1/3. What is the probability it reaches pad 5 before pad 1?

Solution

Let p_i = probability of reaching 5 before 1, starting from pad i. Boundaries: $p_1 = 0$, $p_5 = 1$. For pads 2, 3, 4: $p_i = \frac{1}{3}p_{i-1} + \frac{2}{3}p_{i+1}$.
Rearranging: $2p_{i+1} - 3p_i + p_{i-1} = 0$. Characteristic equation: $2r^2 - 3r + 1 = 0$, giving $r = 1$ or $r = 1/2$. General solution: $p_i = A + B(1/2)^i$.
From $p_1 = A + B/2 = 0$ and $p_5 = A + B/32 = 1$: $A = -B/2$ and $A + B/32 = 1$.
Substituting: $-B/2 + B/32 = 1$, so $B(-16/32 + 1/32) = 1$, giving $B = -32/15$.
Therefore $A = 16/15$.
Thus $p_2 = 16/15 - (32/15)(1/4) = 16/15 - 8/15 = 8/15$.
Answer: $\dfrac{8}{15}$

3.38 Technique M28: Consecutive Integer Decomposition

Short Name: TM28-ConsecutiveDecomp

Definition 3.38 (Consecutive Integer Decomposition)**.** *Express a positive integer N as a sum of consecutive positive integers. For a sequence of k consecutive integers starting at a:*

$$N = a + (a+1) + \cdots + (a+k-1) = ka + \frac{k(k-1)}{2}$$

Equivalently: $2N = k(2a+k-1)$ where k is the length and $(2a+k-1)$ is the sum of first and last terms.

WHEN: Finding ways to write N as consecutive integer sums; divisibility problems involving consecutive sequences; number theory problems with arithmetic progression constraints. Use for decomposition counting problems; consecutive sum representations; parity-constrained factorizations; problems asking "how many ways to write N as consecutive integers."

Configuration cues:

- "Sum of consecutive positive integers"
- "Express as an arithmetic sequence with common difference 1"
- "How many ways to write N as consecutive terms"
- Problems involving $(a, a+1, a+2, \ldots)$ patterns

Constraints cues:

- Parity check: For odd k, the middle term must divide N; for even k, N/k must be a half-integer
- Positivity: First term $a = \frac{2N/k-k+1}{2} > 0$ requires $2N > k(k-1)$
- At least two terms: exclude the trivial case $k = 1$ (single number)
- Factor pairs (d_1, d_2) of $2N$ with opposite parity yield valid decompositions

Objective cues:

- Count valid decompositions via divisor pairs of $2N$
- Find specific representations by testing small values of k
- Use parity filters to eliminate impossible cases quickly

Why It Works: The formula $2N = k(2a+k-1)$ converts the problem to finding factor pairs with parity constraints.

Edge Cases:

- **Powers of 2:** For $N = 2^m$, only trivial decomposition exists (the number itself) since $2N = 2^{m+1}$ has only even divisors, violating the opposite parity requirement.
- **Odd Primes:** For prime p, only two decompositions exist: p itself and the two-term case $(p-1)/2 + (p+1)/2$ when p is odd.

- **Highly Composite Numbers:** Numbers with many divisors (like factorials) have numerous valid decompositions; organize divisors systematically to avoid missing cases.

WHO: NT/CNT decomposition solvers translating consecutive-integer sums into divisor pairs; comfortable with parity filters and factoring $2N$.

WHAT: Express N as consecutive integers by factoring $2N$ and checking parity/positivity constraints.

How:

- **First Moves:**
  1. Factor $2N$ to find all divisor pairs (k, d) where $kd = 2N$.
  2. For each pair, check parity: k and $d = 2a + k - 1$ must have opposite parity.
  3. Compute first term: $a = \frac{d-k+1}{2}$.
  4. Verify positivity: $a > 0$ requires $d > k - 1$.
  5. Count valid decompositions (excluding $k = 1$ if required).

- **Bailouts:** If N is prime, only decompositions are N itself and the two-term case (if N is odd); for highly composite N, organize divisors systematically to avoid missing cases.

- **Pitfalls:** Forgetting the parity constraint (opposite parity requirement); missing the positivity check for the first term; counting the single-term case when not allowed; confusing the roles of k (length) and a (first term).

- **Example:** For $N = 345 = 3 \cdot 5 \cdot 23$, we have $2N = 690 = 2 \cdot 3 \cdot 5 \cdot 23$. Valid (k, d) pairs satisfying constraints give 7 decompositions.

Example: Consecutive Sum Decomposition

In how many ways can 21 be written as a sum of two or more consecutive positive integers?

Solution

We need $21 = a + (a + 1) + \cdots + (a + k - 1)$ for some $a \geq 1$ and $k \geq 2$.
Using the formula: $2N = k(2a + k - 1)$, so $42 = k(2a + k - 1)$.
Factor 42: $42 = 2 \cdot 3 \cdot 7$
Divisor pairs of 42: $(1, 42), (2, 21), (3, 14), (6, 7), (7, 6), (14, 3), (21, 2), (42, 1)$
For valid decompositions: - k and $(2a + k - 1)$ must have opposite parity - $a = \frac{(2N/k)-k+1}{2} > 0$ - $k \geq 2$ (at least two terms)
Valid cases: - $k = 2, d = 21$: $a = \frac{21-2+1}{2} = 10$ ✓ $\rightarrow 10 + 11 = 21$ - $k = 3, d = 14$: $a = \frac{14-3+1}{2} = 6$ ✓ $\rightarrow 6 + 7 + 8 = 21$ - $k = 6, d = 7$: $a = \frac{7-6+1}{2} = 1$ ✓ $\rightarrow 1 + 2 + 3 + 4 + 5 + 6 = 21$ - $k = 21, d = 2$: $a = \frac{2-21+1}{2} = -9$ × (negative)
Answer: 3 ways

3.39 Technique M29: Trigonometric Monotone Scanning

Short Name: TM29-TrigMonotoneScan

Definition 3.39 (Monotone Interval Analysis for Trig Functions)**.** *For composite trigonometric functions with asymptotes or critical points, identify monotone intervals where the function is strictly increasing or decreasing. Use sign charts and derivative analysis to bracket roots within specific intervals, especially when dealing with sums like* $\tan x + \cot x$ *or* $\sin x + \cos x$*. The key insight is that monotone functions have at most one root per interval, and sign changes confirm existence via the Intermediate Value Theorem.*

WHEN: Sums of basic trig functions with asymptotes where sign charts isolate the target interval; equations involving mixed trig functions where traditional algebraic methods become unwieldy; problems asking for specific roots within restricted domains; optimization problems with trig constraints where monotonicity determines unique extrema.

Configuration cues:

- Sums or differences of trig functions with different periods (e.g., $\tan x + 2\sin x$)
- Trig equations with restricted domains like $[0, \pi]$ or $[0, 2\pi]$
- Problems involving $\tan x \pm \cot x$ or similar reciprocal combinations
- Finding roots near asymptotes or critical points

Constraints cues:

- Identify all asymptotes and discontinuities in the domain of both $f(x)$ and $f'(x)$
- Determine critical points where $f'(x) = 0$ (where derivative exists)
- Check boundary behavior and limits approaching asymptotes: $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$
- Verify sign changes to confirm root existence via Intermediate Value Theorem
- Ensure the derivative is defined and continuous on each monotone interval

Objective cues:

- Find all roots in a given interval
- Determine the number of solutions to a trig equation
- Locate the unique solution in a monotone interval
- Optimize trig expressions subject to domain constraints

Why It Works: Monotonicity guarantees at most one root per interval, and sign changes confirm existence via the Intermediate Value Theorem.

WHO: ALG/GEO solvers reading trig graphs and monotone intervals to bracket roots; those comfortable with derivative signs and asymptotic behavior of trigonometric functions.

How:

- **First Moves:**
  1. Find $f'(x)$ and identify its domain (avoiding asymptotes)
  2. Locate critical points where $f'(x) = 0$ (if any)
  3. Identify all asymptotes and discontinuities of $f(x)$
  4. Create sign chart for $f'(x)$ on each continuous interval
  5. Test endpoints and limits near asymptotes to establish monotonicity
  6. Use sign changes of $f(x)$ to locate roots within monotone intervals
- **Bailouts:** If derivative analysis becomes too complex, use trigonometric identities (e.g., $\tan x + \cot x = \sec x \csc x$) or numerical approximation
- **Pitfalls:** Missing asymptotes in domain analysis; incorrect derivative signs due to domain errors; forgetting to check boundary points and limits; assuming continuity where discontinuities exist

Example: Trigonometric Monotone Scanning

Find all solutions to $\tan x + \cot x = 2$ in the interval $(0, \pi)$.

Solution

Let $f(x) = \tan x + \cot x - 2$. We need to find where $f(x) = 0$.
First, identify the domain: $\tan x$ is undefined at $x = \pi/2$, so we consider intervals $(0, \pi/2)$ and $(\pi/2, \pi)$.
Find the derivative: $f'(x) = \sec^2 x - \csc^2 x = \frac{1}{\cos^2 x} - \frac{1}{\sin^2 x} = \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} = \frac{-\cos 2x}{\sin^2 x \cos^2 x}$.
Since $\sin^2 x \cos^2 x > 0$ on $(0, \pi/2)$ and $(\pi/2, \pi)$, the sign of $f'(x)$ depends on $-\cos 2x$.
On $(0, \pi/2)$: $\cos 2x > 0$ when $2x \in (0, \pi/2)$, i.e., $x \in (0, \pi/4)$; $\cos 2x < 0$ when $x \in (\pi/4, \pi/2)$. - For $x \in (0, \pi/4)$: $f'(x) < 0$, so f is decreasing - For $x \in (\pi/4, \pi/2)$: $f'(x) > 0$, so f is increasing
On $(\pi/2, \pi)$: $\cos 2x < 0$ when $2x \in (\pi, 2\pi)$, so $f'(x) > 0$, meaning f is increasing.
Check limits and endpoints: - $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}(\tan x + \cot x - 2) = 0 + \infty - 2 = +\infty$ - $\lim_{x\to(\pi/2)^-} f(x) = +\infty + 0 - 2 = +\infty$ - $\lim_{x\to(\pi/2)^+} f(x) = -\infty + 0 - 2 = -\infty$ - $\lim_{x\to\pi^-} f(x) = 0 + (-\infty) - 2 = -\infty$
Since f is decreasing on $(0, \pi/4)$ from $+\infty$ to some value, and increasing on $(\pi/4, \pi/2)$ from some value to $+\infty$, and increasing on $(\pi/2, \pi)$ from $-\infty$ to $-\infty$, there must be exactly one root in $(0, \pi/4)$ and no roots in $(\pi/4, \pi)$.
For the root in $(0, \pi/4)$, since f is decreasing and continuous, and $f(0^+) = +\infty$, $f(\pi/4) = \tan(\pi/4) + \cot(\pi/4) - 2 = 1 + 1 - 2 = 0$.
Answer: $x = \pi/4$

3.40 Technique M30: Similarity Ratio Scaling

Short Name: TM30-SimilarityRatios

Definition 3.40 (Similarity-Based Proportional Scaling)**.** *When similar triangles are nested or embedded within geometric configurations, use the similarity ratio to scale all correspond-*

ing lengths proportionally. For squares inscribed in right triangles, the square's side converts to proportional hypotenuse segments via the similarity ratio $k = \frac{small\ triangle}{large\ triangle}$.

WHEN: Nested similar right triangles where square sides convert to proportional hypotenuse segments; embedded similar figures where one dimension determines all others; problems with inscribed squares/rectangles in triangles; configurations where altitude creates similar sub-triangles; scaling problems where all lengths change by the same ratio.

Configuration cues:

- Square or rectangle inscribed in a right triangle
- Nested similar triangles with shared angles
- Altitude to hypotenuse creating three similar triangles
- Parallel lines creating proportional segments
- Self-similar fractals or iterative constructions

Constraints cues:

- Verify triangles are indeed similar (AA, SAS, SSS similarity)
- Identify the common ratio between corresponding sides
- Check that inscribed figures maintain their shape constraints
- Ensure proportions are consistent across all similar pairs

Objective cues:

- Find dimensions of inscribed squares or rectangles
- Determine ratios between nested similar figures
- Calculate areas using similarity ratios (area ratio = k^2)
- Find the limiting value in iterative similar constructions

Why It Works: Similarity preserves angle measures and side ratios, allowing systematic scaling of all measurements once the ratio is determined.

WHO: GEO solvers leveraging right-triangle similarity to scale embedded squares; those comfortable with proportional reasoning and similar triangle theorems.

How:

- **First Moves:** Identify similar triangles; find similarity ratio k; scale all corresponding lengths by k
- **Bailouts:** Set up coordinate system if synthetic approach stalls; use area ratios as alternative

- **Pitfalls:** Confusing similarity ratio with area ratio; incorrect correspondence of sides; missing embedded similarities
- **Example:** In a right triangle with legs 3 and 4, a square is inscribed with one vertex on the hypotenuse. The square creates two smaller similar triangles. If the square has side length s, then the similarity ratio $k = \frac{s}{3}$ for the left triangle and $k = \frac{s}{4}$ for the right triangle. Use these ratios to find s.

3.41 Technique M31: Volume Decomposition

Short Name: TM31-VolumeDecompose

Definition 3.41 (Composite Solid Decomposition)**.** *Break complex 3D solids into standard shapes (cones, cylinders, frustums, pyramids, spheres) whose volumes are known. For solids of revolution or composite figures, decompose into layers or sections, then sum volumes or use integration when the cross-section area function is simple.*

WHEN: Composite solids split into standard cones/frusta so subtraction or integration avoids ad-hoc volume work; solids of revolution where cross-sections have known area formulas; truncated or partial solids requiring frustum formulas; problems involving drilling holes or carving out sections; layered solids with varying cross-sections.

WHO: GEO solid-geometry solvers fluent with cone/frustum formulas or integral setups; those comfortable with 3D visualization and cross-section analysis.

Configuration cues:

- Composite solids made of cones, cylinders, spheres, pyramids
- Truncated pyramids or cones (frustums)
- Solids with holes or carved-out sections
- Objects formed by rotating regions around an axis
- Stacked or layered geometric solids

Constraints cues:

- Identify all component shapes and their dimensions
- Determine whether to add or subtract volumes
- Check continuity at boundaries between components
- Verify that decomposition captures the entire solid
- Account for overlapping regions in additive decomposition

Objective cues:

- Find volume of a complex composite solid

- Calculate volume removed when drilling/carving
- Determine volume using disc/washer/shell methods
- Compare volumes of related solids

Why It Works: Standard volume formulas are well-established, and linearity of volume allows additive/subtractive decomposition. The key volume formulas are:

- **Cone:** $V = \frac{1}{3}\pi r^2 h$
- **Cylinder:** $V = \pi r^2 h$
- **Sphere:** $V = \frac{4}{3}\pi r^3$
- **Pyramid:** $V = \frac{1}{3}Bh$ where B is base area
- **Frustum of cone:** $V = \frac{\pi h}{3}(R^2 + Rr + r^2)$ where R, r are radii of top/bottom
- **Frustum of pyramid:** $V = \frac{h}{3}(B_1 + \sqrt{B_1 B_2} + B_2)$ where B_1, B_2 are base areas

How:

- **First Moves:**
  1. Sketch the solid and identify all component shapes
  2. Calculate volumes of individual components using standard formulas
  3. For additive decomposition: $V_{\text{total}} = \sum V_{\text{components}}$
  4. For subtractive decomposition: $V_{\text{remaining}} = V_{\text{outer}} - V_{\text{removed}}$
- **Bailouts:** If decomposition unclear, try integration with cross-sections; use Cavalieri's principle for equal cross-section areas; consider coordinate geometry approach
- **Pitfalls:** Incorrect frustum formula application; missing overlap regions in additive cases; sign errors in subtraction; wrong axis of revolution; double-counting shared volumes
- **Example:** A hemisphere of radius 3 with a cylindrical hole of radius 1 drilled through its center. Volume $= \frac{2}{3}\pi(3)^3 - \pi(1)^2(6) = 18\pi - 6\pi = 12\pi$.

3.42 Technique M32: MediantBound (Farey Sequence Bounds)

Short Name: TM32-MediantBound

Definition 3.42 (Mediant Bound)**.** *For fractions $\frac{a}{b}$ and $\frac{c}{d}$ with $\gcd(a, b) = \gcd(c, d) = 1$, the mediant $\frac{a+c}{b+d}$ lies strictly between them and has minimal denominator among all fractions in the interval $(\frac{a}{b}, \frac{c}{d})$ when $\frac{a}{b}$ and $\frac{c}{d}$ are Farey neighbors.*

WHEN: Adjacent fraction bounds minimizing denominators; rational approximation problems requiring the "simplest" fraction in an interval; Farey sequence applications; continued fraction convergents.

WHO: NT Farey-bound analysts working with rational approximations, Diophantine inequalities, or best rational bounds.

Configuration cues:

- Finding rational numbers between given bounds with minimal denominator
- Approximating irrationals with controlled denominator growth
- Farey sequence neighbors and Ford circle tangencies

Constraints cues:

- Given fractions must be in reduced form ($\gcd(a, b) = \gcd(c, d) = 1$)
- For minimal denominator property, fractions should be Farey neighbors
- Target value must lie in the open interval $(\frac{a}{b}, \frac{c}{d})$

Objective cues:

- Find the fraction with smallest denominator in a given interval
- Approximate irrational numbers with rationals of bounded denominator
- Construct Farey sequences or continued fraction convergents

Why It Works: The mediant preserves the ordering (lies between the bounds) and has minimal denominator among all fractions in the interval when the bounds are Farey neighbors. This follows from the fundamental property that $\gcd(a + c, b + d) = 1$ when $\gcd(a, b) = \gcd(c, d) = 1$ and $ad - bc = \pm 1$ (Farey condition).

How:

- **First Moves:** Given bounds $\frac{a}{b} < x < \frac{c}{d}$, compute mediant $\frac{a+c}{b+d}$
- **Iterate:** Continue taking mediants to refine bounds as needed
- **Bailouts:** Use continued fractions for irrational targets; apply Stern-Brocot tree navigation
- **Pitfalls:** The minimal denominator property requires Farey neighbor condition; without this, other fractions may have smaller denominators

3.43 Technique M33: GCD-LCM Bounds

Short Name: TM33-GCDLCMBounds

Definition 3.43 (GCD-LCM Relationship Bounds)**.** *For integers a, b with $\gcd(a, b) = g$ and $\text{lcm}(a, b) = L$, we have $ab = gL$. This constrains the possible values of a and b when g and L are known, particularly for consecutive divisors or pairs with shared prime factors.*

WHEN: Neighbor divisors need shared primes; constraining pairs of integers via their GCD/LCM relationship; divisibility screening in number theory problems.

WHO: NT divisibility screeners analyzing relationships between divisors, particularly when prime factorizations interact.

Configuration cues:

- Problems involving pairs of numbers with specified GCD or LCM
- Divisor chains where consecutive terms share factors
- Optimization under GCD/LCM constraints

Constraints cues:

- Given values for $\gcd(a, b)$ and $\text{lcm}(a, b)$ must be consistent
- For consecutive divisors, they must share at least one prime factor
- The relationship $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$ must hold

Objective cues:

- Find all pairs (a, b) with given GCD and LCM
- Determine constraints on consecutive divisors in a sequence
- Optimize expressions involving GCD/LCM relationships

Why It Works: The fundamental identity $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$ provides a constraint that reduces the search space. When combined with prime factorization analysis, this allows systematic enumeration of valid pairs and identification of shared prime factors in consecutive divisors.

How:

- **First Moves:** Use $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$ to establish bounds
- **Factor analysis:** Express via prime factorizations to see min/max relationships
- **Bailouts:** Check small cases; use divisor lattice structure for systematic enumeration
- **Pitfalls:** Not all combinations of GCD and LCM are achievable; need to verify consistency

3.44 Technique M34: Cofunction Solving

Short Name: TM34-CofunctionSolve

Definition 3.44 (Cofunction Identity for Equation Solving)**.** *When trigonometric equations involve sine and cosine with complementary arguments, use the cofunction identity* $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$ *to convert the equation to a form involving only sine (or only cosine), often reducing to manageable forms like* $\sin x + \cos x = k$*.*

WHEN: Equations of the form $\sin(f(x)) = \cos(g(x))$ where $f(x) + g(x)$ simplifies nicely (often to $\frac{\pi}{2}$); equations with mixed sine/cosine that can be converted using complementary angle relationships; problems where arguments naturally sum to $\frac{\pi}{2}$ or multiples thereof; trig equations that resist standard algebraic approaches but have complementary structure.

Configuration cues:

- Equations like $\sin\left(\frac{\pi}{2}\cos x\right) = \cos\left(\frac{\pi}{2}\sin x\right)$
- Mixed sine/cosine equations where arguments are related by $\frac{\pi}{2}$
- Problems where $\sin(\alpha) = \cos(\beta)$ and $\alpha + \beta$ has a nice form
- Trig equations with nested compositions involving complementary angles
- Equations of form $\sin(90^\circ - x) = \cos(x)$ type relationships
- Problems asking to solve $a\sin x + b\cos x = c$ after transformation

Constraints cues:

- Check domain restrictions after applying cofunction identity
- When converting $\sin\theta = \sin\phi$, remember both cases: $\theta = \phi + 2n\pi$ and $\theta = \pi - \phi + 2n\pi$ for integer n
- Verify solutions against original equation (squaring may introduce extraneous roots)
- Consider range constraints when using inverse trig functions

Objective cues:

- Reduce complex trig equation to simpler form like $\sin x + \cos x = k$
- Eliminate one trig function in favor of another
- Convert nested trig compositions to single-variable equations
- Solve for all values in a given interval

Why It Works: The cofunction identity exploits the complementary relationship between sine and cosine, allowing conversion between the two functions when their arguments sum to $\frac{\pi}{2}$. This often simplifies complex equations dramatically. The key insight is that $\sin x + \cos x = \sqrt{2}\sin(x + \frac{\pi}{4})$, which has known range and periodicity.

WHO: ALG/GEO trig equation resolvers comfortable with complementary angle relationships and function composition.

How:

- **First Moves:**
  1. Apply cofunction identity: $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$ or $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$
  2. Rewrite equation with single trig function
  3. For $\sin x + \cos x = k$ forms, recall: $\sin x + \cos x = \sqrt{2}\sin(x + \frac{\pi}{4}) = \sqrt{2}\cos(x - \frac{\pi}{4})$
  4. Use standard solving techniques for resulting equation
  5. Back-substitute and verify all solutions
- **Alternative Approaches:**
  - **Direct substitution:** For $\sin(f(x)) = \cos(g(x))$, use $\cos(g(x)) = \sin(\frac{\pi}{2} - g(x))$ to get $\sin(f(x)) = \sin(\frac{\pi}{2} - g(x))$
  - **Linear combination method:** For $a\sin x + b\cos x = c$, use $a\sin x + b\cos x = \sqrt{a^2 + b^2}\sin(x + \alpha)$ where $\tan\alpha = \frac{b}{a}$
  - **Weierstrass substitution:** For complex rational trig equations, use $t = \tan\frac{x}{2}$ with $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$
  - **Graphical analysis:** Plot both sides to identify intersection points and verify algebraically
- **Range Analysis:** When dealing with $\sin x \pm \cos x = k$:
  - $\sin x + \cos x = \sqrt{2}\sin(x + \frac{\pi}{4}) \in [-\sqrt{2}, \sqrt{2}]$ (maximum at $x = \frac{\pi}{4} + 2n\pi$)
  - $\sin x - \cos x = \sqrt{2}\sin(x - \frac{\pi}{4}) \in [-\sqrt{2}, \sqrt{2}]$ (maximum at $x = \frac{3\pi}{4} + 2n\pi$)
  - $\cos x - \sin x = \sqrt{2}\cos(x + \frac{\pi}{4}) \in [-\sqrt{2}, \sqrt{2}]$ (maximum at $x = -\frac{\pi}{4} + 2n\pi$)
  - For $a\sin x + b\cos x = c$: range is $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$ when $a, b \neq 0$
  - Always check if $|k| \leq \sqrt{2}$ for $\sin x \pm \cos x = k$ to determine if solutions exist
- **Bailouts:** If cofunction doesn't simplify, try sum-to-product formulas; parametric substitution; graphical analysis
- **Pitfalls:**
  - Missing solutions when $\sin\theta = \sin\phi$ (remember both $\theta = \phi + 2n\pi$ and $\theta = \pi - \phi + 2n\pi$)
  - Introducing extraneous solutions when squaring (always verify against original equation)

  - Incorrect handling of domain/range, especially with inverse trig functions
  - Forgetting that $\sin x + \cos x$ has maximum $\sqrt{2}$, not 2
  - Not considering all cases when arguments don't naturally sum to $\frac{\pi}{2}$

- **Example:** See AMC 2021 12A Problem 19 solution below

Example: Cofunction Solving

How many solutions does $\sin\left(\frac{\pi}{2}\cos x\right) = \cos\left(\frac{\pi}{2}\sin x\right)$ have in $[0, \pi]$?

Solution

Using the cofunction identity $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$:

$$\sin\left(\frac{\pi}{2}\cos x\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{2}\sin x\right)$$

Since $\sin\theta = \sin\phi$ implies either:

- Case 1: $\theta = \phi + 2n\pi$, giving us:

$$\frac{\pi}{2}\cos x = \frac{\pi}{2} - \frac{\pi}{2}\sin x + 2n\pi$$

$$\cos x + \sin x = 1 + 4n$$

  Since $\sin x + \cos x = \sqrt{2}\sin(x + \frac{\pi}{4}) \in [-\sqrt{2}, \sqrt{2}]$, we have $-\sqrt{2} \leq 1 + 4n \leq \sqrt{2}$. This requires $n = 0$, giving $\sin x + \cos x = 1$.

  Squaring: $\sin^2 x + 2\sin x\cos x + \cos^2 x = 1$

  Since $\sin^2 x + \cos^2 x = 1$: $1 + 2\sin x\cos x = 1 \Rightarrow \sin(2x) = 0$

  This gives $2x = k\pi$ for integer k, so $x = 0, \frac{\pi}{2}, \pi$ in $[0, \pi]$

  Checking original constraint $\sin x + \cos x = 1$: - $x = 0$: $\sin 0 + \cos 0 = 0 + 1 = 1$ ✓ - $x = \frac{\pi}{2}$: $\sin\frac{\pi}{2} + \cos\frac{\pi}{2} = 1 + 0 = 1$ ✓ - $x = \pi$: $\sin\pi + \cos\pi = 0 + (-1) = -1$ ×

- Case 2: $\theta = \pi - \phi + 2n\pi$, giving us:

$$\frac{\pi}{2}\cos x = \pi - \left(\frac{\pi}{2} - \frac{\pi}{2}\sin x\right) + 2n\pi$$

$$\frac{\pi}{2}\cos x = \pi - \frac{\pi}{2} + \frac{\pi}{2}\sin x + 2n\pi$$

$$\frac{\pi}{2}\cos x = \frac{\pi}{2} + \frac{\pi}{2}\sin x + 2n\pi$$

$$\cos x - \sin x = 1 + 4n$$

  Since $\cos x - \sin x = \sqrt{2}\cos(x + \frac{\pi}{4}) \in [-\sqrt{2}, \sqrt{2}]$, we have $-\sqrt{2} \leq 1 + 4n \leq \sqrt{2}$. This requires $n = 0$, giving $\cos x - \sin x = 1$.

  Using $\cos x - \sin x = \sqrt{2}\cos(x + \frac{\pi}{4}) = 1$:

$$\cos(x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}} = \cos\left(\frac{\pi}{4}\right)$$

  This gives $x + \frac{\pi}{4} = \pm\frac{\pi}{4} + 2k\pi$ for integer k, so: - $x + \frac{\pi}{4} = \frac{\pi}{4} + 2k\pi \Rightarrow x = 2k\pi$ - $x + \frac{\pi}{4} = -\frac{\pi}{4} + 2k\pi \Rightarrow x = -\frac{\pi}{2} + 2k\pi$

  In $[0, \pi]$: $x = 0$ (from first case) and $x = \frac{3\pi}{2}$ (from second case, but $\frac{3\pi}{2} > \pi$).

  Verifying $x = 0$: $\cos 0 - \sin 0 = 1 - 0 = 1$ ✓

Combining both cases: $x \in \{0, \frac{\pi}{2}\}$.
Answer: 2 solutions

Example: Linear Combination Method

Solve $\sin x + \cos x = 1$ for $x \in [0, 2\pi]$.

Solution

Using the linear combination method:

$$\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right) = 1$$

So $\sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \sin\left(\frac{\pi}{4}\right)$.
This gives:

- $x + \frac{\pi}{4} = \frac{\pi}{4} + 2n\pi \Rightarrow x = 2n\pi$
- $x + \frac{\pi}{4} = \pi - \frac{\pi}{4} + 2n\pi = \frac{3\pi}{4} + 2n\pi \Rightarrow x = \frac{\pi}{2} + 2n\pi$

In $[0, 2\pi]$: $x = 0, \frac{\pi}{2}, 2\pi$.
Verifying:

- $x = 0$: $\sin 0 + \cos 0 = 0 + 1 = 1$ ✓
- $x = \frac{\pi}{2}$: $\sin\frac{\pi}{2} + \cos\frac{\pi}{2} = 1 + 0 = 1$ ✓
- $x = 2\pi$: $\sin 2\pi + \cos 2\pi = 0 + 1 = 1$ ✓

Answer: $x = 0, \frac{\pi}{2}, 2\pi$

Example: When Cofunction Doesn't Apply

Solve $\sin(2x) = \cos(3x)$ for $x \in [0, \pi]$.

Solution

Since $2x+3x = 5x \neq \frac{\pi}{2}$ (in general), cofunction identity doesn't directly apply. Instead, use:

$$\cos(3x) = \sin\left(\frac{\pi}{2} - 3x\right)$$

So $\sin(2x) = \sin\left(\frac{\pi}{2} - 3x\right)$.
This gives:

- $2x = \frac{\pi}{2} - 3x + 2n\pi \Rightarrow 5x = \frac{\pi}{2} + 2n\pi \Rightarrow x = \frac{\pi}{10} + \frac{2n\pi}{5}$
- $2x = \pi - \left(\frac{\pi}{2} - 3x\right) + 2n\pi = \frac{\pi}{2} + 3x + 2n\pi \Rightarrow -x = \frac{\pi}{2} + 2n\pi \Rightarrow x = -\frac{\pi}{2} - 2n\pi$

In $[0, \pi]$: $x = \frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}$.
Verifying against original equation $\sin(2x) = \cos(3x)$:

- $x = \frac{\pi}{10}$: $\sin\left(\frac{\pi}{5}\right) = \cos\left(\frac{3\pi}{10}\right)$ Using cofunction: $\cos\left(\frac{3\pi}{10}\right) = \sin\left(\frac{\pi}{2} - \frac{3\pi}{10}\right) = \sin\left(\frac{\pi}{5}\right)$ ✓
- $x = \frac{\pi}{2}$: $\sin\pi = \cos\left(\frac{3\pi}{2}\right) \Rightarrow 0 = 0$ ✓
- $x = \frac{9\pi}{10}$: $\sin\left(\frac{9\pi}{5}\right) = \cos\left(\frac{27\pi}{10}\right)$ Using cofunction: $\cos\left(\frac{27\pi}{10}\right) = \sin\left(\frac{\pi}{2} - \frac{27\pi}{10}\right) = \sin\left(\frac{\pi}{2} - \frac{27\pi}{10} + 2\pi\right) = \sin\left(\frac{\pi}{2} - \frac{7\pi}{10}\right) = \sin\left(\frac{9\pi}{5}\right)$ ✓

Answer: $x = \frac{\pi}{10}, \frac{\pi}{2}, \frac{9\pi}{10}$

3.45 Technique M35: Constructive Polynomial Example

Short Name: TM35-ConstructivePolyExample

Definition 3.45 (Constructive Polynomial Building)**.** *Given constraints on degree, coefficients, and behavior (remainders upon division, values at points, or root properties), explicitly construct a polynomial satisfying all conditions. Often the minimal degree polynomial is unique.*

WHEN: Configuration: problems requiring a specific polynomial with prescribed remainders when divided by given polynomials; minimal degree polynomials satisfying multiple congruences; existence proofs via explicit construction; Constraints: degree bounds; coefficient restrictions (integer, real, or specific values); remainder conditions modulo various divisors; Objective: build the unique or minimal polynomial meeting all requirements; prove existence by construction; find sum of coefficients or evaluate at specific points. Use for "find the polynomial of least degree such that..."; Chinese Remainder Theorem for polynomials; interpolation with side constraints; problems where $P(x) \equiv r_1(x) \pmod{q_1(x)}$ and $P(x) \equiv r_2(x) \pmod{q_2(x)}$.

Configuration cues:

- "Find the polynomial of minimal/least degree satisfying..."
- Multiple remainder conditions when dividing by different polynomials

- Constraints on both degree and specific coefficients (especially constant term)
- Existence questions answered by explicit construction

Constraints cues:

- Minimal degree requirement forces unique or highly constrained solutions
- Remainder polynomials must have degree less than divisor polynomials
- When divisors are coprime, use polynomial Chinese Remainder Theorem
- Integer coefficient requirements may further constrain the solution

Objective cues:

- Construct the explicit polynomial satisfying all conditions
- Prove uniqueness of the minimal degree solution
- Compute specific values or coefficient properties of the constructed polynomial

Why It Works: Systematic construction ensures all constraints are met while maintaining minimal degree, often yielding a unique solution.

WHO: ALG polynomial construction specialists; solvers comfortable with undetermined coefficients, polynomial division, and the polynomial Chinese Remainder Theorem.

How:

- **First Moves:**
  1. Write $P(x) = Q_1(x) \cdot D_1(x) + R_1(x)$ for each divisor-remainder pair.
  2. For minimal degree, assume quotients are linear (or constant if possible).
  3. Set up the system: if $P(x) = (x^2 + x + 1)Q_1(x) + (x + 2)$ and $P(x) = (x^2 + 1)Q_2(x) + (2x + 1)$.
  4. Use undetermined coefficients: let $Q_1(x) = ax + b$, $Q_2(x) = ax + c$ (same leading coefficient).
  5. Solve by comparing coefficients or evaluating at strategic points (0, 1, -1).
- **Bailouts:** If linear quotients don't work, try quadratic; use polynomial CRT if divisors are coprime; for uniqueness, show degree bound forces the solution.
- **Pitfalls:** Assuming quotients have different leading coefficients when they must match; forgetting to verify all original constraints; missing the uniqueness argument.
- **Example:** Finding $P(x)$ with $P(x) \equiv x + 2 \pmod{x^2 + x + 1}$ and $P(x) \equiv 2x + 1 \pmod{x^2 + 1}$ yields unique minimal $P(x) = x^3 + 2x^2 + 3x + 3$.

Example: Minimal Polynomial with Two Remainder Conditions

Find the polynomial $P(x)$ of minimal degree such that when $P(x)$ is divided by $x^2 + x + 1$, the remainder is $x + 2$, and when $P(x)$ is divided by $x^2 + 1$, the remainder is $2x + 1$.

Solution

Let $P(x) = (x^2 + x + 1)Q_1(x) + (x + 2) = (x^2 + 1)Q_2(x) + (2x + 1)$.
For minimal degree, assume $Q_1(x) = ax+b$ and $Q_2(x) = ax+c$ (same leading coefficient for $P(x)$ to be well-defined).
Expanding: $P(x) = ax^3 + (a + b)x^2 + (a + b)x + (b + 2)$ Also: $P(x) = ax^3 + cx^2 + (2 + a)x + (c + 1)$
Comparing coefficients: - x^2: $a + b = c$ - x^1: $a + b = 2 + a \Rightarrow b = 2$ - x^0: $b + 2 = c + 1 \Rightarrow c = b + 1 = 3$
From $a + b = c$: $a + 2 = 3 \Rightarrow a = 1$.
Therefore: $P(x) = (x^2 + x + 1)(x + 2) + (x + 2) = x^3 + 2x^2 + 3x + 3$.
Answer: $P(x) = x^3 + 2x^2 + 3x + 3$

3.46 Technique M36: Legendre Symbol and Quadratic Reciprocity

Short Name: TM36-LegendreSymbol

Definition 3.46 (Legendre Symbol)**.** *For prime p and integer a with $\gcd(a, p) = 1$, the Legendre symbol is:*

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p \end{cases}$$

The Legendre symbol is undefined when $p \mid a$.

Euler's Criterion: *For prime p and integer a with $\gcd(a, p) = 1$:*

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$$

Quadratic Reciprocity: *For distinct odd primes p and q:*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$$

Supplementary Laws:

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases} \tag{28}$$

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8} \\ -1 & \text{if } p \equiv \pm 3 \pmod{8} \end{cases} \tag{29}$$

Gauss's Lemma: *For prime p and integer a with $\gcd(a,p)=1$, let $S=\{a,2a,3a,\ldots,\frac{p-1}{2}a\}$ and let n be the number of elements in S whose least positive residue modulo p is greater than $p/2$. Then:*

$$\left(\frac{a}{p}\right)=(-1)^n$$

Connection to Trigonometry: *For distinct odd primes p and q:*

$$\frac{\sin(qc)\sin(2qc)\cdots\sin(\frac{p-1}{2}qc)}{\sin(c)\sin(2c)\cdots\sin(\frac{p-1}{2}c)}=\left(\frac{q}{p}\right)$$

where $c=\frac{2\pi}{p}$.

WHEN: Configuration: products of trigonometric expressions with arguments in arithmetic progression modulo primes; quadratic residue determination; Gauss sum evaluations; Constraints: prime moduli; expressions involving $\sin(k\pi/p)$ or $\cos(k\pi/p)$ products; questions about solvability of $x^2\equiv a \pmod p$; Objective: evaluate trigonometric products via number theory; determine quadratic character; simplify expressions using reciprocity laws. Use for products like $\prod_{k=1}^{(p-1)/2}\sin(k\pi/p)$; determining if a is a square modulo p; Eisenstein-style trigonometric identities; simplifying ratios of sine/cosine products.

Configuration cues:

- Products of sines/cosines with arguments that are multiples of π/p or $2\pi/p$ for prime p
- Questions asking whether $x^2\equiv a \pmod p$ has solutions
- Ratios of trigonometric products that telescope or simplify via number theory

Constraints cues:

- Prime modulus required for Legendre symbol; composite moduli use Jacobi symbol
- Legendre symbol undefined when $\gcd(a,p)\neq 1$ (i.e., when $p\mid a$)
- For trigonometric applications, arguments must form arithmetic progressions modulo the period
- Quadratic reciprocity only applies to distinct odd primes
- Gauss's lemma most effective for small primes or when other methods are cumbersome

Objective cues:

- Evaluate complex trigonometric products without direct computation
- Determine quadratic residue status efficiently
- Convert geometric/trigonometric problems to number-theoretic ones

Why It Works: Connects algebraic properties of quadratic equations modulo primes to geometric properties of regular polygons and roots of unity.

WHO: NT/ALG specialists comfortable with quadratic reciprocity, Gauss sums, and connections between number theory and trigonometry; problems involving regular polygon geometry or cyclotomic polynomials.

How:

- **First Moves:**
  1. Check that $\gcd(a, p) = 1$; if not, the Legendre symbol is undefined.
  2. For trigonometric products, identify the prime p and rewrite using $c = 2\pi/p$.
  3. Try supplementary laws first: $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.
  4. For other values, use quadratic reciprocity: $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right) \cdot (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$.
  5. Apply Euler's criterion: $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.
  6. For small primes, consider Gauss's lemma as an alternative.
- **Bailouts:** If direct evaluation is complex, use Euler's criterion; for composite moduli, switch to Jacobi symbol; for trigonometric applications, use the connection formula.
- **Pitfalls:** Forgetting the (-1) factors in reciprocity; misapplying to composite moduli; sign errors in supplementary laws; assuming Legendre symbol is defined when $\gcd(a, p) \neq 1$.
- **Example:** $\left(\frac{3}{11}\right) = \left(\frac{11}{3}\right) \cdot (-1)^{(11-1)(3-1)/4} = \left(\frac{2}{3}\right) \cdot (-1)^5 = (-1) \cdot (-1) = 1$ (since $2 \equiv -1 \pmod{3}$ and $(-1)^{(3-1)/2} = -1$).

Example: Trigonometric Product via Legendre Symbol

Evaluate the product $\prod_{k=1}^{5} \sin\left(\frac{k\pi}{11}\right)$ using the connection to Legendre symbols.

Solution

We can use the connection between Legendre symbols and trigonometric products. For prime $p = 11$, we have the identity:

$$\prod_{k=1}^{5} \sin\left(\frac{k\pi}{11}\right) = \frac{\sqrt{11}}{2^5}$$

This follows from the general formula for prime p:

$$\prod_{k=1}^{(p-1)/2} \sin\left(\frac{k\pi}{p}\right) = \frac{\sqrt{p}}{2^{(p-1)/2}}$$

For $p = 11$:

$$\prod_{k=1}^{5} \sin\left(\frac{k\pi}{11}\right) = \frac{\sqrt{11}}{2^5} = \frac{\sqrt{11}}{32}$$

Verification using Legendre symbols: The connection to Legendre symbols comes through Gauss sums. For a prime p, the product of sines is related to the quadratic character of the field $\mathbb{F}_p$. Specifically, the product $\prod_{k=1}^{(p-1)/2} \sin(k\pi/p)$ can be expressed in terms of the Legendre symbol $\left(\frac{-1}{p}\right)$ and other quadratic characters.

For $p = 11$, we have $\left(\frac{-1}{11}\right) = (-1)^{(11-1)/2} = (-1)^5 = -1$, which confirms that 11 is not a sum of two squares, and the product formula gives us the stated result.

Answer: $\frac{\sqrt{11}}{32}$

Example: Legendre Symbol Calculation

Compute $\left(\frac{17}{23}\right)$ using quadratic reciprocity.

Solution

We'll use quadratic reciprocity to compute $\left(\frac{17}{23}\right)$.
Since both 17 and 23 are distinct odd primes, we can apply quadratic reciprocity:

$$\left(\frac{17}{23}\right)\left(\frac{23}{17}\right) = (-1)^{\frac{17-1}{2}\cdot\frac{23-1}{2}} = (-1)^{8\cdot 11} = (-1)^{88} = 1$$

Therefore: $\left(\frac{17}{23}\right) = \left(\frac{23}{17}\right)$
Now we need to compute $\left(\frac{23}{17}\right)$. Since $23 \equiv 6 \pmod{17}$:

$$\left(\frac{23}{17}\right) = \left(\frac{6}{17}\right) = \left(\frac{2}{17}\right)\left(\frac{3}{17}\right)$$

Using the supplementary law for 2:

$$\left(\frac{2}{17}\right) = (-1)^{(17^2-1)/8} = (-1)^{(289-1)/8} = (-1)^{36} = 1$$

since $17 \equiv 1 \pmod 8$.
For $\left(\frac{3}{17}\right)$, we use quadratic reciprocity again:

$$\left(\frac{3}{17}\right)\left(\frac{17}{3}\right) = (-1)^{\frac{3-1}{2}\cdot\frac{17-1}{2}} = (-1)^{1\cdot 8} = 1$$

So $\left(\frac{3}{17}\right) = \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right)$ since $17 \equiv 2 \pmod 3$.
Since $2 \equiv -1 \pmod 3$ and $(-1)^{(3-1)/2} = (-1)^1 = -1$:

$$\left(\frac{2}{3}\right) = -1$$

Therefore:

$$\left(\frac{17}{23}\right) = \left(\frac{23}{17}\right) = \left(\frac{2}{17}\right)\left(\frac{3}{17}\right) = 1\cdot(-1) = -1$$

Answer: -1 (17 is a quadratic non-residue modulo 23)

3.47 Technique M37: Constructive Polynomial Example

Short Name: TM37-ConstructivePolyExample

Definition 3.47 (Constructive Polynomial Building)**.** *Given constraints on degree, coefficients, and behavior (remainders upon division, values at points, or root properties), explicitly construct a polynomial satisfying all conditions. Often the minimal degree polynomial is unique.*

WHEN: Configuration: problems requiring a specific polynomial with prescribed remainders when divided by given polynomials; minimal degree polynomials satisfying mul-

tiple congruences; existence proofs via explicit construction; Constraints: degree bounds; coefficient restrictions (integer, real, or specific values); remainder conditions modulo various divisors; Objective: build the unique or minimal polynomial meeting all requirements; prove existence by construction; find sum of coefficients or evaluate at specific points. Use for "find the polynomial of least degree such that..."; Chinese Remainder Theorem for polynomials; interpolation with side constraints; problems where $P(x) \equiv r_1(x) \pmod{q_1(x)}$ and $P(x) \equiv r_2(x) \pmod{q_2(x)}$.

Configuration cues:

- "Find the polynomial of minimal/least degree satisfying..."
- Multiple remainder conditions when dividing by different polynomials
- Constraints on both degree and specific coefficients (especially constant term)
- Existence questions answered by explicit construction

Constraints cues:

- Minimal degree requirement forces unique or highly constrained solutions
- Remainder polynomials must have degree less than divisor polynomials
- When divisors are coprime, use polynomial Chinese Remainder Theorem
- Integer coefficient requirements may further constrain the solution

Objective cues:

- Construct the explicit polynomial satisfying all conditions
- Prove uniqueness of the minimal degree solution
- Compute specific values or coefficient properties of the constructed polynomial

Why It Works: Systematic construction ensures all constraints are met while maintaining minimal degree, often yielding a unique solution.

WHO: ALG polynomial construction specialists; solvers comfortable with undetermined coefficients, polynomial division, and the polynomial Chinese Remainder Theorem.

How:

- **First Moves:**
  1. Write $P(x) = Q_1(x) \cdot D_1(x) + R_1(x)$ for each divisor-remainder pair.
  2. For minimal degree, assume quotients are linear (or constant if possible).
  3. Set up the system: if $P(x) = (x^2 + x + 1)Q_1(x) + (x + 2)$ and $P(x) = (x^2 + 1)Q_2(x) + (2x + 1)$.
  4. Use undetermined coefficients: let $Q_1(x) = ax + b$, $Q_2(x) = ax + c$ (same leading coefficient).
  5. Solve by comparing coefficients or evaluating at strategic points (0, 1, -1).
  6. Verify the constructed polynomial satisfies all original constraints.

- **Bailouts:** If linear quotients don't work, try quadratic; use polynomial CRT if divisors are coprime; for uniqueness, show degree bound forces the solution; if coefficients become unwieldy, evaluate at special points to simplify.
- **Pitfalls:** Assuming quotients have different leading coefficients when they must match; forgetting to verify all original constraints; missing the uniqueness argument; not checking that remainder degrees are valid.
- **Example:** Finding $P(x)$ with $P(x) \equiv x + 2 \pmod{x^2 + x + 1}$ and $P(x) \equiv 2x + 1 \pmod{x^2 + 1}$ yields unique minimal $P(x) = x^3 + 2x^2 + 3x + 3$.

Example: Minimal Polynomial with Two Remainder Conditions

Find the polynomial $P(x)$ of minimal degree such that when $P(x)$ is divided by $x^2 + x + 1$, the remainder is $x + 2$, and when $P(x)$ is divided by $x^2 + 1$, the remainder is $2x + 1$.

Solution

Step 1: Set up the structure. Let $P(x) = (x^2+x+1)Q_1(x)+(x+2) = (x^2+1)Q_2(x)+(2x+1)$.
Step 2: Assume minimal degree quotients. For minimal degree, try $Q_1(x) = ax+b$ and $Q_2(x) = cx+d$. Since both expressions equal $P(x)$, the leading coefficients must match, so $a = c$.
Step 3: Expand both forms.

$$\begin{aligned} P(x) &= (x^2+x+1)(ax+b)+(x+2) && (30)\\ &= ax^3+bx^2+ax^2+bx+ax+b+x+2 && (31)\\ &= ax^3+(a+b)x^2+(a+b+1)x+(b+2) && (32) \end{aligned}$$

$$\begin{aligned} P(x) &= (x^2+1)(ax+d)+(2x+1) && (33)\\ &= ax^3+dx^2+ax+d+2x+1 && (34)\\ &= ax^3+dx^2+(a+2)x+(d+1) && (35) \end{aligned}$$

Step 4: Compare coefficients.

- x^2 coefficient: $a+b = d$
- x^1 coefficient: $a+b+1 = a+2 \Rightarrow b = 1$
- x^0 coefficient: $b+2 = d+1 \Rightarrow d = b+1 = 2$

From $a+b = d$: $a+1 = 2 \Rightarrow a = 1$.
Step 5: Construct the polynomial. $P(x) = (x^2+x+1)(x+1)+(x+2) = x^3+x^2+x^2+x+x+1+x+2 = x^3+2x^2+3x+3$
Step 6: Verify.

- $(x^3+2x^2+3x+3) = (x^2+x+1)(x+1)+(x+2)$ ✓
- $(x^3+2x^2+3x+3) = (x^2+1)(x+2)+(2x+1)$ ✓

Answer: $P(x) = x^3+2x^2+3x+3$

Example: Polynomial with Fixed Values

Find the polynomial $P(x)$ of degree at most 3 such that $P(0) = 1$, $P(1) = 3$, $P(2) = 7$, and $P(3) = 13$.

Solution

Method 1: Undetermined Coefficients Let $P(x) = ax^3 + bx^2 + cx + d$.
From the conditions:

$$P(0) = d = 1 \tag{36}$$
$$P(1) = a + b + c + d = 3 \tag{37}$$
$$P(2) = 8a + 4b + 2c + d = 7 \tag{38}$$
$$P(3) = 27a + 9b + 3c + d = 13 \tag{39}$$

Substituting $d = 1$:

$$a + b + c = 2 \tag{40}$$
$$8a + 4b + 2c = 6 \Rightarrow 4a + 2b + c = 3 \tag{41}$$
$$27a + 9b + 3c = 12 \Rightarrow 9a + 3b + c = 4 \tag{42}$$

Solving this system:

- From equations (1) and (2): $3a + b = 1$
- From equations (2) and (3): $5a + b = 1$
- Therefore: $2a = 0 \Rightarrow a = 0$
- Then $b = 1$ and $c = 1$

Method 2: Lagrange Interpolation Since we need exactly degree 3 or less with 4 points, and the solution from Method 1 gives degree 2, we have:
$P(x) = x^2 + x + 1$
Verification:

- $P(0) = 0 + 0 + 1 = 1$ ✓
- $P(1) = 1 + 1 + 1 = 3$ ✓
- $P(2) = 4 + 2 + 1 = 7$ ✓
- $P(3) = 9 + 3 + 1 = 13$ ✓

Answer: $P(x) = x^2 + x + 1$

3.48 Technique M38: Pell Equation

Short Name: TM38-PellEquation

Definition 3.48 (Pell Equation and Quadratic Diophantine Equations)**.** *The Pell equation $x^2 - Dy^2 = 1$ (where D is a non-square positive integer) and related forms $x^2 - Dy^2 = N$ appear when:*

- *Figurate numbers (triangular, pentagonal) must also be perfect squares*
- *Rational approximations to $\sqrt{D}$ are needed*
- *Recurrence relations arise from quadratic constraints*

Solutions connect to continued fractions of $\sqrt{D}$, with fundamental solution generating all others via $(x_1 + y_1\sqrt{D})^n$.

WHEN: When figurate numbers must also satisfy square conditions, revealing Pell-type recurrences; when triangular numbers $t_n = \frac{n(n+1)}{2} = k^2$ leads to $(2n+1)^2 - 8k^2 = 1$; when pentagonal or hexagonal numbers intersect with squares; when finding integer solutions to $ax^2 + bxy + cy^2 = d$ reduces to Pell form; when rational approximations with controlled denominators are needed.

WHO: NT analysts fluent in quadratic Diophantine structure and continued fractions; solvers comfortable with recurrence generation from fundamental solutions and recognizing when quadratic forms reduce to Pell equations.

Configuration cues:

- Triangular numbers that are perfect squares: $\frac{n(n+1)}{2} = k^2$
- Pentagonal numbers that are squares: $\frac{n(3n-1)}{2} = k^2$
- Problems asking for the n-th figurate number with a special property
- Quadratic Diophantine equations $x^2 - Dy^2 = N$ with small $|N|$
- Sequences where a_{n+1} relates to a_n via a quadratic recurrence

Constraints cues:

- D must not be a perfect square (else trivial or no solutions)
- For negative Pell ($x^2 - Dy^2 = -1$), solutions exist only for specific D
- Initial solution generates all others multiplicatively
- Continued fraction period determines solution structure

Objective cues:

- Find all integer solutions to a quadratic Diophantine equation
- Determine the n-th figurate number with a square property
- Establish non-existence of solutions for certain forms
- Generate recurrence relations for special number sequences

Why It Works: The multiplicative structure of solutions to Pell equations, combined with continued fraction periodicity, provides a complete characterization of all solutions. The fundamental solution acts as a generator for the infinite family of solutions.

How:

- **First Moves:**
  1. Transform the problem into standard Pell form $x^2 - Dy^2 = N$
  2. For triangular-square: $(2n+1)^2 - 8k^2 = 1$ with $D = 8$
  3. Find fundamental solution via continued fraction convergents of $\sqrt{D}$
  4. Generate larger solutions: if (x_0, y_0) is fundamental for $x^2 - Dy^2 = 1$, then $(x_n, y_n) = (x_0 + y_0\sqrt{D})^n$
- **Bailouts:** Use small cases to identify pattern; check if D is square (no non-trivial solutions); verify N divides $\gcd(x^2, Dy^2)$ for solvability
- **Pitfalls:** Missing the transformation to standard form; forgetting that negative Pell may have no solutions; not recognizing when to use conjugate solutions; calculation errors in large number multiplication
- **Example:** For triangular squares after $t_1 = 1, t_8 = 36, t_{49} = 1225$, the pattern follows from $(2n+1)^2 - 8k^2 = 1$ with fundamental solution $(3, 1)$. The fourth solution is $(3 - 2\sqrt{2})^4 = 577 - 408\sqrt{2}$, giving $n = 288$, $k = 204$, and $t_{288} = 41616$.

Example: Triangular Numbers that are Perfect Squares

Find the fourth smallest triangular number that is also a perfect square.

Solution

Step 1: Set up the equation. For $t_n = \frac{n(n+1)}{2} = k^2$, we get:

$$n(n+1) = 2k^2$$

Step 2: Transform to Pell form. Completing the square on the left:

$$(2n+1)^2 - 1 = 8k^2$$

$$(2n+1)^2 - 8k^2 = 1$$

This is a Pell equation with $D = 8$.
Step 3: The fundamental solution is $(x_1, y_1) = (3, 1)$ (meaning $2n + 1 = 3, k = 1$, so $n = 1$).
Step 4: Generate solutions using $(3 + 2\sqrt{2})^m$:

$$m = 1 : (3, 1) \to n = 1, t_1 = 1 = 1^2 \tag{43}$$
$$m = 2 : (17, 12) \to n = 8, t_8 = 36 = 6^2 \tag{44}$$
$$m = 3 : (99, 35) \to n = 49, t_{49} = 1225 = 35^2 \tag{45}$$
$$m = 4 : (577, 204) \to n = 288, t_{288} = 41616 = 204^2 \tag{46}$$

Answer: The fourth triangular square is $t_{288} = 41616$.

3.49 Technique M42: Focus-Directrix Projections

Short Name: TM42-FocusDirectrix

Definition 3.49 (Focus-Directrix Properties)**.** *A parabola is the locus of points P equidistant from a fixed point (focus F) and a fixed line (directrix ℓ):*

- ***Definition:*** *$PF = d(P, \ell)$ for all points P on the parabola*
- ***Standard form:*** *For parabola $y = ax^2$ with vertex at origin: - Focus: $F = (0, \frac{1}{4a})$ - Directrix: $y = -\frac{1}{4a}$ - General: $(x - h)^2 = 4p(y - k)$ has focus $(h, k + p)$ and directrix $y = k - p$*
- ***Reflection property:*** *Rays parallel to axis reflect through focus*
- ***Tangent property:*** *Tangent at P bisects the angle between the focal ray PF and the line through P parallel to the axis of symmetry*

WHEN: Directrix-foot constructions turn focus-distance equalities into solvable right triangles; parabola problems with tangent lines or optimization; when reflection properties of parabolas apply; locus problems involving equal distances; problems asking for parabola equations given geometric constraints.

WHO: GEO parabola specialists projecting to the directrix; students comfortable converting between different parabola representations and using the distance formula effectively.

Configuration cues:

- Parabola defined by focus and directrix
- Tangent lines to parabolas with angle conditions
- Reflection problems involving parabolic mirrors or paths
- Distance optimization on parabolas
- "Shortest path" problems involving parabola and external point
- Light ray or projectile problems with parabolic reflectors
- Locus of points equidistant from point and line
- Parabola problems with mixed constraints (point on parabola, tangent conditions)

Constraints cues:

- The axis of the parabola is perpendicular to the directrix
- Vertex lies midway between focus and directrix
- For vertex at origin with vertical axis: $x^2 = 4py$ where p is the directed distance from vertex to focus
- Focal parameter (latus rectum) has length $|4p|$

- All parabolas are similar (differ only by scaling and position)

Objective cues:

- Find equation of parabola from focus-directrix data
- Locate focus or directrix from parabola equation
- Determine tangent lines using reflection property
- Optimize distances using the equidistance property
- Find intersection points using distance conditions
- Prove collinearity or concurrency using focus properties

How:

- **First Moves:**
  1. Identify or establish coordinate system: - Place vertex at origin if possible - Align axis with coordinate axis - Or place directrix as $x = 0$ or $y = 0$
  2. Find focus F and directrix equation
  3. For point P on parabola: - Drop perpendicular to directrix, call foot D - Use $PF = PD$
  4. Apply reflection or tangent properties as needed
- **Basic Formulas and Properties:**
  - **Standard forms:** - Vertical parabola $(x-h)^2 = 4p(y-k)$: Focus at $(h, k+p)$, directrix $y = k-p$ - Horizontal parabola $(y-k)^2 = 4p(x-h)$: Focus at $(h+p, k)$, directrix $x = h-p$ - Vertex form $y = a(x-h)^2 + k$: Focus at $(h, k + \frac{1}{4a})$, directrix $y = k - \frac{1}{4a}$
  - **Distance formula application:** - Point (x, y) on parabola: $\sqrt{(x-h)^2 + (y-k-p)^2} = |y - (k-p)| = |y - k + p|$
  - **Parametric form:** $x = 2pt$, $y = pt^2$ (vertex at origin, opens upward)
- **Tangent and Reflection Properties:**
  - **Tangent at point (x_0, y_0):** - For vertical parabola $y = ax^2$: tangent is $y = 2ax_0 \cdot x - ax_0^2$, slope $= 2ax_0$ - For horizontal parabola $x = ay^2$: tangent is $x = 2ay_0 \cdot y - ay_0^2$, slope $= \frac{1}{2ay_0}$ - Parametric form: $x = 2pt$, $y = pt^2$ has tangent with slope t at point $(2pt, pt^2)$
  - **Reflection property:** - Incoming ray parallel to axis reflects to pass through focus - Angle of incidence = angle of reflection (with axis)
- **Problem-Solving Strategies:**

  - **Equation finding:** Use definition $PF = d(P, \text{directrix})$ with point (x, y)
  - **Tangent problems:** - Use reflection: tangent bisects angle between focal ray and vertical - Or use calculus: derivative gives slope
  - **Optimization:** - Shortest path from external point to focus via parabola - Use reflection to straighten the path
  - **Intersection problems:** - Set up distance equations - Use the fact that parabola points satisfy $PF = PD$

- **Common Applications:**
  - **Parabolic mirrors:** All parallel rays focus at F
  - **Projectile motion:** Path is parabola with vertical directrix
  - **Satellite dishes:** Signal collection at focus
  - **Bridge cables:** Approximately parabolic under uniform load

- **Advanced Properties:**
  - **Latus rectum:** Chord through focus perpendicular to axis has length $|4p|$
  - **Optical property:** All rays from focus reflect parallel to axis
  - **Polar form:** With focus at origin: $r = \frac{2p}{1-\cos\theta}$
  - **Coordinate-free:** Tangent at P bisects angle between focal ray PF and line through P parallel to axis
  - **Subnormal:** Projection of normal onto axis has constant length $2p$

- **Bailouts:**
  - If focus-directrix unclear, convert to vertex form first
  - Use parametric form: $x = 2pt$, $y = pt^2$ for computational ease
  - Apply coordinate geometry with distance formula directly
  - For tangent problems, use implicit differentiation
  - Consider special positions (vertex, endpoints of latus rectum)

- **Pitfalls:**
  - Confusing vertex form $y = a(x-h)^2 + k$ with standard form $(x-h)^2 = 4p(y-k)$
  - Sign errors: p is positive if focus is above directrix (for vertical parabola)
  - Forgetting that distance to directrix is perpendicular distance
  - Using wrong formula when parabola opens horizontally vs vertically
  - Mixing up focus-vertex distance (p) with focal parameter ($4p$)
  - Not recognizing when problem is asking for locus (which is a parabola)
  - Assuming vertex is at origin when it's not

- **Example:** A parabola has focus $F = (0, 3)$ and directrix $y = -1$. Find the equation of the parabola and determine the point where a ray parallel to the axis of symmetry reflects through the focus.

  Solution: - Vertex is at $(0, 1)$ (midpoint of focus and directrix) - Parameter $p = 2$, so equation is $x^2 = 8(y-1)$ or $x^2 = 8y-8$ - For a ray parallel to the axis (vertical ray), it reflects through the focus - Any vertical ray $x = c$ intersects the parabola at $(c, \frac{c^2+8}{8})$ - The tangent at this point has slope $\frac{c}{4}$ - By reflection property, the angle between the incoming vertical ray and the tangent equals the angle between the tangent and the line to the focus - This gives us the reflection point $(0, 1)$ where the vertical ray through the vertex reflects to the focus

Works Well With: T2-CoordGeo (setting up equations); T8-CompleteSquare (converting forms); Calculus techniques (for tangents); T17-GeoProb (for probability on parabolas); Optimization techniques.

Typical Difficulty: Problems 15-20 on AMC 12; more common in AIME. Focus-directrix definition rarely appears before Problem 15.

Special Cases:

- Parabola with horizontal axis (roles of x and y switch)
- Degenerate case: focus on directrix gives a line
- Confocal parabolas (same focus, different directrices)
- Orthoptic of parabola is the directrix

Success Rate: 70-80

Category Classification: Medium-Probability - Important for parabola problems but less frequent than vertex-form manipulation. Essential for certain optimization and reflection problems.

3.50 Technique M43: Curvature and Tangency Analysis

Short Name: TM43-CurvatureTangency

Definition 3.50 (Curvature-Based Tangency)**.** *Two curves are tangent at a point if they share the same tangent line (first derivatives equal) and optionally the same curvature (second derivatives equal for higher-order tangency).*

Order of Contact:

- ***First-order tangency:*** *At point (x_0, y_0): $f(x_0) = g(x_0)$ and $f'(x_0) = g'(x_0)$*
- ***Second-order tangency (osculation):*** *Additionally $f''(x_0) = g''(x_0)$*
- ***nth-order contact:*** *First n derivatives match: $f^{(k)}(x_0) = g^{(k)}(x_0)$ for $k = 0, 1, \ldots, n$*

WHEN: When derivative/curvature comparisons certify multiplicity of shared points for curves; determining tangency between curves; finding points of osculation; designing smooth transitions between curves.

WHO: ALG/CALC intersection analysts confirming tangency via derivatives or curvature; engineers ensuring continuity in design.

Configuration cues:

- Two curves touching at exactly one point
- Optimization problems with tangency constraints
- Envelope problems
- Osculating circles or higher-order contact
- Finding common tangent lines to multiple curves
- Tangency in polar coordinates (matching $r(\theta)$ and $\frac{dr}{d\theta}$)
- Space curves requiring matching of tangent vectors
- Tangent planes to surfaces (3D extension)
- Bezier curve continuity conditions
- Roller coaster or road design (continuous curvature)

Key Formulas:

- **Curvature (Cartesian):** $\kappa = \frac{|f''(x)|}{(1+(f'(x))^2)^{3/2}}$
- **Curvature (Parametric):** For $(x(t), y(t))$: $\kappa = \frac{|x'y''-y'x''|}{(x'^2+y'^2)^{3/2}}$
- **Tangency conditions:** $f(a) = g(a)$ (intersection), $f'(a) = g'(a)$ (tangency)
- **Osculation condition:** Additionally $f''(a) = g''(a)$
- **Implicit curves:** For $F(x, y) = 0$ and $G(x, y) = 0$: $\nabla F \parallel \nabla G$ at tangency
- **Discriminant for tangency:** For $f(x) = g(x)$, discriminant of $f(x)-g(x) = 0$ equals zero

How:

- **First Moves:**
  - Set functions equal for intersection points: $f(x) = g(x)$
  - Set first derivatives equal for tangency: $f'(x) = g'(x)$
  - Check second derivatives for osculation: $f''(x) = g''(x)$
  - For implicit curves $F(x, y) = 0$ and $G(x, y) = 0$: verify $\nabla F \parallel \nabla G$

  - For parametric curves: ensure parameter correspondence when comparing derivatives
  - Verify existence of derivatives at the point of interest
  - For vertical tangents: consider using dx/dy instead of dy/dx

- **Bailouts:**
  - Use discriminant method for algebraic curves
  - Parametric approach for complex curves
  - Implicit differentiation for curves not easily solved for y
  - Lagrange multipliers for constrained optimization with tangency
  - Numerical methods (Newton-Raphson) for transcendental equations
  - Geometric constructions for simple cases (e.g., tangent from external point to circle)

- **Pitfalls:**
  - Forgetting to verify that point lies on both curves
  - Missing multiple tangency points
  - Confusing geometric tangency with algebraic multiplicity
  - Not accounting for vertical tangents where dy/dx is undefined
  - Missing tangency at singular points, cusps, or self-intersections
  - Assuming tangency implies local non-intersection (curves can cross at tangent points)
  - In parametric curves: comparing derivatives at different parameter values
  - Overlooking complex or imaginary tangency points in algebraic curves

Common Applications:

- **Circle tangent to parabola:** Solve system with tangency conditions
- **Common tangent to two circles:** External and internal tangent lines
- **Line tangent to conic sections:** Use discriminant equals zero
- **Envelope of family of curves:** Eliminate parameter from $F(x, y, t) = 0$ and $\frac{\partial F}{\partial t} = 0$
- **Smooth path design:** Ensure C^1 continuity (tangency) or C^2 continuity (matching curvature)

Example: Find the line tangent to both the parabola $y = x^2$ and the circle $x^2 + y^2 = 1$.
Solution: Let the tangent line be $y = mx + b$. For tangency to both curves, we need:

1. Line tangent to parabola: The system $y = x^2$ and $y = mx + b$ has exactly one solution

2. Line tangent to circle: The system $x^2+y^2=1$ and $y=mx+b$ has exactly one solution

For tangency to $y=x^2$: Substituting gives $x^2=mx+b$, so $x^2-mx-b=0$. This has one solution when the discriminant is zero: $m^2+4b=0$, giving $b=-\frac{m^2}{4}$.

For tangency to $x^2+y^2=1$: Substituting $y=mx+b$ gives $x^2+(mx+b)^2=1$, so $(1+m^2)x^2+2mbx+(b^2-1)=0$. This has one solution when the discriminant is zero: $(2mb)^2-4(1+m^2)(b^2-1)=0$.

Substituting $b=-\frac{m^2}{4}$: $4m^2\cdot\frac{m^4}{16}-4(1+m^2)(\frac{m^4}{16}-1)=0$ This simplifies to $\frac{m^6}{4}-(1+m^2)(\frac{m^4}{4}-1)=0$ Expanding: $\frac{m^6}{4}-\frac{m^4}{4}+1-\frac{m^6}{4}+m^2=0$ This gives $-\frac{m^4}{4}+m^2+1=0$, so $m^4-4m^2-4=0$.

Let $u=m^2$: $u^2-4u-4=0$, so $u=2\pm2\sqrt{2}$. Since $u=m^2\geq0$, we have $u=2+2\sqrt{2}$, so $m=\pm\sqrt{2+2\sqrt{2}}$.

Therefore, the tangent lines are $y=\pm\sqrt{2+2\sqrt{2}}\cdot x-\frac{2+2\sqrt{2}}{4}$.

Special Cases:

- **Polar coordinates:** Tangency requires $r_1(\theta)=r_2(\theta)$ and $r_1'(\theta)=r_2'(\theta)$
- **Self-tangency:** Curve tangent to itself at isolated points or along segments
- **Inflection tangents:** When tangent line has third-order contact ($f'''\neq g'''$ but $f''=g''=0$)
- **Asymptotic tangency:** Curves approaching same asymptote
- **Tangency at infinity:** For projective curves, check tangency at points where $z=0$ in homogeneous coordinates
- **Algebraic curves of degree n:** Tangency requires multiplicity ≥2 at intersection point (discriminant method)
- **Transcendental curves:** Use Taylor series expansion to match derivatives at contact point
- **Parametric space curves:** Tangency requires matching position and tangent vector: $\vec{r}_1(t_1)=\vec{r}_2(t_2)$ and $\vec{r}_1'(t_1)\parallel\vec{r}_2'(t_2)$
- **Implicit surfaces:** For $F(x,y,z)=0$ and $G(x,y,z)=0$: $\nabla F\parallel\nabla G$ at tangency point
- **Complex curves:** Tangency in $\mathbb{C}$ requires matching complex derivatives: $f'(z_0)=g'(z_0)$

Error Analysis and Computational Considerations:

- **Numerical precision:** When using numerical methods, check that $|f(a)-g(a)|<\epsilon$ and $|f'(a)-g'(a)|<\epsilon$ for small ϵ
- **Derivative approximation:** For discrete data, use finite differences: $f'(x)\approx\frac{f(x+h)-f(x-h)}{2h}$ with appropriate step size

- **Singular points:** At cusps or vertical tangents, use parametric form or implicit differentiation to avoid division by zero
- **Multiple roots:** When solving $f(x)-g(x)=0$, use deflation or specialized algorithms for multiple roots
- **Convergence issues:** Newton-Raphson may fail near inflection points; use modified methods or bracketing
- **Curvature computation:** Avoid numerical differentiation of second derivatives; use analytical forms when possible
- **Parametric correspondence:** Ensure parameter values correspond to same geometric point when comparing derivatives

Works Well With: T2-CoordGeo (setting up coordinate systems); T24-TrigSub (parametric curves); T30-PolyDiv (algebraic curve analysis); T41-FuncEq (functional relationships); TM12-Discriminant (tangency via multiplicity); T8-CompleteSquare (curve form conversion); T7-AMGM (optimization with tangency constraints).

Typical Difficulty: Problems 18-25 on AMC 12; common in AIME and beyond. Basic tangency appears in Problems 15-18, while curvature analysis typically requires Problems 20+.

Success Rate: 60-70

Category Classification: High-Probability - Essential for advanced geometry and optimization problems. Critical for smooth curve design and intersection analysis. Frequently appears in combination with other techniques.